

CUSPIDAL WEIERSTRASS POINTS ON MODULAR CURVES

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ABSTRACT. We give a result on the asymptotic behavior of the number of groups that occur infinitely often as the torsion subgroup of an elliptic curve over a degree d number field. This motivates us to study cuspidal Weierstrass points on modular curves, especially on the family $X_1(M, N)$ that parametrizes $(\mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z})$ -torsion structures on elliptic curves. We show that with the single exception of $X_1(5, 10)$, when $M \geq 2$ all modular curves $X_1(M, N)$ of genus at least 2 have cuspidal Weierstrass points. Because when $M = N$ the modular automorphism group acts transitively on the cusps, we recover Schoeneberg's Theorem that *all* cusps on $X(N)$ are Weierstrass points. We also find several infinite families of modular curves $X(\Gamma)$ for which the modular automorphism group has more than one orbit on the cusps and for which every cusp is a Weierstrass point. Finally, we study when every Weierstrass point on a modular curve is a cusp and conjecture that this holds for only finitely many modular curves.

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1. INTRODUCTION

1.1. Overview and Provenance. The *Weierstrass points* of a nice curve $X/\mathbb{C}$ of genus at least 2 are a nonempty finite set of points that are deeply related to the geometry of X : e.g. $\mathrm{Aut} X$ acts on the set of Weierstrass points, and this is a faithful action in the nonhyperelliptic case, while in the hyperelliptic case the only nontrivial element in the kernel of the action is the hyperelliptic involution, and thus $\mathrm{Aut} X$ must be finite. On a “general” curve X of genus $g \geq 4$, it is difficult to find the Weierstrass points explicitly. However in 1951 a remarkable paper of Schoeneberg [Sc51] gave a criterion for fixed points of an automorphism of X to be Weierstrass points and applied that criterion to the modular curves $X(N)$, showing in particular that for all $N \geq 7$ – i.e., such that $X(N)$ has genus at least 2 – all the cusps on $X(N)$ are Weierstrass points. On $X(7)$ (the Klein quartic) and $X(8)$ (the Wiman curve) the Weierstrass points are precisely the cusps, while for all $N \geq 9$ Schoeneberg showed there are Weierstrass points with j -invariant 0 or 1728.¹

Schoeneberg's work does not seem to have attracted much attention for about ten years. Then in 1963 Lewittes [Le63] gave some refinements of Schoeneberg's general criterion for Weierstrass points, including recasting it in the appealing form that if a nontrivial automorphism on a Riemann surface has at least 5 fixed points, then all the fixed points are Weierstrass points. Soon after this Lehner–Newman [LN64] and then Atkin [At67] studied whether the cusp at ∞ on $X_0(N)$ is a Weierstrass point. Whereas Lehner–Newman used Schoeneberg's criterion for Weierstrass points

¹Weierstrass points on $X(N)$ were studied in a slightly earlier work of Petersson [Pe50], whose most significant result seems to be that $X(N)$ is never hyperelliptic of genus at least 2.

in terms of automorphisms, Atkin gave what amounts to a generalization of Schoeneberg's criterion that uses non-Galois coverings² $X \rightarrow Y$ to produce Weierstrass points on X . The work of Lehner–Newman and Atkin shows that for most non-squarefree values of N , the cusp at ∞ is a Weierstrass point of $X_0(N)$. In 1978, Ogg showed [Og78] that if N is prime, then no cusp on $X_0(N)$ is a Weierstrass point (and in fact slightly more than this, as we will recall later). Nevertheless, there are no known examples of squarefree N for which any cusp on $X_0(N)$ is a Weierstrass point, and for most squarefree values the problem remains open. After Ogg's work there was no progress on Weierstrass points on modular curves for a long time, until a 2008 work of Choi [Ch08] and a significant 2022 generalization thereof of Jeon [Je22].

The first author has had a longstanding but mostly passive interest in Weierstrass points on modular curves, especially in Ogg's work [Og78] and its refinement by Ahlgren–Ono [AO03], which has recently been activated by work with the second author. Namely, in a study of the asymptotic behavior of the classification of torsion subgroup of elliptic curves over degree d number fields, we used the cusp at 0 on the modular curve $X_1(M, N)$ to construct rational functions of all degrees $d \geq 2g(X_1(M, N))$. If this cusp is not a Weierstrass point, then the bound would improve to all $d \geq g(X_1(M, N)) + 1$, yielding for instance an improvement in the number of torsion subgroups that occur infinitely often in a number field of degree d : see §5. This motivated us to study cuspidal Weierstrass points on the family of curves $X_1(M, N)$, which includes as special cases the families $X(N) = X_1(N, N)$ and $X_1(N) = X_1(1, N)$.

We were thus led to study the above literature, which turned up two somewhat conflicting general principles:

I. Most of the methods to either prove or disprove that cusps on modular curves are Weierstrass points involve some amount of “modular functoriality”: that is, the analysis of Weierstrass points on a modular curve $X(\Gamma)$ usually involves a modular map $X(\Gamma) \rightarrow X(\Gamma')$ for a group $\Gamma \subseteq \Gamma' \subseteq \mathrm{PSL}_2(\mathbb{Z})$. E.g.: (i) to study $X(N)$, Schoeneberg uses the modular automorphism $\sigma = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, and the quotient $X(N) \rightarrow X(N)/\langle \sigma \rangle$ is the modular map $X(N) \rightarrow X_1(N)$. (ii) For a prime $p \nmid N$, Ogg uses the map $X_0(Np) \rightarrow X_0(N)$ in characteristic p to rule out cuspidal Weierstrass points on $X_0(Np)$. (iii) As we will see here, understanding how $\mathrm{Aut}_M X_1(M, N)$ acts on cusps of $X_1(M, N)$ involves the Atkin–Lehner subgroup of $\mathrm{Aut}_M X_0(\frac{N}{M})$.

II. Given a map $\pi : X(\Gamma) \rightarrow X(\Gamma')$ of modular curves and a point P on $X(\Gamma)$, knowing whether P (resp. $\pi(P)$) is a Weierstrass point is usually not directly helpful in determining whether $\pi(P)$ (resp. P) is a Weierstrass point. For instance: for a prime p , we have modular maps $X(p) \rightarrow X_1(p) \rightarrow X_0(p)$. As mentioned above, the cusp ∞ is always a Weierstrass point on $X(p)$ and never a Weierstrass point on $X_0(p)$. In the middle we have $X_1(p)$, on which – outside of finitely many values, by direct computation – it is completely unknown whether ∞ is a Weierstrass point.

Putting these two principles together, we find that it is natural to study Weierstrass points on multiple families of modular curves at once, while understanding that these various families will require separate consideration. The length of this paper is partly explained by the fact that we examine the cuspidality of Weierstrass points on *ten different families* of modular curves.

²In this paper, a **covering** of Riemann surfaces is the same as a finite morphism of nice complex algebraic curves: i.e., what from the topological perspective would be called a *branched covering*. In fact, in §2 we will see that we are rooting for rather than against the branch points.

This paper is partially expository: Section 2 gives background on Weierstrass points on curves over fields of characteristic 0, while Section 3 gives some background on modular curves. (However each of these sections contains some modest new results.) We have written this paper in such a way as to attempt to draw in those who already know about modular curves to the study of their Weierstrass points and also to collect various computational facts about modular curves for future use: e.g. for the ten families of modular curves we consider here, we list all such curves of genera 0, 1, 2 and 3 and all such hyperelliptic curves.

In general it seems that – with the notable exception of some recent work by the “Korean school” including Choi, Im, Jeon and Kim – it has been several decades since the modular curves community has taken a serious interest in Weierstrass points. We think that now could be a good time for a renewed interest in this subject, for several reasons. First, in our day those who study modular curves are interested in much more general families of modular curves than just $X(N)$, $X_1(N)$ and $X_0(N)$: whereas Mazur in 1978 laid out a program to study Galois representations on elliptic curves via rational points on various modular curves, only more recently have mathematicians become deeply interested in, say, Cartan-type modular curves, but now they most certainly are, both theoretically and computationally. Second, Weierstrass points on any curve are fundamentally about k -fold holomorphic differentials on that curve, hence Weierstrass points on a modular curve are fundamentally about modular forms, and computational tools for modular forms have burgeoned in recent years. Finally, we must call attention to the **L-functions and modular forms database (LMFDB)**, in which each of thousands of individual modular curves get their own page. All of the “named families” of modular curves in the LMFDB are considered here.

Finally, let us mention that the founding papers of Schoeneberg [Sc51] and Lewittes [Le63] were heavily group-theoretic: for a finite group G acting effectively by automorphisms on a Riemann surface X , both works studied the representation of G on the space $H^0(X, \Omega_X^1)$ of global holomorphic differentials. The refinements they attained have not found much use in the study of Weierstrass points on modular curves, and one may view Atkin’s work as “taking away the group theory” by giving a method that uses not necessarily Galois covers $X \rightarrow Y$ to produce Weierstrass points on X . However group theory is a primary theme of the present work: the more we understand the cuspidal orbits of the modular automorphism group $\text{Aut}_M X(\Gamma)$, the better equipped we are to either successfully apply the Schoeneberg–Lewittes method or rule out its applicability. For instance, the orbit of the modular automorphism group $\text{Aut}_M X_1(N)$ of $X_1(N)$ on the cusp at ∞ is unramified, and this is the only cuspidal $\text{Aut}_M X_1(N)$ -orbit if and only if N is squarefree. Thus the Schoeneberg–Lewittes method will never show that the cusp at ∞ is a Weierstrass point on $X_1(N)$, which when N is squarefree means that it will never show that any cusp is a Weierstrass point, while for non-squarefree N it may yet apply to cusps outside of the $\text{Aut}_M X_1(N)$ -orbit of ∞ .³ Conversely, Jeon’s result in [Je22] is precisely that (apart from an explicit, small set of exceptions that he analyzes completely) the Schoeneberg–Lewittes method *does* apply to all cusps on $X_1(N)$ not in the $\text{Aut}_M X_1(N)$ -orbit of ∞ . Some of the work in the present paper could be pushed further if we knew more about the structure of $\text{Aut}_M X_1(M, N)$ and its cuspidal orbits.

1.2. Main Results. For the definitions of the modular curves appearing here, see Section 3.1 and [HL25]. For the definitions of $\text{Aut}_M X(\Gamma)$ and $\Gamma_0^*(\frac{N}{M})$, see §3.5. For the definition of Hall cusps and non-Hall cusps, see §4. For a more precise definition of $\mathcal{T}^\infty(d)$, see Section 5.

³One cannot similarly “rule out” Atkin’s method: as we will see, it in principle finds all Weierstrass points on all curves...albeit not necessarily in a useful way.

Theorem 1.1. For $d \in \mathbb{Z}^+$, let $\mathcal{T}^\infty(d)$ be the set of pairs (M, N) of positive integers $M \mid N$ such that $\mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}$ occurs infinitely often as the torsion subgroup of an elliptic curve over some degree d number field. Then

$$(1) \quad \lfloor \sqrt{12d} \rfloor \leq \#\mathcal{T}^\infty(d) \leq 71.637\sqrt{d}.$$

Trying to bring the two coefficients of $\sqrt{d}$ in (1) closer together led us to consider Weierstrass points on the modular curves $X_1(M, N)$.

Theorem 1.2. Let $M, N \in \mathbb{Z}^+$ with $M \mid N$ and $N \neq 4$.

- a) (Kim–Koo [KK00], Zemel [Ze17]) The normalizer of $\Gamma_1(M, N)$ in $\mathrm{PSL}_2(\mathbb{R})$ is $\Gamma_0^*(\frac{N}{M})$, so the modular automorphism group $\mathrm{Aut}_M X_1(M, N)$ of $X_1(M, N)$ is $\Gamma_0^*(\frac{N}{M})/\Gamma_1(M, N)$.
- b) Let $W(\frac{N}{M}) \subseteq \mathrm{Aut}_M X_0(\frac{N}{M})$ be the Atkin–Lehner group. The natural map

$$\pi : X_1(M, N) \rightarrow X_0\left(\frac{N}{M}\right)$$

induces a bijection from the set of $\mathrm{Aut}_M X_1(M, N)$ -orbits of cusps on $X_1(M, N)$ to the set of $W(\frac{N}{M})$ -orbits of cusps on $X_0(\frac{N}{M})$. In particular, $\mathrm{Aut}_M X_1(M, N)$ acts transitively on the cusps of $X_1(M, N)$ if and only if $\frac{N}{M}$ is squarefree.

The $M = 1$ case of Theorem 1.2a) is due to Kim–Koo [KK00] and Lang [La01], independently. In [Ze17, Remark 3.6], Zemel deduces a result equivalent to Theorem 1.2 from several of his other results. We give a proof of Theorem 1.2a) that is a direct adaptation of the argument of Kim–Koo.

The following result is a generalization of Schoeneberg’s Theorem on $X(N)$:

Theorem 1.3. Let $M, N \in \mathbb{Z}^+$ with $2 \leq M \mid N$ be such that $g(X_1(M, N)) \geq 2$.

- a) The curve $X_1(5, 10)$ is **Bring’s curve**, of genus 4. On it, no cusp is a Weierstrass point.
- b) Suppose $(M, N) \neq (5, 10)$. Then:
 - (i) The cusp at ∞ is a Weierstrass point.
 - (ii) If $\frac{N}{M}$ is squarefree, then it follows from Theorem 1.2b) that every cusp on $X_1(M, N)$ is a Weierstrass point.
 - (iii) Suppose that $\mathrm{Aut}_M X_1(M, N)$ has two orbits on the cusps of $X_1(M, N)$: this happens if only if $N = 4bM$ or $8bM$ with b odd and squarefree, or if $N = 9bM$ with b indivisible by 3 and squarefree. Then every cusp on $X_1(M, N)$ is a Weierstrass point.

Theorem 1.4. Let $N \in \mathbb{Z}^+$ be such that $g(X_0(N)) \geq 2$.

- a) (Larcher [La71]) The curve $X_0(64)$ is the **Fermat quartic**, of genus 3. Its Weierstrass points are precisely the 12 cusps, each having weight 2.
- b) For all $N \neq 64$, there is a noncuspidal Weierstrass point on $X_0(N)$.

Theorem 1.5. Let $M, N \in \mathbb{Z}^+$ with $M \mid N$ be such that $g(X_1(M, N)) \geq 2$.

- a) (Jeon [Je15]) The curve $X_1(16)$ has genus 2. Its Weierstrass points are precisely the 6 non-Hall cusps, each having weight 1.
- b) (Schoeneberg [Sc51]) The curve $X(7)$ is the **Klein quartic**, of genus 3. Its Weierstrass points are precisely the 24 cusps, each having weight 1.
- c) (Lewittes [Le63b], Kato–Magaard–Volklein [KMV11]) The curve $X(8)$ is the **Wiman curve**, of genus 5. Its Weierstrass points are precisely the 24 cusps, each having weight 5.
- d) The curve $X_1(3, 12)$ has genus 3. Its Weierstrass points are precisely the 20 cusps, of which 16 have weight 1 and 4 have weight 2.
- e) For all other pairs (M, N) , there is a noncuspidal Weierstrass point on $X_1(M, N)$.

Corollary 1.6. *Let $N \in \mathbb{Z}^{\geq 7}$, so $X(N)$ has genus at least 2. Then $\text{Aut } X(N)$ acts transitively on the Weierstrass points of $X(N)$ if and only if $N \in \{7, 8\}$.*

Proof. Let $N \in \mathbb{Z}^{\geq 7}$. As shown by Bars–Kontogeorgis–Xarles [BKX13], we have

$$\text{Aut } X(N) = \text{Aut}_M X(N) \cong \text{PSL}_2(\mathbb{Z}/N\mathbb{Z}),$$

and thus $\text{Aut } X(N)$ is the Galois group of the Galois covering map $\pi : X(N) \rightarrow X(1)$.

By Theorem 1.5, if $N \in \{7, 8\}$, then the Weierstrass points are precisely the cusps – i.e., precisely the fiber over ∞ of the Galois covering $X(N) \rightarrow X(1)$ – so this action is transitive.

If $N \geq 9$, then by Theorem 1.5 every cusp of $X(N)$ is a Weierstrass point but not all Weierstrass points are cusps. Since $\text{Aut } X(N) = \text{Aut}_M X(N)$, every automorphism of $X(N)$ carries cusps to cusps, so $\text{Aut } X(N)$ has at least two orbits on the set of Weierstrass points. $\square$

When N is prime, Corollary 1.6 is due to Laing–Singer [LS12, Cor. 14]. If for some class of pairs (M, N) we knew that $\text{Aut}_M X_1(M, N) = \text{Aut } X_1(M, N)$ with finitely many exceptions, we would get a version of Corollary 1.6 for that class of curves $X_1(M, N)$. For instance, if $p \geq 23$ and $p \neq 37$ is a prime number, then $\text{Aut}_M X_1(p) = \text{Aut } X_1(p)$ (see Remark 3.12c), and thus on this family of curves the automorphism group does not act transitively on the Weierstrass points.

Theorem 1.7. *Let $N \in \mathbb{Z}^+$.*

- a) *Suppose $N \equiv 2 \pmod{4}$ and $g(X_{\text{ns}}(N)) \geq 2$, so $N \notin \{2, 6, 10\}$. Then every cusp on $X_{\text{ns}}(N)$ is a Weierstrass point.*
- b) *Suppose $N \equiv 2 \pmod{4}$ and $g(X_{\text{ns}}^+(N)) \geq 2$, so $N \notin \{2, 6, 10, 14, 18\}$. Then every cusp on $X_{\text{ns}}^+(N)$ is a Weierstrass point.*
- c) *Suppose $N \equiv 3, 6 \pmod{9}$ and $g(X_{\text{ns}}^*(N)) \geq 2$, so $N \notin \{3, 6, 12, 15, 21, 24, 30, 42\}$. Then every cusp on $X_{\text{ns}}^*(N)$ is a Weierstrass point unless $N = 60$, in which case no cusp is a Weierstrass point.*

The following is an easily stated special case of more general results proved in §7.2.

Theorem 1.8. *Let $p \geq 5$ be prime, let $a \in \mathbb{Z}^{\geq 3}$ be odd, and let $M \in \mathbb{Z}^+$ be prime to p . Then every cusp on $X_0(p^a M)$ is a Weierstrass point.*

Corollary 1.9. *The set of $N \in \mathbb{Z}^+$ for which every cusp on $X_0(N)$ is a Weierstrass point has positive density.*

Schoeneberg showed that for all N such that $X(N)$ has genus at least 2, then *all* the cusps on $X(N)$ are Weierstrass points, but since $\text{Aut}_M X(N)$ acts transitively on the cusps of $X(N)$, this is equivalent to showing that the cusp at ∞ is a Weierstrass point. So far as we know, before now there were only finitely many known modular curves with the following properties: (i) the modular automorphism group does not act transitively on the cusps; and (ii) all cusps are Weierstrass points. The above results exhibit three infinite families of such modular curves: certain curves $X_1(M, N)$ (Theorem 1.3); certain Cartan-type curves $X_{\text{ns}}^+(N)$ and $X_{\text{ns}}^*(N)$ (Theorem 1.7) and certain curves $X_0(N)$ (Theorem 1.8). In the former case, the modular automorphism group has exactly two cuspidal orbits, whereas in the latter two cases the number of orbits of the modular automorphism group on the cusps can grow arbitrarily large.

The work of this paper was done and written up in the first half of 2026. Soon after the results were obtained, the second author obtained an important complement to Theorem 1.3: for $M \mid N$ such that $X_1(M, N)$ has genus at least 2 and $(M, N) \neq (1, 18)$, every *non-principal cusp* on $X_1(M, N)$ – i.e., every cusp not in the $\text{Aut}_M X_1(M, N)$ -orbit of ∞ – is a Weierstrass point. The proof involves extending the notion of **diamond operators** from $X_1(N)$ to $X_1(M, N)$ and is

similar in spirit to Jeon’s proof of Theorem 4.9a) (which is the $M = 1$ special case of this result, although in view of Jeon’s work the second author’s proof assumes that $M \geq 2$). This is part of the second author’s thesis work and will appear shortly as a solo paper.

2. BACKGROUND ON WEIERSTRASS POINTS

In this paper, we work exclusively with **nice** – smooth, projective and geometrically integral – algebraic curves X defined over a field F of characteristic 0. Much of the time we will be working over $F = \mathbb{C}$, and we refer to nice curves over $\mathbb{C}$ as **Riemann surfaces** (thus for us a Riemann surface is compact and connected).

A Riemann surface X of genus at least 2 carries a finite nonempty set of distinguished points called “Weierstrass points.” For a Riemann surface X of any genus g and a point P of X , Riemann–Roch implies the existence of a nonconstant rational function on X of degree at most $g + 1$ that is regular away from P . The point P is a Weierstrass point if we can find such an f of degree at most g . It is easy to see that Weierstrass points do not exist when g is 0 or 1.

Moreover every $P \in X$ has a “Weierstrass weight” $w_P \in \mathbb{Z}^{\geq 0}$ such that $w_P \geq 1$ if and only if P is a Weierstrass point. Although the notion of Weierstrass weight is slightly more technical and perhaps less initially appealing than the notion of Weierstrass point, in practice, the notion of Weierstrass weight is often necessary in order to tell whether a set of known Weierstrass points on a curve is the set of all Weierstrass points. The Weierstrass weight w_P is defined using the set of all “pole orders” at P : i.e., degrees of meromorphic functions on X that are regular away from P . Inside this definition some useful algebra lives: the set of pole orders form a *numerical semigroup*, from which we can recover the genus of X and define the Weierstrass weight w_P . So we begin with some discussion of this basic algebraic structure.

2.1. Numerical Semigroups. Let S be a submonoid of $(\mathbb{Z}^{\geq 0}, +)$: a subset containing 0 and closed under addition. We say that S is a **numerical semigroup** if the $\mathbb{Z}$ -module it generates is all of $\mathbb{Z}$: in other words, if there are $x_1, \dots, x_m \in S$ that are not all divisible by any $d > 1$. A basic fact is that then $\mathbb{Z}^{\geq 0} \setminus S$ is finite [ClFT, Prop. 7.24]. The elements of $\mathbb{Z}^{\geq 0} \setminus S$ are called the **gaps** of S , and accordingly the elements of S are the **non-gaps** of S . The number of gaps is called the **genus** $g(S)$, the smallest positive non-gap is called the **multiplicity** $m(S)$, and the largest gap is called the **Frobenius number** $F(S)$ of S . For all $0 \leq n \leq F(S)$, at most one of n and $F(S) - n$ can lie in S – otherwise

$$F(S) = n + (F(S) - n) \in S$$

– so $g(S) \geq \frac{F(S)+1}{2}$ and thus $F(S) \leq 2g(S) - 1$. It follows that the set of numerical semigroups of a given genus is finite. For $g \in \{0, 1\}$ there is a unique numerical semigroup of genus g ; henceforth we will assume $g \geq 2$.

For a subset S of $\mathbb{Z}^{\geq 0}$, we denote by $\langle S \rangle$ the submonoid it generates. Every numerical subgroup has a unique generating set that is minimal with respect to inclusion.

For a numerical semigroup S of genus g with gaps $1 = b_1 < \dots < b_g$, we define its **weight**

$$|S| := \sum_{i=1}^g (b_i - i).$$

Example 2.1. Let $g \in \mathbb{Z}^{\geq 2}$.

- a) The unique numerical semigroup of genus g and weight 0 is the **ordinary semigroup**

$$W_{g,0} := \{0\} \cup \mathbb{Z}^{\geq g+1} = \langle g+1, g+2, \dots, 2g+1 \rangle,$$

with gaps $1 < 2 < \dots < g$.

- b) The unique numerical subgroup of genus g and weight 1 is the **simple semigroup**

$$W_{g,1} := \{g\} \cup \mathbb{Z}^{\geq g+2} = \langle g, g+2, \dots, 2g+1 \rangle,$$

with gaps

$$1 < \dots < g-1 < g+1.$$

Example 2.2. Let $g \in \mathbb{Z}^{\geq 2}$. We claim that there is a unique numerical semigroup H_g of genus g and multiplicity 2. Indeed, since $2 \in H_g$, also $2, 4, \dots, 2g \in H_g$; the remaining integers in the interval $[1, 2g-1]$ are $1, 3, \dots, 2g-1$, of which there are g , so they must be non-gaps. Thus we must have

$$H_g = \{0, 2, \dots, 2g-2\} \cup \mathbb{Z}^{\geq 2g},$$

which is indeed a numerical semigroup of genus g , called the **hyperelliptic semigroup of genus g** . Its weight is $\sum_{i=1}^g (2i-1-i) = \frac{g^2-g}{2}$. In fact, for any numerical semigroup S of genus g , we have $|S| \leq |H_g|$, with equality if and only if $S = H_g$.

Example 2.3. Let $g \in \mathbb{Z}^{\geq 3}$. We claim that there are two numerical subgroups of genus g with first two positive nongaps 4 and 6. Indeed, the even non-gaps $1 \leq a \leq 2g$ are precisely the $g-1$ numbers $4, 6, 8, \dots, 2g$, so S must have exactly one odd non-gap $3 \leq a \leq 2g-1$. We must have either $a = 2g-1$ or $a = 2g-3$, because if a were any smaller we could add 4 to get another odd non-gap less than $2g$. We call the semigroup that contains $2g-3$ the **light bielliptic semigroup of genus g** :

$$\text{Bl}_g = \langle 4, 6, 2g-1, 2g+1 \rangle.$$

We have $\text{Bl}_3 = W_{g,0}$, $\text{Bl}_4 = W_{g,1}$. For all $g \geq 3$ we have $|\text{Bl}_g| = \frac{(g-2)(g-3)}{2}$. We call the semigroup that contains $2g-1$ the **heavy bielliptic semigroup of genus g** :

$$\text{Bh}_g = \langle 4, 6, 2g-3 \rangle.$$

For all $g \geq 3$ we have $|\text{Bh}_g| = \frac{(g-2)(g-3)}{2} + 2$.

Example 2.4. A numerical semigroup S is **trigonal** if $m(S) = 3$. A trigonal numerical semigroup is uniquely determined by its genus and Frobenius number [Ro05, Cor. 4], so if for $g \geq 2$ and $g \leq F \leq 2g-1$ there is a trigonal numerical semigroup of genus g and Frobenius number F , we denote it by $T_{g,F}$.

Let $g \in \mathbb{Z}^{\geq 3}$ with $g \not\equiv 2 \pmod{3}$. Then $T_{g,2g-1}$ exists: it is $\langle 3, g+1 \rangle$. We have $|T_{g,2g-1}| = \frac{g^2-g}{3}$. It has the largest Frobenius number of any genus g trigonal semigroup, so we abbreviate it to Tm_g .

Example 2.5 (Numerical Semigroups of Very Low Genus).

- a) The numerical semigroups of genus 2 are $W_{2,0}$ and $W_{2,1} = H_2$.
- b) The numerical semigroups of genus 3 are $W_{3,0} = \text{Bl}_3$, $W_{3,1}$, $\text{Bh}_3 = \text{Tm}_3$, and H_3 . Their weights are 0, 1, 2 and 3.
- c) The numerical semigroups of genus 4 are: $W_{4,0}$, of weight 0; $W_{4,1} = \text{Bl}_4$, of weight 1; $T_{4,6} = \langle 3, 7, 8 \rangle$ and $\langle 4, 5, 7 \rangle$, of weight 2; Bh_4 , of weight 3; $\text{Tm}_4 = \langle 3, 5 \rangle$, of weight 4; and H_4 , of weight 6.

For $g \in \mathbb{Z}^{\geq 3}$, let $M(g)$ be the maximal weight of a nonhyperelliptic numerical semigroup of genus g . We want to know about $M(g)$ so as to get a lower bound on the number of Weierstrass points on a nonhyperelliptic curve of genus g , which we will then use to prove that certain modular curves have non-cuspidal Weierstrass points. There is a phenomenon here that we must take g larger to

see. We are indebted to N. Kaplan, who used Python and Gap to compute, for each $5 \leq g \leq 20$, all numerical semigroups of genus g of maximal nonhyperelliptic weight. This is the answer:

Example 2.6. (N. Kaplan)

- $g = 5$: maximal weight 5; attained by Bh_5 , $T_{5,8} = \langle 3, 7, 11 \rangle$.
- $g = 6, 7, 9$: maximal weight $\frac{g^2-g}{3}$; attained by Tm_g .
- $g = 10$: maximal weight $\frac{10^2-10}{3} = 30 = \frac{(10-2)(10-3)}{2} + 2$; attained by Bh_{10} , Tm_{10} .
- $g = 8, 11 \leq g \leq 20$: maximal weight $\frac{(g-2)(g-3)}{2} + 2$; attained by Bh_g .

So starting at $g = 6$, it seems that Tm_g and Bh_g are the only competitors. If so, then since Tm_g is not defined for $g \equiv 2 \pmod{3}$ and $\frac{(g-2)(g-3)}{2} + 2 > \frac{g^2-g}{3}$ for all $g \geq 11$, the results are explained. Moreover we predict that for all $g \geq 11$ the unique nonhyperelliptic genus g semigroup of maximal weight is Bh_g , so in particular that $M(g) = \frac{(g-2)(g-3)}{2} + 2$ for all $g \geq 10$. We will return to this shortly in the context of Weierstrass semigroups.

2.2. The Weierstrass Semigroup and the Weierstrass Divisor. Let X/F be a nice curve of genus $g \geq 2$ defined over a field F of characteristic 0.

We first suppose that F is algebraically closed, and let $P \in X(F)$. The **Weierstrass semigroup** at W_P is the set of $n \in \mathbb{N}$ such that there is a rational function $f \in F(X)$ regular away from P and with $\text{ord}_P(f) = -n$. Constant functions give that $0 \in W_P$, and the positive elements of W_P are the $n \in \mathbb{Z}^{\geq 1}$ such that there is a rational function regular away from P and with a pole of order exactly n at P . Since $\text{ord}_P(fg) = \text{ord}_P(f) + \text{ord}_P(g)$, clearly W_P is a semigroup. A positive integer b is a gap for W_P if and only if there is no rational function on X regular away from P and with a pole of order b at P if and only if the dimensions of the Riemann–Roch spaces of the divisors $(b-1)[P]$ and $b[P]$ coincide: $h^0((b-1)[P]) = h^0(b[P])$. For all $b \in \mathbb{Z}^+$ we have

$$h^0(b[P]) - h^0((b-1)[P]) \in \{0, 1\}.$$

Clearly $h^0(0[P]) = 1$, while by Riemann–Roch, for all $d \in \mathbb{Z}^{\geq 0}$ we have $h^0((2g-1+d)[P]) = g+d$, we deduce the **Weierstrass Gap Theorem**: W_P is a numerical semigroup of genus g .

The **Weierstrass weight** w_P of P is $w_P := |W_P|$, the weight of the Weierstrass semigroup at P . We say P is a **Weierstrass point** if $w_P \geq 1$; this holds if and only if there is a non-gap $b \leq g$, i.e., if there is a nonconstant rational function on X that is regular away from P and having a pole of order at most g at P . Then WE CLAIM that there are only finitely many points $P \in X(F)$ with $w_P \geq 1$, so we may define the **Weierstrass divisor**

$$\mathbb{W}(X) := \sum_{P \in X} w_P [P] \in \text{Div } X.$$

Moreover WE CLAIM the fundamental result that the degree of the Weierstrass divisor is $g^3 - g$. Both of these claims will be explained in more detail and with precise citations in §2.3.

Now suppose that F is any field of characteristic 0, with algebraic closure $\overline{F}$ and absolute Galois group $\mathfrak{g}_F := \text{Aut}(\overline{F}/F)$. Then for all $P \in X(\overline{F})$ and all $\sigma \in \mathfrak{g}_F$ we have $w_{\sigma(P)} = w_P$. It follows that as a divisor on $X_{/\overline{F}}$ we have that $\mathbb{W}(X)$ is $\mathfrak{g}_F$ -invariant and thus defines a divisor on X that we continue to denote by $\mathbb{W}(X)$.

Remark 2.7. Is every numerical semigroup W of genus $g \geq 2$ a Weierstrass semigroup? More precisely, for each such W , is there a Riemann surface X of genus g and a point $P \in X$ such that

$W = W_P$? This question was asked by Hurwitz in 1892, but the first (correct!) example of a non-Weierstrass numerical semigroup W was given by Buchweitz in 1976 [Bu76]. Buchweitz's example has genus 16. Conversely, it is known that for all $2 \leq g \leq 7$, every genus g numerical semigroup is a Weierstrass semigroup. A recent work of Eisenbud–Schreyer [ES26] contains further information on the history of this problem and also constructs a non-Weierstrass numerical semigroup of genus 13, which as of 2026 is the smallest known genus of a non-Weierstrass semigroup.

2.3. Holomorphic Differentials. In this section we use the language of Riemann surfaces. We do so just because we like the exposition in the text of Farkas–Kra [FK]. Everything carries over verbatim to nice curves over an algebraically closed field F of characteristic 0: one can either follow [FK] in this algebraic setting, without any issue, or one can make an *a priori* reduction to the case of $F = \mathbb{C}$, an instance of the “Lefschetz Principle.”

Let X be a Riemann surface of genus $g \geq 2$. There is an important “dual” characterization of Weierstrass points in terms of holomorphic differentials. Indeed, the following is a standard application of Riemann–Roch [FK, III.5.4 and III.5.8]:

Lemma 2.8. *Let X be a Riemann surface, let $P \in X$, and let $i \in \mathbb{Z}^+$.*

- a) *The following are equivalent:*
 - (i) *The integer i is a gap at P : i.e., $i \notin W_P$.*
 - (ii) *There is a nonzero holomorphic differential ω on X such that $\text{ord}_P(\omega) = i - 1$.*
- b) *P is a Weierstrass point if and only if there is a nonzero holomorphic differential ω on X such that $\text{ord}_P(\omega) \geq g$.*
- c) *If the gaps at P are $1 = b_1 < \dots < b_g \leq 2g - 1$, then the possible orders of a holomorphic differential at P are $\{b_i - 1 \mid 1 \leq i \leq g\}$.*

It follows from Lemma 2.8 that there are holomorphic differentials $\omega_1, \dots, \omega_g$ on X such that

$$0 = \text{ord}_P \omega_1 < \dots < \text{ord}_P \omega_g,$$

and then we must have that $\omega_1, \dots, \omega_g$ is a $\mathbb{C}$ -basis of $H^0(X, \Omega_X^1)$. Farkas–Kra call this an “adapted basis,” and such a basis gives the Weierstrass gaps at P – they are $\text{ord}_P(\omega_i) + 1$ for $1 \leq i \leq g$ – and therefore also the Weierstrass weight as

$$w_P = \sum_{i=1}^g (\text{ord}_P \omega_i - i + 1).$$

Indeed w_P can be computed from *any* basis for the holomorphic differentials: let $\underline{\omega} := (\omega_1, \dots, \omega_g)$ be a $\mathbb{C}$ -basis for $H^0(X, \Omega_X^1)$. Let $P \in X$ and let z be a rational function on X such that $\text{ord}_P(z) = 1$. Then there are $\varphi_1, \dots, \varphi_g \in \mathbb{C}(X)$ such that $\omega_i = \varphi_i dz$ for all $1 \leq i \leq g$. Let $M_{\underline{\omega}}(z) \in M_{g,g}(\mathbb{C})$ be the matrix with $M_{\underline{\omega}}(z)(i, j) = (\frac{d}{dz})^{i-1} \varphi_j$, and let

$$\Phi_{\underline{\omega}}(z) := \det M_{\underline{\omega}}(z)(i, j),$$

the **Wronskian**. Then:

Theorem 2.9. *With notation as above, we have:*

- a) *$\Phi_{\underline{\omega}}(z) dz^{\frac{g(g+1)}{2}}$ is a nonzero $\frac{g(g+1)}{2}$ -fold holomorphic differential, i.e., a nonzero element of $H^0(X, \Omega_X^{\frac{g(g+1)}{2}})$.*
- b) *For all $P \in X$, we have $\text{ord}_P \Phi_{\underline{\omega}}(z) = w_P$, the Weierstrass weight of P .*
- c) *The Weierstrass divisor on X has degree $g^3 - g$.*

Proof. a) See [FK, III.5.9]. b) See [FK, III.5.8]. c) By part b), the Weierstrass divisor $\mathbb{W}(X)$ is the divisor of $\Phi_{\underline{\omega}}(z) dz^{\frac{g(g+1)}{2}}$, which has degree $(2g - 2) \cdot \frac{g(g+1)}{2} = g^3 - g$. $\square$

As we will see later, when X is a modular curve and P is a cusp on X , then Theorem 2.9 tells us that we can compute the Weierstrass weight w_P from a basis for the weight 2 cusp forms on Γ , each expressed as a q -expansion at P .

2.4. Irreducibility of the Weierstrass Divisor. Let X/F be a nice curve of genus $g \geq 2$ defined over a field F of characteristic 0. There are two different senses in which the Weierstrass divisor $\mathbb{W}(X)$ may be irreducible: as a divisor, or as a scheme. We say $\mathbb{W}(X)$ is **strongly irreducible** if $\mathbb{W}(X)$ is an irreducible F -divisor on X : this means the absolute Galois group $\mathbf{g}_F$ acts transitively on the Weierstrass points of $X/\overline{F}$ and every such Weierstrass point has weight 1 (equivalently, there are $g^3 - g$ distinct Weierstrass points). We say $\mathbb{W}(X)$ is **irreducible** if $\mathbb{W}(X)$ is an irreducible F -scheme, which since $\mathbb{W}(X)$ is irreducible is equivalent to the underlying topological space having a single point and also equivalent to the action of $\mathbf{g}_F$ on $\overline{F}$ -Weierstrass points being transitive.

Example 2.10. Let X/F be a hyperelliptic curve of genus g , so there is $P \in F[x]$ a separable polynomial of degree $2g + 2$ so that X has affine equation. Each Weierstrass point has weight $\frac{g^2 - g}{2}$, so $\mathbb{W}(X)$ can only be strongly irreducible when $g = 2$, and then weak irreducibility and strong irreducibility coincide. Moreover $\mathbb{W}(X)$ is weakly irreducible if and only if $P \in F[x]$ is an irreducible polynomial. Hilbert Irreducibility implies that over a number field, 100% of hyperelliptic curves (in the sense of height functions on the standard universal family) are weakly irreducible.

Example 2.11. Let $g \geq 2$, and let F be a field of characteristic 0, and let $\mathcal{M}_g$ be the moduli space of genus g curves, viewed as a variety over F : it is quasi-projective and irreducible of dimension $3g - 3$. A **universal curve of genus g** is a curve $X_{/F(\mathcal{M}_g)}$ that maps to the generic point on $\mathcal{M}_g$: such a curve exists and is unique up to isomorphism over the algebraic closure of $F(\mathcal{M}_g)$. It has been proved by Harris ($g = 2$), Canuto ($g = 3$) and Eisenbud–Harris ($g \geq 4$) that the Weierstrass divisor on a universal curve $X_{/F(\mathcal{M}_g)}$ is strongly irreducible. Moreover, if the Weierstrass points on the universal curve are $P_1, \dots, P_{g^3 - g}$, then these authors prove that the Galois group of the splitting field $F(\mathcal{M}_g)(P_1, \dots, P_{g^3 - g})/F(\mathcal{M}_g)$ of the Weierstrass divisor is as large as possible, namely the full symmetric group $S_{g^3 - g}$ on $\{P_1, \dots, P_{g^3 - g}\}$.

Example 2.11 exhibits curves of all genera $g \geq 2$ with strongly irreducible Weierstrass divisors. The only “catch” is that these curves have transcendental moduli: they cannot be defined over number fields. If $(\mathcal{M}_g)_{/\mathbb{Q}}$ is a rational variety, Hilbert Irreducibility applies to show that for “most” $\mathbb{Q}$ -rational points on the moduli space, the corresponding Weierstrass divisor is strongly irreducible. This rationality result is known for $g \in \{2, 3\}$, is unknown for some intermediate cases, and is known to be false for all $g \geq 22$: for these values, $\mathcal{M}_g$ has positive Kodaira dimension. To the best of our knowledge, it is unknown whether genus g curves over $\mathbb{Q}$ with strongly irreducible Weierstrass divisors exist for all g ; we (the authors, at least) do not even know whether for infinitely many g there is a genus g nonhyperelliptic curve over $\mathbb{Q}$ with irreducible Weierstrass divisor.

2.5. Hyperelliptic Curves and Their Weierstrass Points. We first consider hyperellipticity in the context of Riemann surfaces. We say a Riemann surface is **hyperelliptic** if it admits a degree 2 rational function.

Theorem 2.12. *Let X be a Riemann surface of genus $g \geq 2$. The following are equivalent:*

- (i) X is hyperelliptic.
- (ii) At every Weierstrass point P of X , we have $W_P = H_g$.
- (iii) There is a point $P \in X$ for which the Weierstrass semigroup W_P at P is H_g .
- (iv) There are precisely $2g + 2$ Weierstrass points on X .

Proof. Suppose that (i) holds. By Riemann–Hurwitz f has $2g + 2$ fixed points; let P be one. After composing f with an automorphism of $\mathbb{P}^1$, we may assume that $f(P) = \infty$; since f ramifies at P

we find that 2 is a pole order at P , so by Example 2.2 we have $W_P = H_g$, so the Weierstrass weight at P is $\frac{g^2-g}{2}$. The fixed points of f contribute $(2g+2) \cdot \frac{g^2-g}{2} = g^3 - g$ to the Weierstrass divisor, so they are all the Weierstrass points of X . Thus we've shown that (i) implies (ii) and (iv).

Clearly (ii) implies (iii), and if (iii) holds, then $w \in W_P$ means there is a degree 2 rational function on X , so (i), (ii) and (iii) are equivalent and imply (iv). If (iv) holds, then since the Weierstrass divisor has degree $g^3 - g = (2g+2) \cdot \frac{g^2-g}{2}$, then for every Weierstrass point P we have $|W_P| = \frac{g^2-g}{2}$, hence by Example 2.2 we have $W_P = H_g$. $\square$

A striking feature of Theorem 2.12 is the equivalence of conditions (ii) and (iii): in this case, knowing the Weierstrass semigroup at one Weierstrass point determines the Weierstrass semigroups at all Weierstrass points. This is generally not the case for nonhyperelliptic curves, but we will soon see one more case in which something close to this is true.

We now work over a field F of characteristic 0 and take the same definition: a nice curve $X_{/F}$ of genus g is **hyperelliptic** if there is a degree 2 rational function on X . There is then a separable polynomial $f \in F[x]$ of degree $2g+2$ such that X is the smooth projective curve birationally equivalent to the affine curve

$$X^\circ : y^2 = f(x).$$

Moreover X° is smooth and $X \setminus X^\circ$ consists of two points at ∞ (which will be F -rational if and only if the leading coefficient of P is a square in F). If the roots of f in an algebraic closure $\bar{F}$ are $r_1, \dots, r_{2g+2}$, then the Weierstrass points on X are $(r_1, 0), \dots, (r_{2g+2}, 0)$.

An **involution** on a curve $X_{/F}$ is an order 2 automorphism; it gives a degree 2 map $X \rightarrow X/\langle \iota \rangle$. Conversely, because we are not in characteristic 2, every degree two morphism $f : X \rightarrow Y$ of curves induces a quadratic Galois extension $F(X)/F(Y)$ and is thus, up to isomorphism on Y , the quotient by an involution ι on X . We say that an involution ι on X is **hyperelliptic** if $X/\langle \iota \rangle$ is isomorphic to $\mathbb{P}^1$. The Castelnuovo–Severi Inequality [St, Thm. 3.11.3] implies that if X/F is a curve of genus $g \geq 2$, then $X_{/\bar{F}}$ can have at most one hyperelliptic involution, which must therefore lie in the center of $\text{Aut } X_{/\bar{F}}$.

Remark 2.13.

- a) Every curve of genus 0 or 2 is hyperelliptic.
- b) A curve $X_{/F}$ is **geometrically hyperelliptic** if $X_{/\bar{F}}$ is hyperelliptic. Every genus 1 curve is geometrically hyperelliptic. Of course hyperelliptic curves are geometrically hyperelliptic. Conversely, if $g \geq 2$ and $X_{/F}$ is geometrically hyperelliptic, then if ι is a hyperelliptic involution on $X_{/\bar{F}}$, then its uniqueness over $\bar{F}$ forces it to be $\mathbf{g}_F$ -invariant, and thus ι is defined over F . The only snag is that $X/\langle \iota \rangle$ has genus 0 but need not be F -rationally isomorphic to $\mathbb{P}^1$: that is, it may be a pointless conic. In this case a canonical divisor gives a 2-to-1 map $\varphi : X \rightarrow \mathbb{P}^{g-1}$, hence a hyperplane in $\mathbb{P}^{g-1}$ cuts out a degree $g-1$ divisor on the conic $\varphi(X)$, so if g is even then $\varphi(X)$ has odd index and thus is isomorphic to $\mathbb{P}^1$: in even genus, geometrically hyperelliptic curves are hyperelliptic. Similarly, if $X_{/F}$ is geometrically hyperelliptic and has an F -rational point P , then the image of P on $X/\langle \iota \rangle$ is F -rational, so the conic is isomorphic to $\mathbb{P}^1$.

2.6. Interlude on Isotropy Groups. Let G be a finite group acting effectively by automorphisms on the Riemann surface X . For $P \in X$, we define the **isotropy group**

$$G_P := \{g \in G \mid gP = P\}.$$

Let $\pi : X \rightarrow X/G$ be the quotient map, and let $\bar{P} := \pi(P)$. Then the set-theoretic fiber $\pi^{-1}(\bar{P})$ is isomorphic as a G -set to the coset space G/G_P , and the isotropy groups of the elements of $\pi^{-1}(\bar{P})$ are precisely the conjugates of G_P .

It is a basic fact that if k has characteristic 0, every isotropy group G_P is cyclic. The Lefschetz Principle reduces this to the case of Riemann surfaces, for which a complex analytic proof is given in [FK, §III.7.7]. One can also reduce it to algebraic number theory: let $\bar{Q}$ be a point of X/G different from $\bar{P}$. Then the ring A of rational functions regular away from $\bar{Q}$ is a Dedekind domain for which the maximal ideals are naturally in bijection with the points of $X \setminus \{\bar{Q}\}$ and the residue field of every maximal ideal is $\mathbb{C}$. Let B be the integral closure of A in the field $\mathbb{C}(X)$, so B is also a Dedekind domain whose maximal ideals are in bijection with the points of $X \setminus \pi^{-1}(\bar{P})$ and each of whose residue fields is $\mathbb{C}$. Then the isotropy group G_P is the inertia group $I(P|\bar{P})$; since the residue characteristic is 0, the ramification is tame, so G_P is cyclic. We deduce:

Lemma 2.14. *Let G be a finite group acting effectively by automorphisms on a Riemann surface. Suppose $g_1, g_2 \in G$ have the same order but generate different subgroups. Then the fixed point sets of g_1 and g_2 are disjoint.*

Proof. If P is a common fixed point of g_1 and g_2 , the isotropy group G_P would contain two different cyclic subgroups of the same order and hence not be cyclic. $\square$

We also have the following basic fact about group actions:

Lemma 2.15. *Let G be a group acting on a set X . Suppose $g \in G$ is such that $\langle g \rangle$ is a normal subgroup of G . Then the set $\text{Fix}(g)$ of fixed points of g is a union of G -orbits.*

Proof. For any $g, \sigma \in G$, we have $\text{Fix}(\sigma g \sigma^{-1}) = \sigma \text{Fix}(g)$, so

$$\bigcup_{\sigma \in G} \text{Fix}(\sigma g \sigma^{-1}) = G \cdot \text{Fix}(g)$$

is a union of G -orbits. If moreover $\langle g \rangle$ is normal in G , then for all $\sigma \in G$ we have that $\langle \sigma g \sigma^{-1} \rangle = \langle g \rangle$, so – each of g and $\sigma g \sigma^{-1}$ being a power of the other – they have the same fixed points. Thus:

$$\text{Fix}(g) = \bigcup_{\sigma \in G} \text{Fix}(\sigma g \sigma^{-1}) = G \cdot \text{Fix}(g). \quad \square$$

2.7. Weierstrass Points on Bielliptic and Trigonal Curves. A nice curve X/F over a field of characteristic 0 is **bielliptic** if it admits a degree 2 map $f : X \rightarrow E$ to an elliptic curve (i.e., a genus 1 curve with an F -rational point). As above, f must be the quotient by an involution ι , and such involutions are called **bielliptic**. No curve of genus 0 is bielliptic; a genus 1 curve over an algebraically closed field is bielliptic but a genus 1 curve over a non-algebraically closed field may not be. From now on we restrict to the case of $g \geq 2$.

Castelnuovo–Severi implies if $g \geq 6$, X can have at most one bielliptic involution, which must therefore lie in the center of X , and also that X cannot be both hyperelliptic and bielliptic if $g \geq 4$.

Example 2.16.

- a) As we will see later in Example 6.1, the modular curve $X_0(37)$ has genus 2 and is both hyperelliptic and bielliptic. Its bielliptic fixed points are not Weierstrass points. Indeed, in genus 2 bielliptic fixed points are never Weierstrass points: genus 2 curves are hyperelliptic, the Weierstrass points are precisely the hyperelliptic fixed points, and by Lemma 2.14 no point of a Riemann surface of genus at least 2 is simultaneously fixed by more than one involution. Henceforth we restrict to curves of genus at least 3.

- b) The modular curves $X_0(30)$, $X_0(33)$, $X_0(35)$, $X_0(39)$, $X_0(40)$ and $X_0(48)$ all have genus 3 and, as curves over $\mathbb{Q}$, are both hyperelliptic and bielliptic.

Theorem 2.17. *Let X be a Riemann surface of genus $g \geq 3$.*

- a) *Let ι be a bielliptic involution on X . Then ι has $2g - 2$ fixed points. For each fixed point P of ι , the Weierstrass semigroup is either the light bielliptic semigroup Bl_g or the heavy bielliptic semigroup Bh_g . Thus the Weierstrass weight at P is either $\frac{(g-2)(g-3)}{2}$ or $\frac{(g-2)(g-3)}{2} + 2$. We call P a **light bielliptic fixed point** in the former case and a **heavy bielliptic fixed point** in the latter case. It follows that:*
- (i) *If $g \geq 4$, then every bielliptic fixed point is a Weierstrass point.*
 - (ii) *If $g = 3$, then a light bielliptic fixed point is not a Weierstrass point, while a heavy bielliptic fixed point is a Weierstrass point of weight 2.*
- b) *If $g \geq 8$ and P is a Weierstrass point with W_P equal to either Bh_g or Bl_g , then P is the fixed point of a (necessarily unique) bielliptic involution.*

Proof. a) Taking the quotient by ι gives a degree 2 map $f : X \rightarrow E$ for some elliptic curve E . It is immediate from Riemann–Hurwitz that f has $2g - 2$ fixed points; let P be one. By Riemann–Roch, for $d \in \{2, 3\}$ there is a degree d rational function $g_d : E \rightarrow \mathbb{P}^1$ regular away from $f(P)$ and with a pole of order d at $f(P)$. Thus $g_2 \circ f$ shows that 4 is a pole order at P , while $g_3 \circ f$ shows that 6 is a pole order at P . If $g \geq 4$ then X is not hyperelliptic, so 2 is not a pole order at P . If $g = 3$ then X may be hyperelliptic, but as above P cannot simultaneously be the fixed point of a hyperelliptic and a bielliptic involution, so by Theorem 2.12, 2 is not a pole order at P . By Example 2.3, we conclude that W_P is either Bh_g or Bl_g , and the rest is immediate.

b) This was shown by Kato [Ka79, §2] – the result stated there is a bit weaker, but the argument given establishes the general case. See also [Ko83, Prop. 2.9]. $\square$

In view of Theorem 2.17 we call a bielliptic involution on a Riemann surface of genus $g \geq 3$ **totally light** if all its bielliptic fixed points are light, **totally heavy** if all of its bielliptic fixed points are heavy, and **mixed** if it had both light and heavy bielliptic fixed points. For each $g \geq 6$ Park [Pa98] constructs Riemann surfaces of genus g admitting totally light, totally heavy and mixed bielliptic involutions (in fact he shows that the number s of heavy bielliptic fixed points can take any value between 0 and $2g - 2$ except possibly for 2 and $2g - 3$).

Theorem 2.17b) does not in general hold if $g \leq 7$: see [Co88, Cor. 21]. In other words – beware! – for these genera, a bielliptic Weierstrass semigroup need not come from a bielliptic fixed point or even from a bielliptic curve. When $g = 3$, a point P has Weierstrass semigroup Bl_3 if and only if it is *not* a Weierstrass point, and then of course P need not be a bielliptic fixed point. In the case $W_P = \text{Bh}_3$, the assertion is: a genus 3 Riemann surface with a point of Weierstrass weight 2 need not be bielliptic. A specific example is the plane quartic

$$C : Y^4 + X^3Z + XY^2Z + XZ^3 + YZ^3 + Z^4.$$

Let X be a bielliptic Riemann surface of genus $g \geq 2$. If $g \geq 6$, then as mentioned above the bielliptic involution ι on X is unique, hence the $2g - 2$ fixed points of ι cannot be all the Weierstrass points. In view of Park’s work, knowing that a Weierstrass semigroup W_P is either Bh_g or Bl_g does not determine W_Q at every other point, but at *many* other points it leaves just two possibilities.

For $3 \leq g \leq 5$ the possibility of multiple bielliptic involutions makes things more interesting:

- Suppose $g = 5$. Each bielliptic involution has 8 fixed points, each of which has Weierstrass weight 3 or 5, and Lemma 2.14 implies that distinct involutions have no fixed points in common,

so if a denotes the number of bielliptic involutions on X , then the various bielliptic fixed points contribute at least $24a$ to the degree of the Weierstrass divisor, which has total degree $5^3 - 5 = 120$, so there are at most 5 bielliptic involutions. Also a cannot be 4 [KMV11, Lemma 2.6]. Thus there are two possible ways for every Weierstrass point to be a bielliptic fixed point: either there are three totally heavy bielliptic involutions or five totally light bielliptic involutions. In Example 8.3 we will see that the modular curve $X(8)$ has three totally heavy bielliptic involutions. This is in fact the unique genus 5 Riemann surface with this property [KMV11, Thm. 3.5], called the **Wiman curve** and also characterized among genus 5 Riemann surfaces by having more than 160 automorphisms [KMV11, Cor. 3.6]. Moreover the genus 5 Riemann surfaces with five totally light bielliptic involutions form a two-dimensional family [Va86], called **Humbert curves**. Within this family is the **special Humbert curve** H , the unique genus 5 curve with 160 automorphisms [KMV11, Prop. 2.10].

- Suppose $g = 4$. Each bielliptic involution has 6 fixed points, and a light bielliptic fixed point has weight 1; since the Weierstrass divisor has total degree $4^3 - 4 = 60$, now there can be at most 10 bielliptic involutions. Later we will see that the modular curve $X_1(5, 10)$ attains this bound, and its 60 light bielliptic fixed points are precisely the Weierstrass points.
- Suppose $g = 3$. Now a light bielliptic fixed point is not a Weierstrass point, so we cannot argue as above. Instead: a Riemann surface of genus 3 has at most 168 automorphisms, and all automorphism groups of genus 3 curves are known, so we can bound the number of bielliptic involutions by the largest possible number of involutions on a genus three Riemann surface, which turns out to be 21. In fact, on a nonhyperelliptic Riemann surface of genus 3, every involution is a bielliptic involution [Ac, Lemma 5.10]. Later we will see that the modular curve $X(7)$, which is nonhyperelliptic of genus 3, has 21 (necessarily bielliptic) involutions, but all are totally light, hence their fixed points are not Weierstrass points. The modular curve $X_0(64)$ has 15 bielliptic involutions, of which 12 are totally light and three are totally heavy, so the 12 heavy bielliptic fixed points are precisely the Weierstrass points on $X_0(64)$.

A Riemann surface X is **trigonal** if it admits a degree 3 map $f : X \rightarrow \mathbb{P}^1$. The Castelnuovo–Severi inequality implies that if the genus is at least 3, X cannot be both hyperelliptic and trigonal, and if the genus is at least 5, X can have at most one trigonal map up to an automorphism of $\mathbb{P}^1$ and cannot be both bielliptic and trigonal. Contrary to the bielliptic case, what is clear is that if the Weierstrass semigroup at a point P of X is trigonal – i.e., 3 is the least positive non-gap at P – then f admits a trigonal map that is totally ramified at P .

Now suppose that X is nonhyperelliptic of genus 4. By Brill–Noether, X is trigonal. In fact, by [Ka80, Lemma 1] X admits either 1 or 2 trigonal maps. More precisely, X admits a unique trigonal map if and only if there is a divisor D of degree 3 with $h^0(D) = 2$ and such that $2D$ is a canonical divisor (the divisor of a nonzero meromorphic differential). Recall from Example 2.5 that the unique numerical semigroup of genus 4 and weight 4 is $\text{Tm}_4 = \langle 3, 5 \rangle$. In [Ka80, §3], Kato observes that as a consequence of Riemann–Roch, if P is a Weierstrass point of X of weight 4 then $h^0(3P) = 2$ and $6P$ is a canonical divisor, so in this case the trigonal map is unique. He also shows [Ka80, Thm. 2] that the number of weight 4 Weierstrass points on X is at most 6, and if equality holds then X is **cyclic trigonal**, i.e., there is a degree 3 Galois covering $f : X \rightarrow \mathbb{P}^1$.

2.8. The Maximum Weight of a Non-Hyperelliptic Weierstrass Semigroup. For $g \in \mathbb{Z}^{\geq 3}$, let $M_W(g)$ be the maximum weight of a Weierstrass point on a nonhyperelliptic Riemann surface of genus g . Thus $M_W(g) \leq M(g)$, and Remark 2.7 gives equality if $g \leq 7$. A work of Kato

[Ka79] computes $M_W(g)$ in all cases and gives even more information than that. The following result combines his results with some information from Section 2.1.

Theorem 2.18.

- a) For all $g \in \mathbb{Z}^{\geq 3}$ with $g \not\equiv 2 \pmod{3}$, there is a Riemann surface X of genus g and $P \in X$ with $W_P = \text{Tm}_g$, so $|W_P| = \frac{g(g-1)}{3}$.
- b) For all $g \in \mathbb{Z}^{\geq 3}$, there is a Riemann surface X of genus g and $P \in X$ with $W_P = \text{Bh}_g$, so $|W_P| = \frac{(g-2)(g-3)}{2} + 2$.
- c) Let S be a nonhyperelliptic semigroup of genus $g \in \{3, 4, 6, 7, 9\}$. Then $|S| \leq \frac{g(g-1)}{3}$, with equality if and only if $S = \text{Tm}_g$. (Note: $\text{Tm}_3 = \text{Bh}_3$.)
- d) Let S be a nonhyperelliptic semigroup of genus 5. Then $|S| \leq 5$, with equality if and only if $S \in \{\text{Bh}_5, T_{5,8} = \langle 3, 7, 11 \rangle\}$.
- e) Let S be a nonhyperelliptic semigroup of genus 10. Then $|S| \leq 30$, with equality if and only if $S \in \{\text{Bh}_{10}, \text{Tm}_{10}\}$.
- f) Let X be a Riemann surface of genus $g \in \{8\} \cup \mathbb{Z}^{\geq 11}$, and let $P \in X$. Then $|W_P| \leq |\text{Bh}_g|$, with equality if and only if $W_P = \text{Bh}_g$.

Remark 2.19. In his influential paper [Ka79], Kato works in the context of Riemann surfaces rather than abstract numerical semigroups, for instance making use of Riemann–Roch and Clifford’s Theorem. But these two appeals can manifestly be replaced by arguments that work in any numerical semigroup. We strongly suspect that Kato’s arguments actually show that $M_W(g) = M(g)$ for all $g \geq 3$ and thereby compute $M(g)$ as well, affirming the pattern that we saw in Section 2.1. Not having checked all the details, we will not insist upon this.

2.9. The Minimum Number of Weierstrass Points on a Non-Hyperelliptic Curve. For $g \in \mathbb{Z}^{\geq 3}$, let $L(g)$ be the minimum number of Weierstrass points on a nonhyperelliptic Riemann surface of genus g . Theorem 2.18 yields an immediate lower bound on $L(g)$:

$$(2) \quad L(g) \geq \frac{g^3 - g}{M_W(g)}.$$

However, as was shown by Keem–Martens [KM10], the upper bound of (2) is only sharp when $g \in \{3, 5\}$. Their opening salvo is the observation that for $g = 8$ or $g \geq 11$, equality would require that $\frac{(g-2)(g-3)}{2} + 2 \mid g^3 - g$, and one can easily see that this forces $g = 15$.

Here is an argument – perhaps a bit simpler than the one given in [KM10] – that eliminates all $g \notin \{3, 5\}$. Suppose that equality holds in (2).

Step 1: Suppose that $g \in \{8\} \cup \mathbb{Z}^{\geq 11}$. By Theorems 2.18 and 2.17, every Weierstrass point of X is a heavy bielliptic fixed point. Since $g \geq 6$, the bielliptic involution is unique and has $2g - 2$ fixed points, contradicting the fact that there are at least $2g + 2$ Weierstrass points. This argument also rules out $g = 10$ if one of the Weierstrass semigroups is Bh_{10} .

Step 2: If $g \in \{4, 6, 7, 9\}$, then Theorem 2.18 implies that all of the Weierstrass points have semigroup Tm_g , and when $g = 10$ this is the remaining case to consider. Thus each Weierstrass point is a total ramification point of a trigonal map. If $g \geq 5$, the trigonal map is unique, hence its total ramification points on X are uniquely determined. By Riemann–Hurwitz, the number of such points is at most $g + 2 < 2g + 2$, again a contradiction. If $g = 4$ we are trying to rule out that X has 15 Weierstrass points, all of weight 4, but as mentioned in §2.7, Kato showed that there are in fact at most 6 Weierstrass points of weight 4.

The genera 3 and 5 cannot be eliminated: as already mentioned, the Wiman curve of genus 5 has 24 Weierstrass points of weight $M_W(5) = 5$: we will see it later as the modular curve $X(8)$. The Fermat quartic of genus 3 has 12 Weierstrass points of weight $M_W(3) = 2$: we will see it later

as the modular curve $X_0(64)$. Remarkably, Keems–Martens show that there is exactly one further Riemann surface that has all its Weierstrass points of weight $M_W(g)$: the genus three **special Kuribayashi–Komiya curve**

$$(3) \quad KK : X^4 + Y^4 + Z^4 + 3X^2Y^2 + 3Y^2Z^2 + 3Z^2X^2 = 0.$$

Thus for all $g \notin \{3, 5\}$, we can modestly improve (2) to:

$$(4) \quad L(g) \geq \left\lfloor \frac{g^3 - g}{M_W(g)} \right\rfloor + 1.$$

We expect that the vast literature on Weierstrass points of Riemann surfaces must give more information on $L(g)$ than this, but we are not ourselves aware of it. Fortunately these results will be sufficient for our applications to producing non-cuspidal points on modular curves.

Remark 2.20. Let $g \in \mathbb{Z}^{\geq 2}$. The unique genus g semigroup of maximal weight is H_g . Excluding this semigroup, we found (at least for Weierstrass semigroups, but we believe in general) that for $g \geq 11$ the unique remaining genus g semigroup of maximal weight is Bh_g . We can ask to continue this process: a reasonable guess for the unique third largest weight for all sufficiently large g is Bl_g , since $|Bl_g| = |Bh_g| - 2$. Excluding this as well corresponds to excluding both hyperelliptic and bielliptic curves, which is natural because – as we will see – not only are there only finitely many modular curves that are either hyperelliptic or bielliptic, but by recent work of Zywinia [Zy] *all* of them are now known.

More generally, we will later recall that for all $\gamma \in \mathbb{Z}^{\geq 2}$ there are only finitely many modular curves X admitting a rational function of degree less than γ . This motivates asking: for $g, \gamma \in \mathbb{Z}^{\geq 2}$, what is the largest weight of a numerical semigroup of genus g and multiplicity at least γ ?

2.10. A Criterion for Weierstrass Points. In general, given a Riemann surface X of genus $g \geq 4$, it seems difficult to find the Weierstrass points explicitly. For $g \geq 3$, the generic Riemann surface of genus g has no nontrivial automorphisms – more precisely, the moduli space $\mathcal{M}_g$ of Riemann surfaces of genus g is an irreducible quasi-projective variety over $\mathbb{C}$ of dimension $3g - 3$, and the locus in $\mathcal{M}_g$ of surfaces admitting a nontrivial automorphism is proper and Zariski-closed. However, most examples of Riemann surfaces X with some or all Weierstrass points explicitly known have nontrivial automorphisms, making use of a technique going back to Schoeneberg [Sc51] and applied by him to the modular curves $X(N)$, which admit $\mathrm{PSL}_2(\mathbb{Z}/N\mathbb{Z})$ as a group of automorphisms. Lewittes give a closely related result in an appealing group-theoretic form.

Theorem 2.21 (Schoeneberg, 1951). *Let X be a Riemann surface of genus $g_X \geq 2$, let $\sigma \in \mathrm{Aut} X$ of order d , let $Y := X/\langle \sigma \rangle$ of genus g_Y , and let $q : X \rightarrow Y$ be the corresponding Galois covering map. If $g_Y \neq \lfloor \frac{g_X}{d} \rfloor$, then every fixed point of σ is a Weierstrass point of X .*

Theorem 2.22 (Lewittes, 1963). *Let X be a Riemann surface of genus at least 2, let $\sigma \in \mathrm{Aut} X$ of order d , let $Y := X/\langle \sigma \rangle$, and let $q : X \rightarrow Y$ be the corresponding Galois covering map. Suppose that some fixed point of σ is not a Weierstrass point of X . Then $g_Y = \lfloor \frac{g_X}{d} \rfloor$ and the number of σ -fixed points is less than 5.*

The proof of Theorem 2.21 involves an analysis of the representation of G on the space of holomorphic differentials of X , and in fact Theorems 2.21 and 2.22 are both corollaries of more elaborate results that we will not need here.

The condition $g_Y \neq \lfloor \frac{g_X}{d} \rfloor$ in Schoeneberg’s Theorem is implied by the inequality $g_X \geq d(g_Y + 1)$. It turns out that assuming this stronger genus inequality, the requirement that the map $q : X \rightarrow Y$ is Galois can be dropped. This was first observed by Atkin in [At67] and was the key breakthrough in his paper. One may say though that Atkin had the full proof but not the full statement: his

treatment takes place as part of an argument involving cuspidal Weierstrass points on $X_0(N)$.

The following result extracts the parts of the work of Schoeneberg–Lewittes and Atkin that we want to use and has a simple proof.

Theorem 2.23. *X and Y be Riemann surfaces, of genera g_X and g_Y . Let $f : X \rightarrow Y$ be a finite morphism of degree $d \geq 2$. Let $\mathcal{T} \subseteq X$ be the set of totally ramified points of f – i.e., points P of X such that $f^{-1}(f(P)) = \{P\}$ – and let $t = \#\mathcal{T}$.*

- a) *If σ is an automorphism of X of order $d \geq 2$ and we take f to be the quotient map $X \rightarrow X/\langle\sigma\rangle$, then $\mathcal{T}$ is precisely the set of σ -fixed points of X .*
- b) *(Atkin [At67]) If $g_X \geq d(g_Y + 1)$, then every element of $\mathcal{T}$ is a Weierstrass point.*
- c) *(Limonov [Li14, Thm. 3]) If $t \geq 5$, then every element of $\mathcal{T}$ is a Weierstrass point.*

Proof. a) This is clear.

b) Suppose $g_X \geq d(g_Y + 1)$, let $P \in \mathcal{T}$, and let $\bar{P} := f(P)$. By Riemann-Roch, there is a nonconstant rational function $g : Y \rightarrow \mathbb{P}^1$ with a pole only at $\bar{P}$ of order at most $g_Y + 1$. Then $g \circ f : X \rightarrow \mathbb{P}^1$ has a pole of order at most $d(g_Y + 1) \leq g_X$ at P , and because $f^{-1}(\bar{P}) = \{P\}$, it has poles at no other points of X . Therefore P is a Weierstrass point of X .

c) The Riemann-Hurwitz formula states

$$(2g_X - 2) = d \cdot (2g_Y - 2) + \sum_{P \in X} (e_P - 1),$$

so using that $t \geq 5$, we get

$$\begin{aligned} g_X &= dg_Y - (d - 1) + \frac{1}{2} \sum_{P \in X} (e_P - 1) \\ &\geq dg_Y + \left\lceil \left(\frac{t}{2} - 1 \right) (d - 1) \right\rceil \geq dg_Y + \left\lceil \frac{3}{2} (d - 1) \right\rceil \geq d(g_Y + 1). \end{aligned}$$

Thus we may apply part b). □

Remark 2.24.

- a) As we saw, in Theorem 2.23, part b) implies part c). Now suppose that every ramification point of f is totally ramified – as holds when f is Galois of prime degree, so in particular when $d = 2$. Then the proof shows that part c) implies part b). Still under this assumption, if $\lfloor \frac{g_X}{d} \rfloor \neq g_Y$ and $t \leq 4$, then we have $t = 0$ (a trivial case) or $t = 1$. Guerrero analyzed the case of Galois covers $f : X \rightarrow Y$ with $t = 1$: he showed that d cannot be a prime power [Gu83] and that every element of $\mathcal{T}$ is a Weierstrass point unless $d = 6$ and $g_X = 6g_Y + 1$ [Gu81]. Thus when f is a Galois covering of prime order, the results of Schoeneberg, Lewittes, Atkin and Limonov are all equivalent. When $f : X \rightarrow Y$ has points that are ramified but not totally ramified, Theorem 2.23b) is stronger than Theorem 2.23c).
- b) On any Riemann surface X of genus at least 2, Theorem 2.23b) finds all Weierstrass points in a tautological way: by definition, if P is a Weierstrass point on X there is a map $f : X \rightarrow \mathbb{P}^1$ of degree at most g_X such that f is totally ramified at P .

Here is a quantitative refinement of Theorem 2.23c):

Theorem 2.25. *Let X and Y be Riemann surfaces, of genera g_X and g_Y . Let $f : X \rightarrow Y$ be a covering map of degree $d \geq 2$. Let $\mathcal{T} \subseteq X$ be the set of totally ramified points of f , and let $t = \#\mathcal{T}$.*

- a) *If t is even and at least 6, then each $P \in \mathcal{T}$ has Weierstrass weight*

$$w_P \geq \frac{(d - 1)(t - 2)(t - 4)}{8}.$$

b) If $t \geq 5$ is odd and d is odd, then each $P \in \mathcal{T}$ has Weierstrass weight

$$w_P \geq \frac{(d-1)(t-3)^2}{8}.$$

c) If $t \geq 5$ is odd and d is even, then each $P \in \mathcal{T}$ has Weierstrass weight

$$w_P \geq \frac{(d-1)(t-3)(t-5)}{8} + \frac{d(t-3)}{4}.$$

Proof. Since $t \geq 5$, the proof of Theorem 2.23 shows that $g_X \geq d(g_Y + 1)$. Riemann–Hurwitz gives

$$2(g_X - 1) = d(2g_Y - 2) + (d-1)t + B,$$

where $(d-1)$ is the contribution from the elements of $\mathcal{T}$ and $B \geq 0$ is the contribution from the other ramification points of f (if any). Let $P \in \mathcal{T}$.

Step 1: Suppose that t is even and that $f(P)$ is *not* a Weierstrass point of Y . Then for all $j \in \mathbb{Z}^{\geq 1}$ there is $g \in \mathbb{C}(Y)^\times$ with polar divisor $(g_Y + j)[f(P)]$. Then $g \circ f \in \mathbb{C}(X)^\times$ has degree $d(g_Y + j)$, so $d(g_Y + h) \in W_P$. Given this, the Weierstrass weight at P is minimized in the case where the g_X gaps at P are the first g_X positive integers not of the form $d(g_Y + j)$, so to prove a lower bound on w_P we may assume that these are the gaps at P . Then $d(g_Y + 1)$ is the smallest positive element of W_P , so the first gaps at P are $1, \dots, d(g_Y + 1) - 1$; these are precisely the gaps that do not contribute to w_P . Thus the number of gaps that do contribute to w_P is

$$m := g_X - d(g_Y + 1) + 1 = \frac{(d-1)(t-4)}{2} + \frac{B}{2}.$$

We will compute the contribution to w_P coming from the first $\frac{(d-1)(t-4)}{2}$ of these gaps, which again gives a lower bound on w_P . For $1 \leq i \leq \frac{t-4}{2}$, the $d-1$ gaps in $(d(g_Y + i), d(g_Y + i + 1))$ each contribute i to w_P , giving

$$w_P \geq (d-1) \sum_{i=1}^{\frac{t-4}{2}} i = \frac{(d-1)(t-2)(t-4)}{8}.$$

Step 2: Suppose that t is even and that $f(P)$ is a Weierstrass point of Y , and let $\alpha_1, \dots, \alpha_{g_X}$ be the first g_X nongaps at $\bar{P}$. Then for all $1 \leq j \leq g_X$ we have that $d\alpha_j$ is a nongap at P , so w_P is minimized when the gaps at P are the first g_X positive integers not of the form $d\alpha_j$. Since for all $1 \leq j \leq g$ we have $d\alpha_j \leq d(g_Y + j)$ the contribution to w_P is at least as large as in Step 1.

Step 3: Suppose that t and d are both odd. As seen in Step 2, we may assume that $f(P)$ is not a Weierstrass point of Y . In this case we still have $\frac{(d-1)(t-4)}{2} \in \mathbb{Z}$. Writing $\frac{(d-1)(t-4)}{2} = (d-1) \cdot \frac{t-5}{2} + \frac{d-1}{2}$, we get

$$w_P \geq (d-1) \sum_{i=1}^{\frac{t-5}{2}} i + \frac{d-1}{2} \cdot \frac{t-3}{2} = \frac{(d-1)(t-3)^2}{8}.$$

Step 4: Suppose that t is odd and d is even. Again we may assume that $f(P)$ is not a Weierstrass point of Y . In this case, since $m \in \mathbb{Z}$ we have $B \geq 1$, so

$$m \geq \frac{(d-1)(t-4) + 1}{2}.$$

Writing $\frac{(d-1)(t-4)+1}{2} = (d-1) \cdot \frac{t-5}{2} + \frac{d}{2}$, we get

$$w_P \geq (d-1) \sum_{i=1}^{\frac{t-5}{2}} i + \frac{d}{2} \cdot \frac{t-3}{2} = \frac{(d-1)(t-3)(t-5)}{8} + \frac{d(t-3)}{4}.$$

□

Remark 2.26.

- a) When $n = 2$, Theorem 2.25 was proved by Torres [To94], then by Towse [To97], and then by Watanabe [Wa98]. Using Watanabe's proof, Perez del Pozo [PdP06] proved Theorem 2.25 when f is a Galois covering (which includes the $d = 2$ case). The proof we gave of Theorem 2.25 is extremely close to Perez del Pozo's proof, except that in the case where f is a Galois covering, t is odd and d is even, he observes that $B \geq d - \alpha$, where α is the largest odd divisor of d , which leads to the improved bound

$$w_P \geq \frac{(d-1)(t-3)(t-5)}{8} + \frac{t-3}{2} \cdot \frac{2d-\alpha-1}{2}.$$

- b) Equality holds in Theorem 2.25 if f is the quotient by a hyperelliptic involution. It also holds if f is the quotient by a bielliptic involution and P is a light bielliptic fixed point.
- c) In [PdP06], Perez del Pozo showed [PdP06, Prop. 1] that if d and t are both even and $t \geq 6$, there is a degree d automorphism of a Riemann surface with t fixed points, at least one of which has Weierstrass weight $\frac{(d-1)(t-2)(t-4)}{8}$. He also showed [PdP06, Prop. 2] that if d and t are both odd and $t \geq 5$, there is a degree d automorphism of a Riemann surface with t fixed points, at least one of which has Weierstrass weight $\frac{(d-1)(t-3)^2}{8}$.

Finally, here and in the next subsection we will briefly consider the positive characteristic case. Let k be any algebraically closed field. For a nice curve $X_{/k}$ of genus g , a point $P \in X$ is a **W-point** if X admits a nonconstant rational function regular away from P and having a pole of order at most g at P . Thus if k has characteristic 0, W-points are precisely Weierstrass points.

Theorem 2.27 (Wayman). *Let X be a nice curve defined over an algebraically closed field k , and let $1 \neq \sigma \in \text{Aut } X$ have at least five fixed points. Then every σ -fixed point is a W-point.*

Proof. See [Wa78]. □

2.11. A Group-Theoretic Supplement. We now present a group-theoretic supplement to Lewittes's Theorem, a mild generalization of a result given (without proof) in [MV06, Remark 3.2].

Lemma 2.28. *Let G be a group acting on a set X . Let $x \in X$, and let H be a subgroup of the stabilizer G_x of x in G . Let Y be the set of H -fixed points in the G -orbit of x , and let $N(H)$ be the normalizer of H in G . Then⁴*

$$\#Y = [N(H) : N(H) \cap G_x].$$

Proof. It suffices to show that $N(H)$ acts transitively on Y ; then Y is isomorphic as an $N(H)$ -set to $N(H)/N(H)_x = N(H)/(N(H) \cap G_x)$.

First we check that $N(H)$ acts on Y . Let $g \in N(H)$, $y \in Y$ and $h \in H$: we must show that $hgy = gy$. Because $g \in N(H)$, there is $h' \in H$ such that $hg = gh'$, and thus $hgy = gh'y = gy$.

Next we check that the action is transitive: by definition of Y , there is $g \in G$ such that $y = gx$. For all $h \in H$ we have $gx = y = hy = hgx$, so $(g^{-1}hg)x = x$, so $g^{-1}hg \in H$, so $g \in N(H)$. □

Theorem 2.29. *Let k be algebraically closed, and let $\pi : X \rightarrow Y$ be a Galois covering of nice curves over k , with automorphism group G . Let $y \in Y$ be a branch point for π . Let $x \in \pi^{-1}(y)$, and let H be a subgroup of G_x . Let $\mathcal{T}_y$ be the set of H -fixed points of X lying in the fiber over y .*

- a) *We have $\#\mathcal{T}_y = [N(H) : N(H) \cap G_x]$.*
- b) *If $[N(H) : N(H) \cap G_x] \geq 5$, then every element of $\mathcal{T}_y$ is a W-point of X .*

⁴Here we have an equality of cardinal numbers, although we will only use the case where G is finite.

Proof. a) We apply Lemma 2.28 to the group G acting on the set-theoretic fiber $\pi^{-1}(y)$. b) First suppose that $H = \langle \sigma \rangle$ is cyclic. Then by part a) and Theorem 2.27, every element of $\mathcal{T}_y$ is a W -point of X . Now assume that H is not cyclic, and choose a nontrivial $\sigma \in H$. Then every H -fixed point is a σ -fixed point, so by part a) there are at least 5 σ -fixed points, hence by Wayman's Theorem every σ -fixed point is a W -point, and thus every H -fixed point is a W -point. $\square$

Remark 2.30.

- a) We suspect that some version of Theorem 2.29 is a folkloric result. Our first experience with it was in [El99, §2.1], in which Elkies describes the three ramified orbits of $\mathrm{PSL}_2(\mathbb{Z}/7\mathbb{Z})$ acting on the Klein quartic $X(7)$. He does not state the result there, but he seems to know it. The characteristic 0 case of Theorem 2.29 appears as [MV06, Remark 3.2], in which they say: “The following group theoretic translation does not appear to be explicitly in the literature.” In their version H is cyclic, which in characteristic 0 it must be since it is a subgroup of the cyclic group G_x . Our reason for extending this result to positive characteristic is to clarify that it does not really have anything to do with cyclic subgroups.
- b) Carrying the simple idea of the proof of Theorem 2.29b) to its conclusion: if for some choice of $H \subseteq G_x$ we have $[N(H) : N(H) \cap G_x] \leq 4$, then we should try again with H replaced by one of its prime order subgroups. While a larger choice of H may make the quantity $[N(H) : N(H) \cap G_x]$ smaller, it may also make it easier to compute: e.g. with $H = G_x$, we are computing the order of the group $N(H)/H$.
- c) We have counted $\mathcal{T}_y$, the number of H -fixed points *in the fiber over y* , rather than $\mathcal{T}$, the number of H -fixed points overall. Of course $\mathcal{T}_y \subseteq \mathcal{T}$, so if $\#\mathcal{T}_y \geq 5$ then certainly $\#\mathcal{T} \geq 5$, but in general it can happen that $\#\mathcal{T}$ is at least 5 while $\#\mathcal{T}_y$ is not. In order to have $\mathcal{T} \supsetneq \mathcal{T}_y$, we would need to have another ramification point $y' \neq y$ and $x' \in \pi^{-1}(y')$ such that the stabilizer H' of x' contains H . We will see an example of this shortly.

We now return to characteristic 0: let G be a finite group of order d acting effectively by automorphisms on a Riemann surface X of genus $g_X \geq 2$, and let $\pi : X \rightarrow X/G =: Y$ be the quotient map. Let $C_1, \dots, C_r$ be a system of nontrivial cyclic subgroups of G , of orders $c_1, \dots, c_r$, such that for every nontrivial cyclic subgroup C of G there is a unique $1 \leq i \leq r$ such that C is conjugate to C_i . Thus for every branch point y of π , the fiber $\pi^{-1}(y)$ is isomorphic as a G -set to G/C_i for a unique i . Let a_i denote the number of $y \in Y$ such that $\pi^{-1}(y) \cong G/C_i$. Letting g_Y be the genus of Y , the Riemann–Hurwitz formula now reads

$$(5) \quad g_X - 1 = d(g_Y - 1) + \frac{d}{2} \sum_{i=1}^r a_i \left(1 - \frac{1}{c_i}\right).$$

For a finite group G , we say $(\alpha, \beta, \gamma) \in G^3$ **form a triple** in G if $\alpha\beta\gamma = 1$ and $G = \langle \alpha, \beta \rangle$. The associated **order triple** is $(\#\langle \alpha \rangle, \#\langle \beta \rangle, \#\langle \gamma \rangle)$. By the Riemann Existence Theorem [Vo, Thm. 2.13], a triple yields a **Galois Belyĭ map**, a Galois cover $X \rightarrow \mathbb{P}^1$ ramified over three points, with the isotropy groups over these points being the conjugates of $\langle \alpha \rangle$ (resp. of $\langle \beta \rangle$, resp. of $\langle \gamma \rangle$). Conversely, given a Galois Belyĭ map with Galois group G , the monodromy around the three ramification points forms a triple in G , well-defined up to simultaneous conjugation.

Example 2.31. Suppose $G = S_5$. Then we may take $C_{2,1} = \langle (12) \rangle$, $C_{2,2} = \langle (12)(34) \rangle$, $C_3 = \langle (123) \rangle$, $C_4 = \langle (1234) \rangle$, $C_5 = \langle (12345) \rangle$ and $C_6 = \langle (12)(345) \rangle$, and we get

$$(6) \quad g_X - 1 = 120(g_Y - 1) + 30a_{2,1} + 30a_{2,2} + 40a_3 + 45a_4 + 48a_5 + 50a_6.$$

For small g_X , (6) is very restrictive: if $g_X \leq 30$, then $g_Y = 0$. And we cannot have $g_X \in \{2, 3\}$.

- a) When g_X is even, we cannot have $a_{2,1} = a_6 = 0$ or $a_{2,2} = a_4 = 0$. Indeed, if $C_{2,1}$ acts freely on X and $X' = X/C_{2,1}$ has genus $g_{X'}$, then $g_X - 1 = 2(g_{X'} - 1)$ is even, contradiction.

So there is $P \in X$ with $C_{2,1} \subsetneq G_P$. Since the square of a 4-cycle has cycle type $(2, 2)$, it follows that G_P is conjugate to (and, in fact, equal to) C_6 . A similar argument rules out $a_{2,2} = a_4 = 0$.

- b) When $g_X = 4$, we must have $a_{2,1} + a_{2,2} = a_4 = a_5 = 1$. Since g_X is even, we must have $a_{2,1} = 1$ and $a_{2,2} = 0$. Since

$$\alpha_1 := (12), \beta_1 := (2345), \gamma_1 := (12543)$$

forms a triple in S_5 , such a Riemann surface exists. In fact there is a unique (up to isomorphism) Riemann surface B of genus 4 with automorphism group S_5 – **Bring’s curve** – and every other genus 4 curve has fewer automorphisms. In Example 8.10 we will see that the modular curve $X_1(5, 10)$ has genus 4 and 120 automorphisms, hence $X_1(5, 10) \cong B$.

There is a unique G -orbit on B of size 24. There are six subgroups conjugate to C_5 , hence Orbit-Stabilizer gives $[S_5 : NC_5] = 6$, hence $\#NC_5 = 20$ so $[NC_5 : C_5] = 4$ and each conjugate of C_5 has 4 fixed points in this orbit. Later on in Example 8.10 we will see that these 24 points are precisely the cusps on $X_1(5, 10)$.

There is a unique G -orbit on B of size 30. There are 15 subgroups conjugate to C_4 , hence $\#NC_4 = 8$ and $[NC_4 : C_4] = 2$ and each conjugate of C_4 has 2 fixed points.

The isotropy subgroups of the elements of the G -orbit of size 60 are the 10 conjugates of $C_{2,1}$, which tells us that $\#NC_{2,1} = 12$, so each conjugate of C_1 has $[NC_{2,1} : C_{2,1}] = 6$ fixed points, all of which are Weierstrass points. By Lemma 2.14, these 10 sets of 6 fixed points are pairwise disjoint, so Bring’s curve B has 60 Weierstrass points that form a complete orbit under $\text{Aut } B$. By Riemann–Hurwitz, the quotient of B by each of the ten subgroups conjugate to $C_{2,1}$ has genus 1: thus the genus 4 curve B has 10 bielliptic involutions.

- c) When $g_X = 6$, there are two solutions: $a_{2,1} + a_{2,2} = a_4 = a_6 = 1$ and $a_3 = 2, a_4 = 1$. Since 6 is even, the latter can’t occur. But the former can:

$$\alpha_2 := (14)(23), \beta_2 := (1542), \gamma_2 := (132)(45)$$

form a triple in S_5 . In fact there is a unique genus 6 curve with automorphism group isomorphic to S_5 , the **Wiman sextic** W . The curve W is not (so far as we know) a modular curve in our sense; rather it is the analogue of a modular curve with $\text{PSL}_2(\mathbb{Z})$ replaced by the cocompact arithmetic triangle group $\Delta(2, 4, 6)$: see [CV19, p. 42].

On W we have one G -orbit of size 10, whose isotropy groups are the 10 subgroups conjugate to C_6 . Thus $\#NC_6 = 12$, so each 4-cycle has $\#NC_6/C_6 = 2$ fixed points. We have one G -orbit of size 60, whose isotropy groups are the order 2 subgroups generated by the 15 double transpositions $(ab)(cd)$ in S_5 . Thus $\#NC_{2,2} = 8$, so each double transposition has $\#NC_{2,2}/C_{2,2} = 4$ fixed points. However, as above, the G -orbit of size 30 is formed by a pair of fixed points for each of the 15 conjugates of C_4 , and in each case the two fixed points of $(acbd)$ are also fixed points of $(acbd)^2 = (ab)(cd)$, so overall each of the 15 double transpositions has 6 fixed points. This identifies 90 Weierstrass points on W . We don’t know whether these are all the Weierstrass points.

- d) The elements

$$\alpha_3 := (12), \beta_3 := (13245), \gamma_3 := (143)(25)$$

form a triple in S_5 , yielding a curve X with genus 9 and $a_{2,1} = 1$, hence having 60 Weierstrass points coming from the G -orbit of size 60. The elements

$$\alpha_4 := (123), \beta_4 := (2543), \gamma_4 := (1345)$$

form a triple in S_5 , yielding a curve X with genus 11 and $a_3 = 1, a_4 = 2$. There are 10 subgroups conjugate to C_3 , so $\#NC_3 = 12$ and $\#NC_3/C_3 = 4$, so the G -orbit of size 40 consists of 4 fixed points of each of the order 3 subgroups of S_5 . Each cyclic order 4

subgroup has two fixed points from each of the two orbits of size 30, so overall has four fixed points. So we did not produce any Weierstrass points. The elements

$$\alpha_5 := (12)(34), \beta_5 := (135)(24), \gamma_5 := (154)(23)$$

form a triple in S_5 , yielding a curve X with genus 11 and $a_{2,2} = 1$, $a_6 = 2$. This time each subgroup conjugate to either $C_{2,2}$ or C_6 has 4 fixed points on X , so again we did not produce any Weierstrass points.

- e) According to the LMFDB there are precisely five Riemann surfaces X of genus $2 \leq g_X \leq 15$ on which S_5 acts effectively by automorphisms, so we have seen them all above. Since $S_5 \cong \mathrm{PGL}_2(\mathbb{Z}/5\mathbb{Z})$, the Riemann surfaces of genus up to 24 with an effective action by $G \cong S_5$ such that $X \rightarrow X/G$ is a Galois Belyi map appear in [CV19, Table 10.5], and there are some further cases in which ramified G -orbits produce Weierstrass points.

Some of Example 2.31 was informed by the recent work [BDH24] of Braden–Disney–Hogg, which gives a “deep dive” on Bring’s curve in the spirit of Elkies’s classic paper [El99] on the Klein quartic. In particular (6) appears in the proof of [BDH24, Prop. 3.3]. The authors of that paper go on to write down the 60 Weierstrass points on Bring’s curve quite explicitly. Our own discussion of the Weierstrass points on Bring’s curve and (some of the) Weierstrass points on the Wiman sextic is rather different and meant to emphasize the group-theoretic approach.

3. BACKGROUND ON MODULAR CURVES

3.1. Basic Definitions. For a commutative ring R , we denote by $\mathrm{SL}_2(R)$ the ring of 2×2 matrices with coefficients in R and unit determinant, and we denote by $\mathrm{PSL}_2(R)$ the quotient by the central subgroup $\{\pm 1\}$ of $\mathrm{SL}_2(R)$.⁵

In this paper a **Fuchsian group** is a discrete subgroup $\Gamma \subseteq \mathrm{PSL}_2(\mathbb{R})$ with finite hyperbolic covolume $v(\Gamma)$ (more properly, this defines a “Fuchsian group of the first kind,” but we will not consider any other kind of Fuchsian groups here). For a Fuchsian group Γ , we define $Y(\Gamma) := \Gamma \backslash \mathcal{H}$ and $X(\Gamma) := \Gamma \backslash \overline{\mathcal{H}}$, i.e., $Y(\Gamma)$ together with the cusps of Γ . Thus $X(\Gamma)$ is a Riemann surface – equivalently, the set of $\mathbb{C}$ -points of a nice algebraic curve over $\mathbb{C}$ – while $Y(\Gamma) = X(\Gamma)$ if Γ is co-compact and otherwise $Y(\Gamma)$ is the set of $\mathbb{C}$ -points of a nonsingular integral affine algebraic curve over $\mathbb{C}$. If $\Gamma' \subseteq \Gamma \subseteq \mathrm{PSL}_2(\mathbb{R})$ with $[\Gamma : \Gamma'] < \infty$, then Γ is a Fuchsian group if and only if Γ' is a Fuchsian group and $v(\Gamma') = [\Gamma : \Gamma']v(\Gamma)$, and conversely if $\Gamma' \subseteq \Gamma$ are Fuchsian groups, then Γ' has finite index in Γ . If $\Gamma_1 \subseteq \Gamma_2 \subseteq \mathrm{PSL}_2(\mathbb{R})$ are Fuchsian groups, there is a natural finite map

$$f : X(\Gamma_1) \rightarrow X(\Gamma_2)$$

of degree $[\Gamma_2 : \Gamma_1]$, that is a Galois covering if and only if Γ_1 is a normal subgroup of Γ_2 .

A Fuchsian group is **maximal** if it is not properly contained in any other Fuchsian group. The modular group $\mathrm{PSL}_2(\mathbb{Z})$ is a Fuchsian group of the first kind; its hyperbolic covolume is $\frac{\pi}{3}$, the minimum covolume of any Fuchsian group with cusps [K, Exc. 4.10], while the minimum covolume of a Fuchsian group without cusps is $\frac{\pi}{21}$ [K, Exc. 4.9]. It follows that $\mathrm{PSL}_2(\mathbb{Z})$ is maximal and that every Fuchsian group is contained in a maximal Fuchsian group. The j -invariant gives an

⁵Beware: the center of $\mathrm{SL}_2(R)$ is the set of scalar matrices $\begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix}$ such $a \in R^\times$ satisfies $a^2 = 1$. So when R is a field, the center is just $\{\pm 1\}$ and $\mathrm{PSL}_2(R)$ is the quotient of $\mathrm{SL}_2(R)$ by its full center. For a more general ring R – and even for $\mathbb{Z}/N\mathbb{Z}$ when e.g. N has more than one odd prime factor – we may not be quotienting out by the entire center. Because of this we need **not** have $\mathrm{PSL}_2(R_1 \times R_2) \cong \mathrm{PSL}_2(R_1) \times \mathrm{PSL}_2(R_2)$.

isomorphism from $Y(\mathrm{PSL}_2(\mathbb{Z}))$ to the affine line $\mathbb{A}^1$ and thus from $X(\mathrm{PSL}_2(\mathbb{Z}))$ to the projective line $\mathbb{P}^1$. For $N \in \mathbb{Z}^+$, we define the **principal congruence subgroup**

$$\Gamma(N) := \mathrm{Ker}(\mathrm{PSL}_2(\mathbb{Z}) \rightarrow \mathrm{PSL}_2(\mathbb{Z}/N\mathbb{Z})).$$

A **congruence subgroup** of $\mathrm{PSL}_2(\mathbb{Z})$ is a subgroup Γ that contains $\Gamma(N)$ for some $N \in \mathbb{Z}^+$; the least such N is called the **level** of Γ . A **modular curve** is the Riemann surface $X(\Gamma)$ associated to a congruence subgroup Γ .

Every subgroup Γ of $\mathrm{PSL}_2(\mathbb{Z})$ can of course be canonically lifted to a subgroup of $\mathrm{SL}_2(\mathbb{Z})$: namely, if $q : \mathrm{SL}_2(\mathbb{Z}) \rightarrow \mathrm{PSL}_2(\mathbb{Z})$ is the quotient map $m \mapsto \pm m$, then $q^{-1}(\Gamma)$ is such a lift and is clearly the unique lift containing -1 . However, in many cases this is not necessarily the most natural or useful lift to $\mathrm{SL}_2(\mathbb{Z})$. In this paper we consider the following families of modular curves:

$$X_0(N), X_1(N), X(N), X_1(M, N), X_{\mathrm{sp}}(N), X_{\mathrm{sp}}^+(N), X_{\mathrm{ns}}(N), X_{\mathrm{ns}}^+(N), X_{\mathrm{ns}}^*(N), X_{S_4}(p).$$

We define $\tilde{\Gamma}(N)$ to be the kernel of the (surjective) homomorphism

$$q_N : \mathrm{SL}_2(\mathbb{Z}) \rightarrow \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}),$$

and then a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ is a subgroup containing $\tilde{\Gamma}(N)$ for some $N \in \mathbb{Z}^+$. For all $N \in \mathbb{Z}^+$ we define subgroups

$$\tilde{\Gamma}_0(N), \tilde{\Gamma}_1(N), \tilde{\Gamma}_1(M, N), \tilde{\Gamma}_{\mathrm{sp}}(N), \tilde{\Gamma}_{\mathrm{sp}}^+(N), \tilde{\Gamma}_{\mathrm{ns}}(N), \tilde{\Gamma}_{\mathrm{ns}}^+(N), \tilde{\Gamma}_{\mathrm{ns}}^*(N)$$

of $\mathrm{SL}_2(\mathbb{Z})$, all containing $\tilde{\Gamma}(N)$ – thus each of these subgroups is defined as $q_N^{-1}(H)$ for a subgroup H of $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$ – and for each prime $p > 2$ a subgroup $\tilde{\Gamma}_{S_4}(p)$ of $\mathrm{SL}_2(\mathbb{Z})$ containing $\tilde{\Gamma}(p)$ – thus defined as $q_p^{-1}(H)$ for a subgroup H of $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$. These subgroups do not necessarily contain -1 , their images in $\mathrm{PSL}_2(\mathbb{Z})$ are denoted by $\Gamma_0(N), \Gamma_1(N), \Gamma_1(M, N), \Gamma_{\mathrm{sp}}(N), \Gamma_{\mathrm{sp}}^+(N), \Gamma_{\mathrm{ns}}(N), \Gamma_{\mathrm{ns}}^+(N), \Gamma_{\mathrm{ns}}^*(N)$ and $\Gamma_{S_4}(p)$, and define the modular curves that we consider here.

The definition of each “tilde subgroups” is given in [HL25]: more precisely, in each case they give a subgroup of $\mathrm{GL}_2(\hat{\mathbb{Z}})$ whose intersection with $\mathrm{SL}_2(\mathbb{Z})$ is the desired “tilde subgroup.”

Let us be explicit in some cases: for $N \in \mathbb{Z}^+$, we put

$$\tilde{\Gamma}_0(N) := \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z}) \mid c \equiv 0 \pmod{N} \right\}.$$

This subgroup always contains -1 . For $M, N \in \mathbb{Z}^+$ with $M \mid N$, we put

$$\tilde{\Gamma}_1(M, N) := \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{Z}) \mid a, d \equiv 1 \pmod{N}, b \equiv 0 \pmod{M}, c \equiv 0 \pmod{N} \right\}.$$

This subgroup contains -1 if and only if $N \leq 2$. We put

$$\tilde{\Gamma}_1(N) := \tilde{\Gamma}_1(1, N),$$

and we also have

$$\tilde{\Gamma}(N) = \tilde{\Gamma}_1(N, N).$$

For a prime $p > 2$, the group $\mathrm{PGL}_2(\mathbb{Z}/p\mathbb{Z})$ contains a subgroup that is isomorphic to S_4 and this subgroup is unique up to conjugacy. Fix one such subgroup H and let H' be its intersection with $\mathrm{PSL}_2(\mathbb{Z}/p\mathbb{Z})$, and view H' as a subgroup of $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$ that contains -1 . Then we put

$$\tilde{\Gamma}_{S_4}(p) := q_p^{-1}(H).$$

This subgroup contains -1 .

We will refer the reader to [HL25] for the definitions of the Cartan-type subgroups.

Remark 3.1. For all $N \in \mathbb{Z}^+$, the canonical $\mathbb{Q}$ -models of $Y_0(N^2)$ and of $Y_{\text{sp}}(N)$ are isomorphic. Indeed, if E is an elliptic curve and C is a cyclic subgroup of order N^2 , let C' be the unique subgroup of C of order N . Then

$$(E, C) \mapsto (E/C', C/C', E[N]/C')$$

is an isomorphism of moduli problems. It follows that the modular curves $X_{\text{sp}}(N)$ and $X_0(N^2)$ are isomorphic as Riemann surfaces and have the same number of cusps.

3.2. Genera of Modular Curves. Genus formulas for the modular curves considered here are given in [HL25]. For our later work we often need to supplement the exact formulas with useful upper and lower bounds.

Proposition 3.2. *Let $N \in \mathbb{Z}^+$.*

a) *(Csirik–Wetherell–Zieve) We have*

$$(7) \quad g(X_0(N)) \geq \frac{N - 5\sqrt{N} - 8}{12}.$$

Equality holds in (7) if and only if $N = p^2$ for a prime $p \equiv 1 \pmod{12}$.

b) *For all $N \geq 5$, we have*

$$g(X_1(N)) \geq \frac{N^2}{4\pi^2} - \frac{1}{4}N^{1+\frac{1.5379 \log 2}{\log \log N}} + 1.$$

Proof. a) See [CWZ00].

b) We recall (see e.g. [HL25]) that for $1 \leq N \leq 4$, we have $g(X_1(N)) = 0$. For all $N \geq 5$, we have

$$g(X_1(N)) = 1 + \frac{N^2}{24} \prod_{p|N} \left(1 - \frac{1}{p^2}\right) - \frac{1}{4} \sum_{d|N} \varphi(d) \varphi\left(\frac{N}{d}\right).$$

Now we have $\prod_{p|N} (1 - \frac{1}{p^2}) > \prod_p (1 - \frac{1}{p^2}) = \frac{6}{\pi^2}$. To complete the proof it suffices to show that $\prod_{d|N} \varphi(d) \varphi(N/d) \leq N^{1+\frac{1.53948 \log 2}{\log \log N}}$. Using the trivial bound $\varphi(d) \varphi(N/d) \leq d \cdot N/d = N$, we get

$$\prod_{d|N} \varphi(d) \varphi(N/d) \leq \sigma_0(N) N,$$

where $\sigma_0(N)$ is the number of divisors of N . Work of Nicolas [Ni88] gives

$$\sigma_0(N) \leq N^{\frac{1.5379 \log 2}{\log \log N}}. \quad \square$$

For a Fuchsian group Γ , we denote by $v_\infty(X(\Gamma))$ the number of cusps on the modular curve $X(\Gamma)$.

Lemma 3.3. *For all $M \mid N$, we have*

$$g(X_1(M, N)) \leq \frac{MN^2}{24}.$$

Proof. From [HL25] we see that $g(X_1(M, N)) = 0$ when $N \leq 4$, and when $N \geq 5$ we have

$$g(X_1(M, N)) = 1 + \frac{MN^2}{24} \prod_{p|N} \left(1 - \frac{1}{p^2}\right) - \frac{v_\infty(X_1(M, N))}{2}$$

and

$$v_\infty(X_1(M, N)) = \frac{1}{2} \sum_{d|N} \varphi\left(\frac{N}{d}\right) \varphi(d) \gcd\left(M, \frac{N}{d}\right).$$

Using the latter formula it is easy to see that (since $N \geq 5$) $X_1(M, N)$ has at least 2 cusps. (Alternately: if $N \geq 2$ then the cusps 0 and ∞ are distinct on $X_0(N)$ hence on any modular curve $X(\Gamma)$ for $\Gamma \subseteq \Gamma_0(N)$, hence on $X_1(M, N)$.) Thus

$$g(X_1(M, N)) \leq 1 + \frac{MN^2}{24} - \frac{v_\infty(X_1(M, N))}{2} \leq \frac{MN^2}{24}. \quad \square$$

In §5.2 we will see that Lemma 3.3 is sharp up to a multiplicative constant.

3.3. Cusps on Modular Curves. Here we just collect a few results for future use.

Lemma 3.4. *If N is squarefree, then the natural map $\pi : X_1(N) \rightarrow X_0(N)$ is unramified over every cusp.*

Proof. If $N \leq 2$ then π is an isomorphism. Otherwise the map π has degree $\frac{\varphi(N)}{2}$ and using e.g. [HL25] we find that when N is squarefree with $\omega(N)$ distinct prime divisors, that

$$v_\infty(\Gamma_1(N)) = \frac{\varphi(N)}{2} 2^{\omega(N)} = \deg(\pi) v_\infty(\Gamma_0(N)),$$

from which the result follows. $\square$

For a positive integer N , we put

$$\psi(N) := [\mathrm{PSL}_2(\mathbb{Z}) : \Gamma_0(N)] = N \prod_{p|N} \left(1 + \frac{1}{p}\right).$$

Lemma 3.5.

a) *For all $N \in \mathbb{Z}^+$, we have*

$$v_\infty(X_0(N)) \leq \psi(N) \sqrt{N} \leq (1 + \log N) \sqrt{N}.$$

b) *For all $N \geq 3$, we have*

$$v_\infty(X_1(N)) \leq \frac{1}{2} N^{1 + \frac{1.5379 \log 2}{\log \log N}}.$$

Proof. a) We have [HL25, Table 1]

$$v_\infty(X_0(N)) = \sum_{d|N} \varphi\left(\gcd\left(d, \frac{N}{d}\right)\right).$$

This is a multiplicative function of N in the sense of classical number theory: if $\gcd(N_1, N_2) = 1$, then $v_\infty(X_0(N_1 N_2)) = v_\infty(X_0(N_1)) \cdot v_\infty(X_0(N_2))$.

Step 1: Let $N = p^a$ be a prime power.

• Suppose that $a = 2k - 1$ is odd. Then

$$\begin{aligned} v_\infty(X_0(p^a)) &= \varphi(1) + \dots + \varphi(p^{k-1}) + \varphi(p^{k-1}) + \dots + \varphi(1) \\ &= 2 \left(\sum_{d|p^{k-1}} \varphi(d) \right) = 2p^{k-1} = \frac{2}{\sqrt{p}} \sqrt{p^a}. \end{aligned}$$

• Suppose that $a = 2k$ is even. Then

$$\begin{aligned} v_\infty(X_0(p^a)) &= \varphi(1) + \dots + \varphi(p^{k-1}) + \varphi(p^k) + \varphi(p^{k-1}) + \dots + \varphi(1) \\ &= 2p^{k-1} + (p-1)p^{k-1} = \left(1 + \frac{1}{p}\right) \sqrt{p^a}. \end{aligned}$$

Since $\frac{2}{\sqrt{p}} < 1 + \frac{1}{p}$, we deduce that

$$v_\infty(X_0(p^a)) \leq (1 + \frac{1}{p})\sqrt{p^a}.$$

Step 2: Now suppose that $N = p_1^{a_1} \cdots p_r^{a_r}$. Then

$$v_\infty(X_0(N)) = \prod_{i=1}^r v_\infty(X_0(p_i^{a_i})) \leq \prod_{i=1}^r (1 + \frac{1}{p_i})\sqrt{p_i^{a_i}} = \frac{\psi(N)}{\sqrt{N}},$$

so to complete the proof it suffices to show that for all $N \in \mathbb{Z}^+$,

$$\psi(N) \leq (1 + \log N)N.$$

Put $M := p_1 \cdots p_r$. Then

$$\begin{aligned} \prod_{p|N} (1 + 1/p) &= \prod_{p|M} (1 + 1/p) = \sum_{d|M} \frac{1}{d} \\ &\leq \sum_{i=1}^M \frac{1}{i} \leq 1 + \int_1^M \frac{dt}{t} = 1 + \log M + 1 \leq 1 + \log N. \end{aligned} \quad \square$$

b) The argument here is the same as in the proof of Proposition 3.2b) above.

Remark 3.6. The bound of Lemma 3.5a) could be improved: for instance, denoting the product of the distinct prime divisors of N by $\text{rad}(N)$, we actually showed that

$$v_\infty(X_0(N)) \leq (1 + \log \text{rad}(N))\sqrt{N}.$$

But because $\prod_p (1 + \frac{1}{p}) = \infty$, it is clear that *we do not have* $v_\infty(X_0(N)) = O(\sqrt{N})$. Similarly, it can be shown that $v_\infty(X_1(N))$ is *not* $O(N)$.

3.4. The Zograf–Abramovich–Kim–Sarnak Theorem. The **modular index** of a finite index subgroup Γ of $\text{PSL}_2(\mathbb{Z})$ is

$$\iota(\Gamma) = [\text{PSL}_2(\mathbb{Z}) : \Gamma] = \deg(X(\Gamma) \rightarrow X(1)).$$

We denote by $\gamma(X(\Gamma))$ the gonality of $X(\Gamma)$, i.e., the least degree of a nonconstant rational function on $X(\Gamma)$. Zograf [Zo87] and Abramovich [Ab96] independently gave a linear lower bound on the gonality $\gamma(X(\Gamma))$ in terms of the modular index. The bounds come in terms of the least positive eigenvalue λ_1 of the hyperbolic Laplacian on the modular curves $X(N)$: Zograf showed

$$\gamma(X(\Gamma)) > \frac{\lambda_1}{24} \iota(\Gamma)$$

and Abramovich showed the same bound with $>$ replaced by $\geq$. (However, Abramovich proved the same bound for Shimura curves using the Jacquet–Langlands correspondence.) It is known that $\lambda_1 \leq \frac{1}{4}$, and Selberg conjectured that equality holds. The best known lower bound on λ_1 is $\lambda_1 \geq \frac{1}{4} - (\frac{7}{64})^2 = \frac{975}{4096}$ due to Kim–Sarnak [Ki03, Appendix 2]. Thus we get:

Theorem 3.7 (Zograf–Abramovich–Kim–Sarnak).

Let Γ be a congruence subgroup of $\text{PSL}_2(\mathbb{Z})$. Then we have:

$$\frac{325}{32768} \iota(\Gamma) < \gamma(X(\Gamma)).$$

Let us now work in the family of all congruence subgroups $\Gamma \subsetneq \mathrm{PSL}_2(\mathbb{Z})$ such that $g(X(\Gamma)) \geq 1$. The Brill–Noether Theorem implies that for a Riemann surface X of genus g , we have $\gamma(X) \leq \frac{g+3}{2}$. Combining this with Theorem 3.7 and Riemann–Hurwitz (and using that $\iota(\Gamma) \geq 2$), we get

$$(8) \quad \frac{325}{32768} \iota(\Gamma) < \gamma(X(\Gamma)) \leq \frac{g(X(\Gamma)) + 3}{2} \leq 2g(X(\Gamma)) \leq 2 + \frac{\iota(\Gamma)}{6} \leq \frac{7}{6} \iota(\Gamma).$$

Thus as we range over such Γ , the quotients $\frac{\iota(\Gamma)}{g(X(\Gamma))}$, $\frac{\iota(\Gamma)}{\gamma(X(\Gamma))}$ and $\frac{g(X(\Gamma))}{\gamma(X(\Gamma))}$ are each bounded above and below by explicit positive constants.

In all of the families of modular curves considered here, it is straightforward to give reasonable lower bounds on the modular index that tend to infinity as the parameter values increase. Thus one application of Theorem 3.7 is to provably enumerate all low genus modular curves in a given family. We record the following for future use:

Remark 3.8 (Low Genus Modular Curves).

a) The modular curve $X_0(N)$ has genus:

$$\begin{aligned} 0 &\iff N \in [1, 10] \cup \{12, 13, 16, 18, 25\}. \\ 1 &\iff N \in \{11, 14, 15, 17, 19, 20, 21, 24, 27, 32, 36, 49\}. \\ 2 &\iff N \in \{22, 23, 26, 28, 29, 31, 37, 50\}. \\ 3 &\iff N \in \{30, 33, 34, 35, 39, 40, 41, 43, 45, 48, 64\}. \end{aligned}$$

b) The modular curve $X_1(N)$ has genus:

$$\begin{aligned} 0 &\iff N \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12\}. \\ 1 &\iff N \in \{11, 14, 15\}. \\ 2 &\iff N \in \{13, 16, 18\}. \\ 3 &\iff N = 20. \end{aligned}$$

c) The modular curve $X(N)$ has genus:

$$\begin{aligned} 0 &\iff N \in \{1, 2, 3, 4, 5\}; \\ 1 &\iff N = 6 \\ 2 &\text{ or } 4 \text{ or } 6 \text{ never} \\ 3 &\iff N = 7 \text{ (Klein quartic)} \\ 5 &\iff N = 8 \text{ (Wiman curve)}. \end{aligned}$$

d) Let $M \mid N$ be positive integers with $1 < M < N$. The modular curve $X_1(M, N)$ has genus:

$$\begin{aligned} 0 &\iff (M, N) \in \{(2, 4), (2, 6), (3, 6)\}. \\ 1 &\iff (M, N) \in \{(2, 8), (4, 8), (3, 9), (2, 10), (2, 12)\}. \\ 2 &\text{ never} \\ 3 &\iff (M, N) = (3, 12). \\ 4 &\iff (M, N) \in \{(2, 14), (5, 10)\}. \end{aligned}$$

e) The modular curve $X_{\mathrm{sp}}(N)$ has genus:

$$\begin{aligned} 0 &\iff N \in [1, 5]. \\ 1 &\iff N \in \{6, 7\}. \\ 2 &\text{ never} \\ 3 &\iff N = 8. \end{aligned}$$

f) The modular curve $X_{\text{sp}}^+(N)$ has genus:

$$0 \iff N \in [1, 7].$$

$$1 \iff N \in \{8, 9\}.$$

$$2 \iff N = 11.$$

$$3 \iff N \in \{10, 13\}.$$

g) The modular curve $X_{\text{ns}}(N)$ has genus:

$$0 \iff N \in [1, 5].$$

$$1 \iff N \in \{6, 7, 8, 10\}.$$

$$2 \iff N = 9.$$

$$3 \iff N = 12.$$

h) The modular curve $X_{\text{ns}}^+(N)$ has genus:

$$0 \iff N \in [1, 10] \cup \{12\}.$$

$$1 \iff N \in \{1, 14, 18\}.$$

$$2 \iff N \in \{15, 16\}.$$

$$3 \iff N \in \{13, 20, 24\}.$$

i) The modular curve $X_{\text{ns}}^*(N)$ has genus:⁶

$$0 \iff N \in [1, 10] \cup \{12, 14, 18\}.$$

$$1 \iff N \in \{11, 15, 21, 22, 24, 30, 42\}.$$

$$2 \iff N \in \{16, 20\}.$$

$$3 \iff N \in \{13, 26\}.$$

j) For a prime $p > 2$, the modular curve $X_{S_4}(p)$ has genus:

$$0 \iff p \in \{3, 5, 7\}.$$

$$1 \iff p = 11.$$

$$2 \text{ never.}$$

$$3 \iff p = 13.$$

⁶When N is odd, we have $X_{\text{ns}}^*(2N) = X_{\text{ns}}^*(N)$.

3.5. The Modular Automorphism Group. For a Fuchsian group Γ , we denote by $\mathfrak{N}\Gamma$ the normalizer of Γ in $\mathrm{PSL}_2(\mathbb{R})$. Then $\mathfrak{N}\Gamma$ is also a Fuchsian group and Γ is a finite index normal subgroup of $\mathfrak{N}\Gamma$. The quotient $\mathfrak{N}\Gamma/\Gamma$ acts effectively by automorphisms on $Y(\Gamma)$, hence also on $X(\Gamma)$, and is called the **modular automorphism group** $\mathrm{Aut}_M(X(\Gamma))$ of $X(\Gamma)$. If Γ has no nontrivial elements of finite order, then the map $\mathcal{H} \rightarrow Y(\Gamma)$ is the universal covering map and thus $\mathrm{Aut}_M X(\Gamma) = \mathrm{Aut} Y(\Gamma)$. If moreover Γ has no cusps – one says that Γ is of **hyperbolic type** – then $X(\Gamma) = Y(\Gamma)$ and thus $\mathrm{Aut}_M X(\Gamma) = \mathrm{Aut} X(\Gamma)$. But all finite index subgroups of $\mathrm{PSL}_2(\mathbb{Z})$ have cusps, so this never applies to the type of modular curves considered here.

Remark 3.9. Let $\mathrm{GL}_2^+(\mathbb{R})$ be the set of all 2×2 real matrices with positive determinant, and let $\mathrm{PGL}_2^+(\mathbb{R})$ be the quotient of $\mathrm{GL}_2^+(\mathbb{R})$ by the subgroup $Z(\mathrm{GL}_2^+(\mathbb{R}))$ of scalar matrices. We have a group homomorphism

$$\varphi : \mathrm{GL}_2^+(\mathbb{R}) \rightarrow \mathrm{PSL}_2(\mathbb{R}), \quad M \mapsto \pm \frac{1}{\sqrt{\det M}} M.$$

Then $Z(\mathrm{GL}_2^+(\mathbb{R}))$ lies in the kernel of φ , and the induced map $\varphi : \mathrm{PGL}_2^+(\mathbb{R}) \rightarrow \mathrm{PSL}_2(\mathbb{R})$ is an isomorphism. Thus we may represent elements of $\mathrm{Aut}_M X(\Gamma)$ by elements of $\mathrm{GL}_2^+(\mathbb{R})$. But if $M \in \mathrm{GL}_2^+(\mathbb{R})$ has rational or even integral entries, the entries of its image in $\mathrm{PSL}_2(\mathbb{R})$ may contain quadratic irrationalities. For this reason the $\mathrm{GL}_2^+(\mathbb{R})$ representation is often cleaner.

Example 3.10. Let $N \in \mathbb{Z}^+$. A **Hall divisor** of N is a positive divisor d of N such that $\gcd(d, \frac{N}{d}) = 1$. Every divisor of N is a Hall divisor if and only if N is squarefree, and if $\omega(N)$ is the number of distinct prime divisors of N , then the number of Hall divisors of N is $2^{\omega(N)}$. For every Hall divisor d of N , there are $x, y, z, w \in \mathbb{Z}$ such that the matrix

$$M_d := \begin{bmatrix} dx & y \\ Nz & dw \end{bmatrix}$$

has determinant d , so by Remark 3.9 determines an element of $\mathrm{PSL}_2(\mathbb{R})$ that we denote by w_d . Then: w_d normalizes $\Gamma_0(N)$ and when $d > 1$ does not lie in $\Gamma_0(N)$, but $w_d^2 \in \Gamma_0(N)$, so w_d determines a modular involution of $X_0(N)$, called the **dth Atkin–Lehner involution**. The involution w_N is also called the **Fricke involution**. When $d = 1$ the set of all such matrices M_1 is precisely the subgroup $\Gamma_0(N) \subseteq \mathrm{SL}_2(\mathbb{Z})$ of matrices that are upper triangular modulo N . In general, for a fixed Hall divisor d , the set of matrices M_d of the above form a full $\Gamma_0(N)$ -coset. Thus as d ranges over all Hall divisors of N , the image in $\mathrm{PSL}_2(\mathbb{R})$ of the set of all such matrices M_d form a subgroup denoted by $\Gamma_0^*(N)$ that normalizes $\Gamma_0(N)$. The **Atkin–Lehner group** is the quotient $\Gamma_0^*(N)/\Gamma_0(N)$; it is isomorphic to $(\mathbb{Z}/2\mathbb{Z})^{\omega(N)}$.

We can view w_d as an automorphism of the moduli problem for $Y_0(N)$, as follows: given a pair $(E, C)_{/F}$ with F a field of characteristic 0 and $C \subseteq E$ an F -rational subgroup scheme that is cyclic of order N , then since d is a Hall divisor of N , if C_d is the unique F -rational subgroup scheme of C of order d and $C_{\frac{N}{d}}$ is the unique F -rational subgroup scheme of C of order $\frac{N}{d}$, then $C = C_d \oplus C_{\frac{N}{d}}$. Then w_d maps (E, C) to $(E/C_d, (E[N/d]/C_d) \oplus C_{\frac{N}{d}})$. This shows that each w_d is defined over $\mathbb{Q}$.

We have

$$W(N) \cong \Gamma_0^*(N)/\Gamma_0(N) \subseteq \mathrm{Aut}_M X_0(N).$$

When neither 4 nor 9 divides N , we have [AS90, p. 321] $\mathfrak{N}\Gamma_0(N) = \Gamma_0^*(N)$. Otherwise $\mathfrak{N}\Gamma_0(N) \supsetneq \Gamma_0^*(N)$ and thus $\mathrm{Aut}_M X_0(N) \supsetneq W(N)$; we call elements of $\mathrm{Aut}_M X_0(N) \setminus W(N)$ **exotic modular automorphisms**. In all cases the group $\mathfrak{N}\Gamma_0(N)$ was computed by Lehner–Newman [LN64, Thm. 1]. The structure of the group $\mathrm{Aut}_M X_0(N)$ was claimed without proof in [AL70, Thm. 8], but with some errors that were later corrected by Akbas–Singer [AS90] and Bars [Ba08]. The general statement is rather complicated and will not be needed, but we will see some specific examples of $\mathrm{Aut}_M X_0(N)$ when N is divisible by 4 or 9 later on.

For all positive integers $M \mid N$, the group $\mathfrak{N}\Gamma_1(M, N)$ was computed by Zemel [Ze17]; when $(M, N) \notin \{(1, 4), (2, 4)\}$ we will show this again following work of Kim–Koo [KK00], who had earlier treated the $M = 1$ case. A recent work of Dose–Lido–Mercuri [DLM22] computes the modular automorphism group for several Cartan-type modular curves including $X_{\text{sp}}(N)$ and $X_{\text{ns}}(N)$.

For a Fuchsian group Γ – and especially for a congruence subgroup of $\text{PSL}_2(\mathbb{Z})$ – one naturally wonders whether $\text{Aut}_M X(\Gamma) = \text{Aut } X(\Gamma)$. Certainly equality does *not* hold if $X(\Gamma)$ has genus 0 or 1: indeed, in these cases $\text{Aut } X(\Gamma)$ is infinite, whereas $\text{Aut}_M X(\Gamma)$ is always finite: for instance $\text{Aut}_M X(1)$ is trivial, while $\text{Aut } X(1) \cong \text{Aut } \mathbb{P}^1 = \text{PGL}_2(\mathbb{C})$. However, as mentioned above, for all $g \in \mathbb{Z}^{\geq 0}$ there are only finitely many modular curves of genus g . Thus one may ask:

Question 3.11. *Does $\text{Aut}_M X(\Gamma) = \text{Aut } X(\Gamma)$ hold for all but finitely many modular curves $X(\Gamma)$?*

Question 3.11 is open; the following results give modest evidence towards an affirmative answer.

Remark 3.12.

- a) Let $N \in \mathbb{Z}^+$ be such that $X_0(N)$ has genus at least 2 (cf. Remark 3.8). By work of Kenku–Momose, Elkies and Harrison, we have $\text{Aut}_M X_0(N) \subsetneq \text{Aut } X_0(N)$ if and only if $N \in \{37, 63, 108\}$.
- b) Let $N \in \mathbb{Z}^{\geq 7}$, so that $X(N)$ has genus at least 2. By work of Bars–Kontogeorgis–Xarles [BKX13], we have

$$\text{Aut } X(N) = \text{Aut}_M X(N) = \text{PSL}_2(\mathbb{Z}/N\mathbb{Z}).$$

- c) Let p be a prime number with $p \geq 23$ and $p \neq 37$. Then Kamienny and Baker have shown that $\text{Aut}_M X_1(p) = \text{Aut } X_1(p)$. The proof uses deep arithmetic results about canonical integral models of $X_0(p)$ and $X_1(p)$ due to Deligne–Rapoport and Katz–Mazur. In an unpublished preprint, F. Momose (now deceased) claimed that $\text{Aut}_M X_1(N) = \text{Aut } X_1(N)$ for all squarefree N such that $X_1(N)$ has genus at least 2, but he also claimed a similar result for modular curves $X_\Delta(N)$ intermediate between $X_0(N)$ and $X_1(N)$ that turned out to be false, and as far as we know his argument for $X_1(N)$ has not been publicly vetted.
- d) Dose–Lido–Mercuri [DLM22] have shown that for all sufficiently large N , all automorphisms of $X_{\text{sp}}(N)$ and $X_{\text{ns}}(N)$ are modular and that for all primes $p \geq 13$, all automorphisms of $X_{\text{sp}}^+(p)$ and $X_{\text{ns}}^+(p)$ are modular.

3.6. Hyperelliptic Modular Curves. The following result is an immediate consequence of Theorem 3.7.

Lemma 3.13. *Let $\Gamma \subseteq \text{PSL}_2(\mathbb{Z})$ be a congruence subgroup. If $\iota(\Gamma) = [\text{PSL}_2(\mathbb{Z}) : \Gamma] \geq 202$, then the Riemann surface $X(\Gamma)$ is not hyperelliptic.*

Since $\text{PSL}_2(\mathbb{Z})$ is finitely generated, it has only finitely many subgroups of any fixed finite index, so Lemma 3.13 implies that there are only finitely many hyperelliptic modular curves. In a 1974 work [Og74] – so long before Theorem 3.7 – Ogg was able not only to prove the finiteness of hyperelliptic modular curves $X_0(N)$ but to find all of them. In previous generations, several mathematicians found all hyperelliptic curves in other families of modular curves.

The state of the art has progressed since then. A recent preprint of Zywinia [Zy] finds *all* congruence subgroups Γ of $\text{PSL}_2(\mathbb{Z})$ such that the Riemann surface $X(\Gamma)$ is hyperelliptic: there are precisely 337 conjugacy classes of such Γ such that $X(\Gamma)$ is hyperelliptic of genus at least 2.

Specializing to the classification of hyperelliptic modular Riemann surfaces in our named families is an immediate consequence of Zywinia’s work....in principle. In practice, there are two technical snags. First, the Cummins–Pauli database only lists the families $X_0(N)$, $X_1(N)$ and $X(N)$ by

name. Of course all these families appear in the LMFDB, but when we search for modular curves in the LMFDB by Cummins–Pauli label, at present the search only returns one curve rather than all curves with this label. In some cases the curve returned is not in one of the named families, but a different such “named” curve appears in the LMFDB with the same Cummins–Pauli label.

Things work better the other way around: if the LMFDB lists a modular curve but does not tell us whether it is geometrically hyperelliptic, then we can make a “Zywina query” by checking whether its Cummins–Pauli label appears in Zywina’s list. In each family Lemma 3.13 reduces us to consider finitely many curves, hence finitely many Zywina inquiries. When combined with prior work and some simple considerations, it turns out that over all the families together we only need to make one Zywina query.

Theorem 3.14 (Hyperelliptic Modular Curves). *In this result, modular curves are considered as Riemann surfaces.*

a) (Ogg [Og74]) *The modular curve $X_0(N)$ is hyperelliptic of genus $g \geq 2$ if and only if:*

$$g = 2 : N \in \{22, 23, 26, 28, 29, 31, 37, 50\}$$

$$g = 3 : N \in \{30, 33, 35, 39, 40, 41, 48\}$$

$$g = 4 : N = 47$$

$$g = 5 : N \in \{46, 59\}$$

$$g = 6 : N = 71.$$

b) (Ishii–Momose [IM91]) *The curve $X_1(N)$ is hyperelliptic of genus $g \geq 2$ if and only if $g = 2$ if and only if $N \in \{13, 16, 18\}$.*

c) *There are no pairs (M, N) with $M \mid N$ and $M \geq 2$ such that $X_1(M, N)$ is hyperelliptic of genus at least 2.*

d) *Among the four families of modular curves $X_{\text{sp}}(N)$, $X_{\text{sp}}^+(N)$, $X_{\text{ns}}(N)$ and $X_{\text{ns}}^+(N)$, the ones that are hyperelliptic of genus at least 2 are precisely: $X_{\text{sp}}^+(11)$, $X_{\text{ns}}(9)$, $X_{\text{ns}}^+(15)$, $X_{\text{ns}}^+(16)$, of genus 2, and $X_{\text{ns}}(12)$, $X_{\text{ns}}^+(20)$, of genus 3.*

e) *The curve $X_{\text{ns}}^*(N)$ is hyperelliptic of genus at least 2 if and only if $N \in \{16, 20\}$; in both cases, we have $g(X_{\text{ns}}^*(N)) = 2$. (Note: $X_{\text{ns}}^*(16) = X_{\text{ns}}^+(16)$.)*

f) *There is no prime $p > 2$ for which $X_{S_4}(p)$ is hyperelliptic of genus at least 2.*

Proof. We only need to prove parts c) through f).

c) By [Ne72, Thm. VII.2], if $X_1(M, N)$ is hyperelliptic, then so is $X_1(N)$ and thus by part b) we have $N \leq 16$ or $N = 18$. Among these curves, the pairs (M, N) such that $g(X_1(M, N)) \geq 2$ are: $(5, 10)$, $(3, 12)$, $(4, 12)$, $(6, 12)$, $(2, 14)$, $(7, 14)$, $(3, 15)$, $(5, 15)$, $(2, 16)$, $(4, 16)$, $(8, 16)$, $(2, 18)$, $(9, 18)$. All of these curves are listed in the LMFDB and they all have $\overline{\mathbb{Q}}$ -gonality at least 3, i.e., they are not hyperelliptic.

d) By Lemma 3.13, we may assume that $\iota(\Gamma) \leq 201$. For $N \in \mathbb{Z}^{\geq 3}$, among the subgroups $\Gamma_{\text{sp}}(N)$, $\Gamma_{\text{sp}}^+(N)$, $\Gamma_{\text{ns}}(N)$ and $\Gamma_{\text{ns}}^+(N)$, the one of smallest modular index is $\Gamma_{\text{ns}}^+(N)$, with $\iota(\Gamma_{\text{ns}}^+(N)) = \frac{\varphi(N)}{2}N$. If $N \geq 43$, then $\frac{\varphi(N)}{2}N \geq 6N \geq 258$, so none of the curves $X_{\text{sp}}(N)$, $X_{\text{sp}}^+(N)$, $X_{\text{ns}}(N)$ and $X_{\text{ns}}^+(N)$ are hyperelliptic. For $N \leq 43$, the LMFDB contains data for all four families of curves and in every case lists whether the curve is hyperelliptic (over $\mathbb{C}$, in particular) or not, with a single exception: the modular curve $X_{\text{ns}}^+(24)$, of genus 3. This curve has Cummins–Pauli label 24AB3, and checking against Zywina’s list,⁷ we find that it is not hyperelliptic.

e) The curves $X_{\text{ns}}^*(16)$ and $X_{\text{ns}}^*(20)$ are both of genus 2, hence are hyperelliptic. We may now

⁷<https://github.com/davidzywina/Modular/blob/main/classification/gonality2.m>

suppose that $g(X_{\text{ns}}^*(N)) \geq 3$ and, by Lemma 3.13, that $\iota(\Gamma_{\text{ns}}^*(N)) \leq 201$. Since $X_{\text{ns}}^*(N) = X_{\text{ns}}^*(2N)$ if N is odd, we may assume that $N \not\equiv 2 \pmod{4}$, which leaves us with

$$N \in \{13, 17, 19, 33, 36, 40, 48, 60\}.$$

For $N \in \{13, 17, 19\}$, since N is prime we have $X_{\text{ns}}^*(N) = X_{\text{ns}}^+(N)$, and none of these curves are hyperelliptic. In each of the remaining 5 cases, the curve $X_{\text{ns}}^*(N)$ appears in the LMFDB: not by name, but as a curve covered by $X_{\text{ns}}^+(N)$. For instance $X_{\text{ns}}^*(60)$ is the modular curve 60.1207.ma.1. In all five cases, the gonality bounds listed there tell us that $X_{\text{ns}}^*(N)$ is not hyperelliptic.

f) Let $p > 2$ be prime. Since $\iota(\Gamma_{S_4}(p)) = \frac{p(p^2-1)}{24}$, by Lemma 3.13, the curve $X_{S_4}(p)$ can only be hyperelliptic if $p \leq 13$. Among these values, $X_{S_4}(p)$ has genus 0 or 1 except for $p = 13$, when the curve has genus 3 and is listed in the LMFDB as nonhyperelliptic. $\square$

Remark 3.15. Each of the hyperelliptic Riemann surfaces in Theorem 3.14 has a canonical $\mathbb{Q}$ -model – since $X_1(M, N)$ is not hyperelliptic for $M \geq 2$, we do not have to choose between the standard and arithmetic models of $X_1(M, N)$. So for each of these 29 curves over $\mathbb{Q}$ of genus $g \geq 2$ that are geometrically hyperelliptic, we can ask whether they are hyperelliptic over $\mathbb{Q}$. By Remark 2.13 this is automatic if $g = 2$ or there is a $\mathbb{Q}$ -rational cusp, which holds for all of the curves except $X_{\text{ns}}(12)$ and $X_{\text{ns}}^+(20)$. In these latter two cases, the LMFDB shows that it is hyperelliptic over $\mathbb{Q}$ and computes a defining polynomial $f(x) \in \mathbb{Q}[x]$. In fact LMFDB computes defining polynomials for all of these curves except $X_0(59)$ and $X_0(71)$. Defining polynomials $f \in \mathbb{Q}[x]$ for all the hyperelliptic modular curves $X_0(N)$ were given by Gonzalez Rovira [GR91].

3.7. Bielliptic Modular Curves. In [Zy], Zywinia also found all congruence subgroups $\Gamma \subseteq \text{PSL}_2(\mathbb{Z})$ such that the Riemann surface $X(\Gamma)$ is bielliptic as well as all Γ such that $X(\Gamma)$ is trigonal. Exactly as we did for geometrically hyperelliptic curves, one could use Zywinia's work to enumerate all bielliptic and all trigonal modular curves in each of our named families. In contrast to the hyperelliptic case, there are now many examples of modular curves that are bielliptic as Riemann surfaces but for which no bielliptic involution is defined over $\mathbb{Q}$. For this and other reasons, knowing that $X(\Gamma)$ is bielliptic or trigonal as a Riemann surface is not as computationally useful in determining the cuspidal part of the Weierstrass divisor as the hyperelliptic case will turn out to be. So we do not carry out this enumeration here, but let's mention a relevant older result:

Theorem 3.16 (Jeon–Kim [JK04], [JK05]). *Let $M \mid N$ be positive integers such that $X_1(M, N)$ has genus $g \geq 2$.*

- a) *The following are equivalent:*
 - (i) *The Riemann surface $X_1(M, N)$ is bielliptic.*
 - (ii) *We have $2 \leq g \leq 6$.*
 - (iii) *(M, N) is one of the following pairs:*

$$g = 2 : (1, 13), (1, 16), (1, 18)$$

$$g = 3 : (7, 7), (3, 12), (1, 20)$$

$$g = 4 : (5, 10), (2, 14).$$

$$g = 5 : (8, 8), (4, 12), (2, 16), (1, 17), (1, 21), (1, 24).$$

$$g = 6 : (1, 22).$$

- b) *When the conditions of part a) hold, at least one bielliptic involution is modular.*

4. PAST RESULTS ON CUSPIDAL WEIERSTRASS POINTS ON MODULAR CURVES

Let $N \in \mathbb{Z}^{\geq 7}$. Then the map $X(N) \rightarrow X(1)$ is a $\mathrm{PSL}_2(\mathbb{Z}/N\mathbb{Z})$ -Galois covering ramified precisely over the points $\infty, 0, 1728$ on the j -line. Thus if the theorems of Schoeneberg and Lewittes of §1.3 can be applied to produce Weierstrass points on $X(N)$, these points will have j -invariant $0, 1728$ or ∞ . Indeed Schoeneberg himself proved the following result:

Theorem 4.1 (Schoeneberg). *Let $N \in \mathbb{Z}^+$, and let $q : X(N) \rightarrow X(1)$ be the natural map, a $\mathrm{PSL}_2(\mathbb{Z}/N\mathbb{Z})$ -Galois covering.*

- a) *If $N = 9$ or $N \geq 11$, then every element of $q^{-1}(1728)$ is a Weierstrass point.*
- b) *If $N \in \{10, 12\}$ or $N \geq 14$, then every element of $q^{-1}(0)$ is a Weierstrass point.*
- c) *If $N \geq 7$, then every cusp on $X(N)$ is a Weierstrass point: equivalently, every element of $q^{-1}(\infty)$ is a Weierstrass point.*

For congruence subgroups Γ of $\mathrm{PSL}_2(\mathbb{Z})$ other than $\Gamma(N)$, in general we do not have that $\mathfrak{N}\Gamma \subseteq \mathrm{PSL}_2(\mathbb{Z})$ – indeed, for all $M \mid N$ with $N \geq 2$ and $M < N$, neither $\mathfrak{N}\Gamma_0(N)$ nor $\mathfrak{N}\Gamma_1(M, N)$ is a subgroup of $\mathrm{PSL}_2(\mathbb{Z})$, so the Schoeneberg-Lewittes method could possibly be used to prove that points on $X(\Gamma)$ not mapping to $\infty, 0, 1728$ on the j -line are Weierstrass points, and we will see examples of this later. Nevertheless, for any such Γ , the cusps of Γ form the most obvious finite nonempty set of “distinguished points” on $X(\Gamma)$, so it is natural to ask how the set of cuspidal points of $X(\Gamma)$ compares to the set of Weierstrass points of $X(\Gamma)$, and throughout the 1960’s and 1970’s there was intense interest in this question. Here are the results:

Theorem 4.2 (Lehner–Newman, 1964). *Let $M \in \mathbb{Z}^+$.*

- a) *Suppose that $g(X_0(4M)) \geq 2$. Then the cusps at 0 and ∞ are Weierstrass points on $X_0(4M)$ except possibly when $4M = 64$ or $4M$ is of the form $4p, 8p, 16p$ or $4pq$ for distinct primes p and q with $q \equiv 3 \pmod{4}$.*
- b) *Suppose that $g(X_0(9M)) \geq 2$. Then the cusps at 0 and ∞ are Weierstrass points on $X_0(9M)$ except possibly when $9M = 81$ or $9M$ is of the form $9p$ or $9pq$ for distinct primes p and q with $q \equiv 2 \pmod{3}$.*

The following result is a generalization:

Theorem 4.3 (Atkin, 1967). *Let N be a positive integer that is not squarefree. Suppose moreover that $g(X_0(N)) \geq 2$: that is, $N \notin \{4, 9, 12, 16, 18, 24, 25, 27, 32, 36, 49\}$.*

- a) *If $N = 81$, then the cusp at ∞ is not a Weierstrass point on $X_0(N)$.*
- b) *If $g(X_0(N)) \geq 2$ and $N \neq 81$, the cusp at ∞ is a Weierstrass point on $X_0(N)$ if none of the following hold:*
 - (i) *For some prime p we have $N = 8p$ or $16p$.*
 - (ii) *For distinct primes p and q , not both $1 \pmod{12}$, we have $N = p^2q$.*
 - (iii) *For distinct primes p, q, r such that neither $x^2 + 1 = 0$ nor $x^2 + x + 1 = 0$ is solvable modulo pqr we have $N = p^2qr$.*

Theorem 4.4 (Ogg, 1978). *For all primes $p \geq 7$, the cusp at ∞ is a Weierstrass point on $X_0(2p^2)$.*

Theorem 4.5 (Ogg, 1978). *Let $M \in \mathbb{Z}^+$ be such that $M \leq 10$ or $M \in \{12, 13, 16, 18, 25\}$, so $X_0(M)$ has genus 0. Let p be a prime number that does not divide M . Then there is no $P \in X_0(pM)(\mathbb{Q})$ that is both a cusp and a Weierstrass point.*

Remark 4.6. It seems to us that Ogg’s proof of Theorem 4.5 does not really use that P is a $\mathbb{Q}$ -rational point. If $P \in X_0(pM)(\mathbb{Q})$ is any Weierstrass point, let $K := \mathbb{Q}_p(P)$ be the extension of $\mathbb{Q}_p$ that it generates, and let R be the valuation ring of K . Whereas Ogg works with the Deligne–Rapoport model of $X_0(pM)$ over $\mathbb{Z}_p$, one can work with the base change of this model to R . The argument is otherwise unchanged.

In [Og78] Ogg says that Atkin sent him a proof of the $M = 1$ case of Theorem 4.5 in a 1973 letter and that Ogg's argument is some kind of algebraic geometric analogue / generalization of Atkin's.

Theorem 4.7 (Ogg, 1978). *Let p be a prime number with $p \geq 17$ and $p \not\equiv 1 \pmod{5}$. Then no cusp of $X_0(11p)$ is a Weierstrass point.*

Taking $M = 8$ or 16 in Ogg's Theorem 4.5 resolves the first of the three infinite families of non-squarefree N left open in Atkin's Theorem 4.3. Also Ogg's theorem 4.4 resolves the $q = 2$ case of the second of the three infinite families of nonsquarefree N left open in Theorem 4.3. To the best of our knowledge, these are *all* the results in the literature that address whether the cusp at ∞ is a Weierstrass point on $X_0(N)$. In particular, case (iii) of theorem 4.3 remains unresolved, as does case (ii) when $q > 2$. Perhaps most importantly, computational evidence suggests that when N is squarefree there are no cuspidal Weierstrass points on $X_0(N)$ – when N is squarefree, $\text{Aut}_M X_0(N)$ acts transitively on the cusps of $X_0(N)$, so some cusp is a Weierstrass point if and only if the cusp at ∞ is a Weierstrass point in this case. However, proving this in general or indeed in any infinite family beyond what was treated in [Og78] has remained open for almost 50 years.

For the curves $X_1(N)$ the situation is yet more dire. If N is squarefree, then again $\text{Aut}_M X_1(N)$ acts transitively on cusps of $X_1(N)$, so some cusp is a Weierstrass point if and only if ∞ is a Weierstrass point. The primes $p \leq 79$ for which ∞ is a Weierstrass point on $X_1(p)$ are

$$17, 19, 29, 31, 41, 47, 59, 61$$

while the primes $q \leq 79$ for which ∞ is *not* a Weierstrass point on $X_1(q)$ are

$$13, 23, 37, 53, 67, 71, 73, 79.$$

We find no clear pattern here, but it may be reasonable to believe that the sets of primes for which ∞ is a Weierstrass point and for which ∞ is not a Weierstrass point are both infinite.

Here are two recent results identifying certain cusps on $X_0(N)$ and $X_1(N)$ as Weierstrass points:

Theorem 4.8 (Choi, 2008). *For all primes $p \geq 11$, the cusp $\frac{1}{2}$ is a Weierstrass point on $X_1(4p)$.*

For the last result we need the concept of a **non-Hall cusp**. Let Γ be a congruence subgroup of $\text{PSL}_2(\mathbb{Z})$ of level N : that is N , is minimal such that $\Gamma(N) \subseteq \Gamma$. Then cusps on $X(\Gamma)$ are in bijection with $\Gamma/\Gamma(N)$ -orbits on pairs $\{(x, y), (-x, -y)\}$ with $(x, y) \in (\mathbb{Z}/N\mathbb{Z})^2$ a primitive vector: that is, $\gcd(x, y, N) = 1$. In this description, the cusp at ∞ on $X(\Gamma)$ is the $\Gamma/\Gamma(N)$ -orbit of $\pm(1, 0)$, while the cusp at 0 is the $\Gamma/\Gamma(N)$ -orbit of $\pm(0, 1)$.

Now suppose that $\Gamma(N) \subseteq \Gamma \subseteq \Gamma_0(N)$, so the action of Γ on primitive vectors carries $\pm(x, y)$ to $\pm(x', uy)$ with $u \in (\mathbb{Z}/N\mathbb{Z})^\times$ and thus preserves the ideal $\langle y, N \rangle$. So to a cusp of Γ represented by $\pm(x, y)$ we can attach an invariant

$$d(P) := \gcd(y, N),$$

and we say that P is a **Hall cusp** if $d = d(P)$ is a Hall divisor of N – i.e., $\gcd(d, \frac{N}{d}) = 1$ – and is a **non-Hall cusp** otherwise. For instance we have $d(0) = 1$ and $d(\infty) = N$, so ∞ and 0 are Hall cusps. Every cusp on $X(\Gamma)$ is Hall if and only if N is squarefree. On $X_0(N)$, the Atkin–Lehner group $W(N) = \Gamma_0(N)^*/\Gamma_0(N) \subseteq \text{Aut}_M X_0(N)$ acts simply transitively on the Hall cusps, while on $X_1(N)$, the modular automorphism group $\text{Aut}_M X_1(N) = \Gamma_0(N)^*/\Gamma_1(N)$ acts simply transitively on the Hall cusps. For other choices of Γ , the modular automorphism group $\text{Aut}_M X(\Gamma)$ may carry Hall cusps to non-Hall cusps: e.g. this is certainly the case on $X(N)$, since $\text{Aut}_M X(N)$ is the Galois group of the covering $q : X(N) \rightarrow X(1)$ and the cusps are the elements of the fiber $q^{-1}(\infty)$, so $\text{Aut}_M X(N)$ acts transitively on the cusps.

In particular, $\frac{1}{2}$ is a non-Hall cusp on $X_1(4p)$, so the following result generalizes Theorem 4.8:

Theorem 4.9 (Jeon, 2022).

- a) Let $N \in \mathbb{Z}^+$ be such that $g(X_1(N)) \geq 2$ and $N \neq 18$. Then every non-Hall cusp on $X_1(N)$ is a Weierstrass point.
- b) Let $N \in \mathbb{Z}^+$ and let p be a prime such that $g(X_0(p^3N)) \geq 2$. The cusp $\frac{1}{p}$ is a Weierstrass point on $X_0(p^3N)$ if and only if p^3N is not 81 nor of the form 8ℓ or 16ℓ for a prime $\ell > 2$.

5. A MOTIVATING RESULT

For $d \in \mathbb{Z}^+$, let $\mathcal{T}^\infty(d)$ be the set of pairs (M, N) of positive integers such that $M \mid N$ and $\mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}$ occurs infinitely often as the torsion subgroup of an elliptic curve in degree d : more precisely, there is an infinite sequence $\{F_n\}_{n=1}^\infty$ of degree d number fields and an infinite sequence $\{(E_n)_{/F_n}\}_{n=1}^\infty$ of elliptic curves such that $j(E_m) \neq j(E_n)$ for all $m \neq n$ and $E_n(F_n)[\text{tors}] \cong \mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}$ for all n .

In this section we will prove the following results and then discuss the prospect of quantitatively improving them:

Theorem 5.1. Let $M, N \in \mathbb{Z}^+$ with $M \mid N$. Let $D \in \mathbb{Z}^+$ be such that:

- (i) $\varphi(M) \mid D$, and
- (ii) $D \geq 2\varphi(M)g(X_1(M, N))$.

Then $(M, N) \in \mathcal{T}^\infty(D)$.

Corollary 5.2. For all $d \in \mathbb{Z}^+$, we have

$$\#\mathcal{T}^\infty(d) \geq \lfloor \sqrt{12d} \rfloor.$$

Corollary 5.2 is in distinct contrast to the case of elliptic curves with complex multiplication (CM). First, for any fixed $d \in \mathbb{Z}^+$ there are only finitely many degree d CM j -invariants, so no torsion subgroup can occur infinitely often in degree d in the CM case. Let us define $\mathcal{T}_{\text{CM}}(d)$ as the set of pairs (M, N) with $M \mid N$ such that there is a degree d number field F and a CM elliptic curve $E_{/F}$ with $E(F)[\text{tors}] \cong \mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}$. Then by [BCP17, Thm. 1.1], for all $\epsilon > 0$, there is a subset $S_\epsilon \subseteq \mathbb{Z}^+$ of lower density at least $1 - \epsilon$ and $B_\epsilon \in \mathbb{Z}^+$ such that for all number fields F with $[F : \mathbb{Q}] \in S_\epsilon$ and all CM elliptic curves $E_{/F}$ we have $\#E(F)[\text{tors}] \leq B_\epsilon$. Moreover the set of $d \in \mathbb{Z}^+$ such that

$$\mathcal{T}_{\text{CM}}(d) = \mathcal{T}_{\text{CM}}(1) = \{(1, 1), (1, 2), (2, 2), (1, 3), (1, 4), (1, 6)\}$$

has a density $\delta \in (0.264, 0.265)$ [BP17, Thm. 1.5].

5.1. Proof of Theorem 5.1 and Corollary 5.2. Let $M, N \in \mathbb{Z}^+$ with $M \mid N$. We need to consider two different rational models of the modular curve $X_1(M, N)$ corresponding to two different open subgroups of $\text{GL}_2(\widehat{\mathbb{Z}})/\pm 1$. The first one is the model $X_1(M, N)_{/\mathbb{Q}}$ given by the subgroup $\pm\Gamma_1(M, N)$ of $\text{GL}_2(\mathbb{Z}/N\mathbb{Z})/\{\pm 1\}$ with $\Gamma_1(M, N) = \dots$. For a field F of characteristic 0, noncuspidal F -rational points on $X_1(M, N)$ correspond to $\bar{F}$ -isomorphism classes of triples (E, P_1, P_2) with $E_{/F}$ an elliptic curve and $P_1, P_2 \in E(F)$ two points with P_1 of order M , P_2 of order N and such that the subgroup $\langle P_1, P_2 \rangle$ of $E(F)[N]$ generated by P_1 and P_2 is isomorphic to $\mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}$. The constant subfield of $\mathbb{Q}(X_1(M, N))$ is $\mathbb{Q}(\zeta_M)$, so this is a regular function field over $\mathbb{Q}$ if and only if $M \leq 2$; in general, $X_1(M, N)_{/\mathbb{C}}$ is isomorphic to a disjoint union of $\varphi(M)$ copies of the Riemann surface $X_1(M, N)$.

The second model of $X_1(M, N)$ was considered by Shimura and is called the “arithmetic model” in the LMFDB; following them, we denote it by $X_{1,\text{arith}}(M, N)$. It is given by the following subgroup of $\text{GL}_2(\mathbb{Z}/N\mathbb{Z})/\pm 1$. Moreover, for a field F of characteristic 0, the noncuspidal F -rational points of $X_{1,\text{arith}}(M, N)$ correspond to triples (E, C, P) with E/F an elliptic curve, $C \subseteq E$ an F -rational cyclic order M subgroup scheme and $P \in E(F)$ a point of order N such that $\langle C, \frac{N}{M}P \rangle = E[M]$. Properties of the Weil e_M -pairing then imply that $C \cong \mu_M$ as group schemes over F . It follows that:

- We have a $\mathbb{Q}$ -rational modular map $\pi : X_{1,\text{arith}}(N, N) \rightarrow X_{1,\text{arith}}(M, N)$.
- The two curves $X_1(M, N)$ and $X_{1,\text{arith}}(M, N)$ become isomorphic over $\mathbb{Q}(\zeta_M)$.

In [Og73, §1], Ogg describes the Galois action on the cusps of $X_{1,\text{arith}}(N, N)$, from which it follows that the cusp 0 is $\mathbb{Q}$ -rational on this curve. It follows that $\pi(0) = 0$ is a $\mathbb{Q}$ -rational cusp on $X_{1,\text{arith}}(M, N)$, hence 0 is a $\mathbb{Q}(\zeta_M)$ -rational cusp on $X_1(M, N)$.

Fix $M \mid N$, and regard $X_1(M, N)$ as a nice curve over $\mathbb{Q}(\zeta_M)$. Let $g := g(X_1(M, N))$, and let $P \in X_1(M, N)(\mathbb{Q}(\zeta_M))$ be the cusp 0. As recalled in §2.1, for all $d \geq 2g$, there is a rational function $f \in \mathbb{Q}(\zeta_M)(X_1(M, N))$ that is regular away from P and has a pole of order d at P , and thus $f : X_1(M, N) \rightarrow \mathbb{P}^1$ has degree d . By [DS17, Lemma 5.1], we have that $(M, N) \in \mathcal{T}^\infty(\varphi(M)d)$. Taking $D := \varphi(M)d$ proves Theorem 5.1.

Applying Theorem 5.1 and Lemma 3.3 with $M = 1$, if $d \in \mathbb{Z}^+$ is at least $\frac{N^2}{12} \geq 2g(X_1(N))$, then $(1, N) \in \mathcal{T}^\infty(d)$, so $(1, N) \in \mathcal{T}^\infty(d)$ for all $1 \leq N \leq \lfloor \sqrt{12d} \rfloor$, proving Corollary 5.2.

5.2. The Prospect of Improvement. Let us first explain why Lemma 3.3, Theorem 5.1 and Corollary 5.2 are sharp up to multiplicative constants. We will use [Fr94, Prop. 2]:

Theorem 5.3 (Frey). *Let F be a number field, and let $X_{/F}$ be a nice curve of gonality γ and an F -rational point. If X has infinitely many closed points of degree d , then $d \geq \frac{\gamma}{2}$.*

For $N \leq 2$, the curve $X_1(M, N)$ has genus 0 hence has infinitely many points over $\mathbb{Q}(\zeta_M) = \mathbb{Q}$. Henceforth we assume $N \geq 3$. Using the inequalities in §3, we find

$$g(X_1(M, N)) \geq \frac{7}{1600} \iota(\Gamma_1(M, N)) = \frac{7}{3200} M \prod_{p \mid N} \left(1 - \frac{1}{p^2}\right) \geq \frac{21}{1600\pi^2} MN^2.$$

Now let $D \in \mathbb{Z}^+$ and $M \mid N$ be such that $(M, N) \in \mathcal{T}^\infty(D)$. Thus there is a degree D number field F and an elliptic curve $E_{/F}$ with $E[M] = E[M](F)$, hence $\mathbb{Q}(\zeta_M) \subseteq F$ and thus $\varphi(M) \mid D$. Applying Theorem 5.3 to $X_1(M, N)_{/\mathbb{Q}(\zeta_M)}$, we get

$$\frac{D}{\varphi(M)} \geq \frac{\gamma(X_1(M, N))}{2} \geq \frac{7}{1600} \iota(\Gamma_1(M, N)) \geq \frac{21}{1600\pi^2} MN^2.$$

Finally, let $d \in \mathbb{Z}^+$. If $(M, N) \in \mathcal{T}^\infty(d)$ and $C := \frac{1600\pi^2}{21}$, then

$$MN^2 \leq M\varphi(M)N^2 \leq Cd.$$

To get an upper bound on the number of such pairs, we rewrite (M, N) as (m, mn) for $m, n \in \mathbb{Z}^+$: then we have

$$(9) \quad m^3 n^2 \leq Cd$$

and thus

$$n \leq \frac{\sqrt{Cd}}{m^{3/2}}.$$

Thus we find

$$\#\mathcal{T}^\infty(d) \leq \sum_{m=1}^{\infty} \frac{\sqrt{Cd}}{m^{3/2}} \leq \zeta(3/2) \sqrt{\frac{1600\pi^2}{21}} \sqrt{d} \leq 71.637\sqrt{d}.$$

So we've determined $\#\mathcal{T}^\infty(d)$ up to a multiplicative factor of roughly 20. We could reduce this discrepancy by finding a rational function on $X_1(M, N)_{/\mathbb{Q}(\zeta_M)}$ of degree $\sim \kappa g(X_1(M, N))$ for some $\kappa < 2$. This is known for $X_1(N)$ for many small values of N : see [DvH14].⁸

Here is one potential improvement: we used the cusp at 0 on $X_1(M, N)$ to get rational functions on $X_1(M, N)_{/\mathbb{Q}(\zeta_M)}$ of all degrees $d \geq 2g(X_1(M, N))$. If we have a $P \in X_1(M, N)(\mathbb{Q}(\zeta_M))$ that is not a Weierstrass point, then the Weierstrass semigroup at P is $\{0\} \cup \mathbb{Z}^{\geq g+1}$ so we get rational functions on $X_1(M, N)_{/\mathbb{Q}(\zeta_M)}$ of all degrees $d \geq g(X_1(M, N)) + 1$, which would improve Theorem 5.1 by a factor of 2 and Corollary 5.2 by a factor of $\sqrt{2}$. It is a “standard guess” that for $M, N \gg 0$, all $\mathbb{Q}(\zeta_M)$ -rational points on $X_1(M, N)$ are cusps: when $M = 1$ this is a famous result of Mazur, and when $M = N = p$ is prime, work of Merel–Stein [MS01] provides evidence for this.

Thus we are led to ask: is the cusp at 0 a Weierstrass point on $X_1(M, N)$? Are there any $\mathbb{Q}(\zeta_M)$ -rational cusps on $X_1(M, N)$ that are not Weierstrass points? We will address these questions in §8.5, §8.6 and §8.7.

6. COMPUTING WEIERSTRASS POINTS

Now we discuss methods to (i) explicitly find all Weierstrass points on certain special curves and (ii) compute Weierstrass weights of cusps on a given modular curve (of small enough level).

6.1. Hyperelliptic Curves. We will now determine the number of cuspidal Weierstrass points on each hyperelliptic modular curve in one of our ten named families, as classified in Theorem 3.14. Recall also Remark 3.15: all of these curves are not just hyperelliptic but hyperelliptic over $\mathbb{Q}$ in their canonical $\mathbb{Q}$ -models. On the computational side, this lets us calculate with defining equations $y^2 = f(x)$ for $f \in \mathbb{Q}[x]$. On the theoretical side, it endows the locus of Weierstrass points with the structure of a $\mathfrak{g}_{\mathbb{Q}}$ -set. It turns out that even if we are only interested in Weierstrass points on $X(\Gamma)$ as a Riemann surface (which – alas – is our primary focus here), this Galois-theoretic information can often be used to rule out certain cusps being Weierstrass points.

We begin with the curves $X_0(N)$, which are the most numerous and perhaps the most interesting examples. For each of the 19 values of N such that $X_0(N)$ is hyperelliptic, we may write $N = pM$ with $p \nmid M$ and $g(X_0(M)) = 0$, so by Theorem 4.5 – together with Remark 4.5 when N is not squarefree – implies that no cusp on $X_0(N)$ is a Weierstrass point. But let us take a closer look at these curves.

From Remark 3.12a) and Theorem 3.14, the unique N for which $X_0(N)$ is hyperelliptic of genus at least 2 and has a nonmodular automorphism is $N = 37$. Otherwise, if N is divisible by neither 4 nor 9, then we have

$$(\text{Aut } X_0(N))(\mathbb{Q}) = \text{Aut } X_0(N) = W(N) \cong (\mathbb{Z}/2\mathbb{Z})^{\omega(N)}.$$

We also have (e.g. by a MAGMA computation)

$$(\text{Aut } X_0(28))(\mathbb{Q}) = \text{Aut } X_0(28) \cong D_6,$$

⁸Also see [DvH14, §4] of their paper for an interesting discussion of patterns in the data supporting conjectures on κ for certain infinite subfamilies.

$$(\operatorname{Aut} X_0(40))(\mathbb{Q}) = \operatorname{Aut} X_0(40) \cong D_4 \times C_2,$$

$$(\operatorname{Aut} X_0(48))(\mathbb{Q}) \cong D_4 \times C_2, \operatorname{Aut} X_0(48) \cong S_4 \times C_2.$$

For all squarefree $N \in \mathbb{Z}^+$, all cusps on $X_0(N)$ are $\mathbb{Q}$ -rational. For nonsquarefree N this may or may not be the case: among the values of N such that $X_0(N)$ is hyperelliptic, all the cusps on $X_0(28)$ and $X_0(40)$ are $\mathbb{Q}$ -rational, but some cusps on $X_0(48)$ and $X_0(50)$ are not.

Example 6.1. For any prime number p , the modular curve $X_0(p)$ has 2 cusps, 0 and ∞ . Theorem 4.5 implies that neither cusp on $X_0(p)$ is a Weierstrass point. If $p \neq 37$, then $(\operatorname{Aut} X_0(p))(\mathbb{Q}) = \operatorname{Aut} X_0(p) = \langle w_p \rangle \cong C_2$. Now we take $p = 37$: the curve $X_0(37)$ has genus 2, so is hyperelliptic, but the hyperelliptic involution ι is *not* a modular automorphism. Indeed, the only nontrivial modular automorphism is w_{37} , which has 2 fixed points, hence $X_0^+(37) := X_0(37)/\langle w_{37} \rangle$ has genus 1.⁹ Thus $X_0(37)$ is bielliptic over $\mathbb{Q}$. A MAGMA computation gives

$$(\operatorname{Aut} X_0(37))(\mathbb{Q}) = \operatorname{Aut} X_0(37) = \langle \iota, w_{37} \rangle \cong C_2 \times C_2,$$

so as a curve over $\mathbb{Q}$, $X_0(37)$ is both hyperelliptic and bielliptic.

Theorem 4.5 implies that neither cusp is a Weierstrass point on $X_0(37)$. Let us see how to use the hyperelliptic structure to deduce this. By [GR91, p. 794], a defining equation for $X_0(37)$ is $y^2 = f(x)$, where

$$f(x) = x^6 - 4x^5 - 40x^4 + 348x^3 - 1072x^2 + 1532x - 860.$$

This sextic is irreducible, so the Weierstrass divisor is irreducible: each Weierstrass point has degree 6. Since the cusps are $\mathbb{Q}$ -rational, once again: no cusp on $X_0(37)$ is a Weierstrass point.

Example 6.2. Let $N \in \{22, 23, 26, 28, 29, 30, 31, 33, 35, 39, 41, 46, 47, 50, 59, 71\}$.

- a) For each of these values of N , by [Og74] some Atkin–Lehner involution $w_d \neq w_4$ on $X_0(N)$ is the hyperelliptic involution. By [Og74, Prop. 3], an Atkin–Lehner involution w_d has cuspidal fixed points if and only if $d = 4$, in which case the unique cuspidal fixed point is $\frac{1}{2}$, so for these values of N , no cusp is a Weierstrass point on $X_0(N)$.
- b) For all of these N , the defining polynomial $f(x) \in \mathbb{Q}[x]$ supplied by [GR91] has no roots in $\mathbb{Q}$.¹⁰ If $N \neq 50$, then all cusps on $X_0(N)$ are $\mathbb{Q}$ -rational, which shows a third time that no cusp is a Weierstrass point on $X_0(N)$.

Example 6.3.

- a) The argument of Example 6.2b) also applies to the $X_0(40)$, which is hyperelliptic of genus 3: all of its cusps are $\mathbb{Q}$ -rational, and the defining polynomial $f(x) = x^8 + 8x^6 - 2x^4 + 8x^2 + 1$ is irreducible over $\mathbb{Q}$, so no cusp is a Weierstrass point.
- b) The curve $X_0(48)$ is hyperelliptic of genus 3, and 8 of its 12 cusps are $\mathbb{Q}$ -rational. We have a degree 2 map $X_0(48) \rightarrow X_0(24)$ and all cusps on $X_0(24)$ are $\mathbb{Q}$ -rational, so the four non- $\mathbb{Q}$ -rational cusps on $X_0(48)$ must each have degree 2 over $\mathbb{Q}$. By [GR91], a defining polynomial for $X_0(48)$ is $f(x) = x^8 + 14x^4 + 1$. This polynomial factors as a product of two degree 4 irreducibles, so every Weierstrass point on $X_0(48)$ has degree 4. Thus no cusp is a Weierstrass point on $X_0(48)$.

Example 6.4. Let $N \geq 5$. By work of Kim–Koo [KK00] and Lang [La01], the modular automorphism group of $X_1(N)$ has size $2^{\omega(N)-1} \varphi(N)$.

⁹In fact, $(X_0^+(37))_{/\mathbb{Q}}$ is a famous elliptic curve: its conductor, 37 is minimal among elliptic curves $E_{/\mathbb{Q}}$ of positive rank.

¹⁰This is not an accident: it is true because neither $-N$ nor $-4N$ is a class number one imaginary quadratic discriminant.

- a) The modular curve $X_1(13)$ has genus 2 and 12 cusps, 6 of which are $\mathbb{Q}$ -rational. Using MAGMA, we find:

$$(\text{Aut } X_1(13))(\mathbb{Q}) \cong C_6, \text{ Aut } X_1(13) \cong D_6.$$

Thus $\text{Aut}_M X_1(13) = \text{Aut } X_1(13)$. Thus $X_1(13)$ has one $\mathbb{Q}$ -rational involution, which must be the hyperelliptic involution. Every nonhyperelliptic involution on a curve of genus 2 is a bielliptic involution, so $X_1(13)$ has six bielliptic involutions, none of which are defined over $\mathbb{Q}$. The LMFDB gives a hyperelliptic model, which after simplification has the form

$$y^2 = f(x) = x^6 + 2x^5 + x^4 + 2x^3 + 6x^2 + 4x + 1.$$

The polynomial $f(x) \in \mathbb{Q}[x]$ is irreducible.

The hyperelliptic involution on $X_1(13)$ is the degree 2 factor of the degree 6 cyclic cover $\pi : X_1(13) \rightarrow X_0(13)$; in other words it is the diamond operator $[5] \in (\mathbb{Z}/13\mathbb{Z})^\times / \{\pm 1\}$ [IM91, Thm. 2.1]; this can also be seen from the LMFDB. By Lemma 3.4, the map π is unramified over the cusps, so no cusp on $X_1(13)$ is a Weierstrass point. In [Je15], Jeon proves this by writing down six points on $\mathcal{H}$ that map to the Weierstrass points on $X_1(13)$.

- b) The modular curve $X_1(16)$ has genus 2 and 14 cusps, of which 6 are $\mathbb{Q}$ -rational. Using MAGMA, we find:

$$(\text{Aut } X_1(16))(\mathbb{Q}) \cong C_4, \text{ Aut } X_1(16) \cong D_4.$$

So as above $X_1(16)$ has one $\mathbb{Q}$ -rational involution, which must be the hyperelliptic involution, and four bielliptic involutions, none of which are defined over $\mathbb{Q}$. Simplifying LMFDB's hyperelliptic model, we get

$$y^2 = x^6 - 2x^5 - x^4 - x^2 + 2x + 1 = (x+1)(x-1)(x^2 - 2x - 1)(x^2 + 1).$$

which shows that $X_1(16)$ has two $\mathbb{Q}$ -rational Weierstrass points and 2 Galois conjugate pairs of quadratic Weierstrass points. A special case of the construction we will make in §8.6 shows that $X_1(16) \rightarrow X(\Gamma_1(8) \cap \Gamma_0(16))$ is a degree 2 map with ramification points the 6 non-Hall cusps; by Riemann-Hurwitz this is the hyperelliptic involution, so these are precisely the Weierstrass points. We conclude that on this curve the Weierstrass points are precisely the non-Hall cusps, as was first shown by Jeon [Je15].

- c) The modular curve $X_1(18)$ has genus 2 and 16 cusps, of which 6 are $\mathbb{Q}$ -rational. Using MAGMA, we find:

$$(\text{Aut } X_1(18))(\mathbb{Q}) \cong C_6, \text{ Aut } X_1(18) \cong D_6,$$

so again $X_1(18)$ has six bielliptic involutions but none defined over $\mathbb{Q}$. Using LMFDB and MAGMA we get a defining equation

$$y^2 = f(x) = x^6 + 4x^5 + 10x^4 + 10x^3 + 5x^2 + 2x + 1.$$

The polynomial $f(x) \in \mathbb{Q}[x]$ is irreducible, so the Weierstrass divisor on $X_1(18)$ is irreducible. The Galois action on cusps on $X_1(N)$ preserves the d -invariant of a cusp defined above, and the number of cusps with any given $d \mid 18$ is always less than 6, so there is no Galois orbit on cusps of size 6. It follows that no cusp on $X_1(16)$ is a Weierstrass point. This was earlier shown by Jeon in [Je15], who again used the work of Ishi-Momose to find six points in the upper half plane mapping to the six Weierstrass points on $X_1(18)$: none of them are cusps.

Thus for $N \in \{13, 16, 18\}$ we find that every automorphism of $X_1(N)$ is modular and that $X_1(N)$ is geometrically bielliptic but not bielliptic over $\mathbb{Q}$.

Example 6.5.

- a) The modular curve $X_{\text{sp}}^+(11)$ has genus 2. It has six cusps, one of which is $\mathbb{Q}$ -rational. Using LMFDB and MAGMA we find that $\mathbb{W}(X_{\text{sp}}^+(11))$ is irreducible: all Weierstrass points have degree 6. Since one of the six cusps is $\mathbb{Q}$ -rational, all have degree at most 5, and thus no cusp on $X_{\text{sp}}^+(11)$ is a Weierstrass point.
- b) The modular curve $X_{\text{ns}}(9)$ has genus 2. It has six cusps, none $\mathbb{Q}$ -rational but all $\mathbb{Q}(\zeta_9)$ -rational. Using LMFDB and MAGMA we get a defining equation

$$y^2 = f(x) = -4x^6 - 3x^4 - 14x^3 - 9x^2 + 4x - 19.$$

In $\mathbb{Q}(\zeta_9)[x]$, the polynomial f factors as a product of two irreducible cubics, so no cusp on $X_{\text{ns}}(9)$ is a Weierstrass point.

- c) The modular curve $X_{\text{ns}}(12)$ is hyperelliptic of genus 3 and has 4 cusps, none $\mathbb{Q}$ -rational. Using LMFDB and MAGMA we find that $\mathbb{W}(X_{\text{sp}}^+(11))$ is weakly irreducible: all Weierstrass points have degree 8. Since each cusp has degree at most 4, no cusp on $X_{\text{ns}}(12)$ is a Weierstrass point.
- d) The modular curves $X_{\text{ns}}^+(15)$ and $X_{\text{ns}}^+(16)$ each have genus 2 and 4 cusps, none $\mathbb{Q}$ -rational. Using LMFDB and MAGMA we find that their Weierstrass divisors are irreducible: each Weierstrass point has degree 6. So no cusp on either of these curves is a Weierstrass point.
- e) We consider $X_{\text{ns}}^+(20)$ and $X_{\text{ns}}^*(20)$ together. The modular curve $X_{\text{ns}}^+(20)$ is hyperelliptic of genus 3 and has 4 cusps, none $\mathbb{Q}$ -rational. The modular curve $X_{\text{ns}}^*(20)$ has genus 2 and 2 cusps, neither $\mathbb{Q}$ -rational. By Riemann–Hurwitz, the degree 2 cover $X_{\text{ns}}^+(20) \rightarrow X_{\text{ns}}^*(20)$ is unramified. Thus $X_{\text{ns}}^+(20)$ has no $\mathbb{Q}(\sqrt{5})$ -rational cusps: if there were, they would come in $\mathfrak{g}_{\mathbb{Q}}$ -orbits of size 2 and would thus map to $\mathbb{Q}$ -rational cusps on $X_{\text{ns}}^*(20)$, which do not exist. A defining polynomial for $X_{\text{ns}}^+(20)$ is

$$f(x) = x^8 - 6x^7 + 7x^6 + 2x^5 - 2x^3 + 7x^2 + 6x + 1 = (x^2 - 4x - 1)(x^6 - 2x^5 - 2x - 1).$$

So the only Weierstrass points on $X_{\text{ns}}^+(20)$ of degree less than 6 have residue field $\mathbb{Q}(\sqrt{5})$, and thus no cusp on $X_{\text{ns}}^+(20)$ is a Weierstrass point. For $X_{\text{ns}}^*(20)$, the LMFDB gives the defining polynomial

$$f(x) = 4x^6 - 12x^5 + 15x^4 - 20x^3 + 15x^2 - 8x + 4 = (x - 2)(x^2 + 1)(x^3 - x^2 + \frac{3}{4}x - \frac{1}{2}).$$

and shows that the unique $\mathbb{Q}$ -rational Weierstrass point is a $\mathbb{Q}(\sqrt{-3})$ -CM point. Thus there are either 0 or 2 cuspidal Weierstrass points on $X_{\text{ns}}^*(20)$. We will see in Example 6.16 that the former is the case: no cusps are Weierstrass points.

6.2. Genus 3 Nonhyperelliptic Curves. Let F be an algebraically closed field of characteristic 0, and let X/F be a nonhyperelliptic curve of genus $g \geq 3$. Then the canonical bundle induces an embedding of X as a degree $2g - 2$ curve in $\mathbb{P}^{g-1}$, so the canonical divisors on X are precisely the intersections of X with hyperplanes in this embedding. By Lemma 2.8b), a point $P \in X(F)$ is a Weierstrass point if and only if some nonzero holomorphic differential ω on X has $\text{ord}_P(\omega) \geq g$, so it follows that P is a Weierstrass point if and only if some hyperplane H intersects X with multiplicity at least g .

Let $g = 3$. By Example 2.5b), each Weierstrass point has weight 1 or 2. By (2), X has at least 12 Weierstrass points. (In fact, Vermeulen showed [Ve83] that when $F = \mathbb{C}$ the number of Weierstrass points on X can be any n with $12 \leq n \leq 24$ except $n \in \{13, 14\}$.) The nonhyperelliptic curve X canonically embeds in $\mathbb{P}^2$ as a plane quartic (conversely, every smooth plane quartic is nonhyperelliptic of genus 3); here hyperplanes are lines. The only line that intersects X at P with multiplicity at least 2 is the tangent line at P , so the Weierstrass points are the **flex points** of the curve, where the tangent line intersects with multiplicity at least 3. Moreover, the Weierstrass

points of weight 2 (if any) are precisely the **hyperflex points** of the curve, where the tangent line intersects with multiplicity 4. If now F is any field of characteristic 0, then all of the above still holds, with the understanding that the flex = Weierstrass points will in general lie in $\overline{F}$.

We now recall a classical result that makes this more explicit: for a degree d plane curve X given by $F(X_1, X_2, X_3) = 0$ its **Hessian curve** is the plane curve cut out by the equation $\det \text{Hess}(F) = 0$, where $\text{Hess}(F)$ is the **Hessian matrix**, the 3×3 matrix with (i, j) entry $\frac{\partial^2 F}{\partial X_i \partial X_j}$. Each entry of the Hessian matrix is homogeneous of degree $d - 2$, so the Hessian curve has degree $3(d - 2)$. Thus, assuming that the curves F and $\det \text{Hess}(F)$ have no component in common, Bézout's Theorem gives that $F \cap \det \text{Hess}(F)$ is a divisor of degree $3d(d - 2)$ on X . When $d = 4$, this intersection yields a divisor of degree 24 that is indeed the Weierstrass divisor on X .

If the defining quartic $F(X, Y, Z) = 0$ of a smooth nonhyperelliptic curve of genus 3 is explicitly known - with coefficients, in say, a number field - then this computation of the Weierstrass divisor can be implemented in the MAGMA programming language. If the coefficients of $F(X, Y, Z)$ lie in a number field, then a routine MAGMA computation will yield the Weierstrass divisor $\mathbb{W}(X)$ as a zero-dimensional scheme, which allows us to compute the Weierstrass divisor and the splitting field of the Galois action on Weierstrass points.

For every genus 3 nonhyperelliptic modular curve X listed in the LMFDB, the database gives a hyperelliptic quartic *and* the j -map as a rational function on this quartic, more specifically as a quotient of two polynomials $\frac{N(X, Y, Z)}{D(X, Y, Z)}$. Thus a point $[X : Y : Z]$ on X is a cusp if and only if $D(X, Y, Z) = 0$. This makes it routine to compute the cuspidal part of $\mathbb{W}(X)$.

The next example is one of the oldest and most famous examples of Weierstrass points on a nonhyperelliptic curve.

Example 6.6. We consider the Fermat quartic

$$F_4 : X^4 + Y^4 - Z^4 = 0$$

as a curve over $\mathbb{Q}$: it is nonhyperelliptic of genus 3. We have

$$(\text{Aut } F_4)(\mathbb{Q}) \cong D_4 \text{ and } G := \text{Aut } F_4 \cong (\mathbb{Z}/4\mathbb{Z})^2 \rtimes S_3.$$

(Here we regard $(\mathbb{Z}/4\mathbb{Z})^2$ as the quotient of $(\mathbb{Z}/4\mathbb{Z})^3$ by the subgroup generated by $(1, 1, 1)$, giving it a natural S_3 -action. Over $\mathbb{C}$ we may take the defining equation to be

$$X^4 + Y^4 + Z^4 = 0,$$

which makes the automorphisms visible.) Every involution of F_4 is bielliptic (cf. §2.7) and thus has 4 fixed points. There are two conjugacy classes of involutions. The first has size 15 and is represented by $(X, Y, Z) \mapsto (Y, X, Z)$. The second consists of the three coordinate reflections

$$(X, Y, Z) \mapsto (-X, Y, Z), (X, Y, Z) \mapsto (X, -Y, Z), (X, Y, Z) \mapsto (X, Y, -Z).$$

The fixed points of $(X, Y, Z) \mapsto (-X, Y, Z)$ are the four projective solutions of $X = 0$, $Y^4 = -Z^4$, two of which are $\mathbb{Q}$ -rational: $[0 : 1 : 1]$ and $[0 : 1 : -1]$. Similarly, $(X, Y, Z) \mapsto (X, -Y, Z)$ has 4 fixed points, two of which are $\mathbb{Q}$ -rational and $(X, Y, Z) \mapsto (X, Y, -Z)$ has 4 fixed points, none of which are $\mathbb{Q}$ -rational. The tangent line at $[0 : 1 : 1]$ is $Y = Z$, whose sole intersection point with F_4 is $[0 : 1 : 1]$. Thus $[0 : 1 : 1]$ is a hyperflex, so has Weierstrass weight 2. Its G -orbit consists of the $4 + 4 + 4$ fixed points of the three coordinate reflections, so F_4 has 12 Weierstrass points, each of weight 2, precisely 4 of which are $\mathbb{Q}$ -rational. In the terminology of §2.7, each of the three coordinate reflections is a totally heavy bielliptic involution, while each of the other 12 involutions is a totally light bielliptic involution.

The modular curves in our named families that are nonhyperelliptic of genus 3 are precisely:

$$X_0(34), X_0(43), X_0(45), X_0(64) \cong X_{\text{sp}}(8), X(7), X_1(3, 12), X_1(20), X_{\text{sp}}^+(13), \\ X_{\text{ns}}^+(13) = X_{\text{ns}}^*(13) = X_{\text{ns}}^*(26), X_{\text{ns}}^+(24), X_{S_4}(13).$$

Example 6.7.

- a) $X_0(34)$ has 4 cusps, each of which is $\mathbb{Q}$ -rational. By Theorem 4.5, no cusp is a Weierstrass point. We can recover this and get a bit more using the defining quartic equation

$$F(X, Y, Z) = X^3Y - X^3Z + X^2Y^2 + XY^3 + 2XY^2Z - XZ^3 - 2Y^4 + Y^3Z + Y^2Z^2 + YZ^3$$

for $X_0(34)$ taken from the LMFDB. A MAGMA computation then shows that $\mathbb{W}(X_0(34))$ is strongly irreducible: there are 24 Weierstrass points, and they form a single Galois orbit. Since each cusp is $\mathbb{Q}$ -rational, this confirms that no cusp is a Weierstrass point.

- b) $X_0(43)$ has 2 cusps, each of which is $\mathbb{Q}$ -rational. The discussion is otherwise identical to that of part a): Theorem 4.5 gives that no cusp is a Weierstrass point. We compute that $\mathbb{W}(X_0(34))$ is strongly irreducible, which confirms this.
- c) $X_0(45)$ has 8 cusps, 4 of which are $\mathbb{Q}$ -rational. Theorem 4.5 shows that the $\mathbb{Q}$ -rational points are not Weierstrass points. Using MAGMA and LMFDB, we find that $\mathbb{W}(X_0(45))$ is strongly irreducible, and thus no cusp is a Weierstrass point, confirming Remark 4.6.

We have $\text{Aut } X_0(45) \cong S_4$ (e.g. [FGL07, §3.2] gives $\text{Aut } X_0(45) \cong \text{PGL}_2(\mathbb{F}_3)$, and we have $\text{PGL}_2(\mathbb{F}_3) \cong S_4$). By [Ba05, Thm. 16] and [RCS24, §2], among nonhyperelliptic curves of genus 3 with fewer than 96 automorphisms, having automorphism group $G \cong S_4$ is necessary and sufficient for G to act transitively on the Weierstrass points (and moreover simply transitively except for the special Kuribayashi–Komiya curve of (3)).

Our next two examples are the only two genus 3 curves with at least 96 automorphisms.

Example 6.8. The modular curve $X_0(64)$ has 12 cusps. In [La71], Larcher claimed that each cusp on $X_0(64)$ is a Weierstrass point of weight 2. This paper contains several errors: Larcher claimed to prove that for all primes $p \geq 3$, the cusps on $X_0(16p)$ are Weierstrass points, which was later contradicted by Theorem 4.5 (and it is Ogg’s proof that is correct). In particular, Larcher claimed that the Weierstrass points on $X_0(48)$ are precisely the 12 cusps, but as we saw in Example ??b), on the one hand no cusp on $X_0(48)$ is a Weierstrass point, and on the other hand $X_0(48)$ is hyperelliptic of genus 3 so has only 8 Weierstrass points.

Nevertheless, Larcher’s claim about $X_0(64)$ is correct: using MAGMA and LMFDB we find that the Weierstrass divisor consists of 12 points, each with multiplicity 2, precisely 4 of which are $\mathbb{Q}$ -rational, and using the map to the j -line supplied by LMFDB, we check that every Weierstrass point is a cusp. Thus on $X_0(64)$ the Weierstrass points are precisely the cusps.

As mentioned in §2.9, in fact $X_0(64)$ is isomorphic to the Fermat quartic, already over $\mathbb{Q}$ if we use the model of Example 6.5, as was shown by Kenku [Ke75, Prop. 2].

Here is a third approach to the Weierstrass points on this curve:

$$G := \text{Aut}_M X_0(64) = \text{Aut } X_0(64) \cong (\mathbb{Z}/4\mathbb{Z})^2 \rtimes S_3,$$

and $\pi : X_0(64) \rightarrow X_0(64)/G$ is a Galois Belyĭ map, with order triple $(2, 3, 8)$ [MSSV02, Table 2]. Thus the smallest G -orbit on $X_0(64)$ has size $\frac{96}{8} = 12$ and every other G -orbit has size at least $\frac{96}{3} = 32$. Since there are at most 24 Weierstrass points, the Weierstrass points must form the G -orbit of size 12 and thus have multiplicity 2. Since there are 12 cusps that are stabilized by G , it follows that the cusps form the G -orbit of size 12. We deduce that the Weierstrass points on $X_0(64)$ are precisely the 12 cusps, each with weight 2, and are acted upon transitively by G .

Example 6.9. The modular curve $X(7)$ is the famous **Klein quartic** curve, with defining equation

$$F(X, Y, Z) = X^3Y + Y^3Z + Z^3X.$$

We have $\text{Aut}_M X(7) \cong \text{PSL}_2(\mathbb{Z}/7\mathbb{Z})$, of order 168. Thus $X(7)$ is a **Hurwitz curve**, a curve X of genus $g \geq 2$ for which equality holds in the Hurwitz bound $\# \text{Aut } X \leq 84(g-1)$. It has 24 cusps. Each of the “obvious” $\mathbb{Q}$ -rational points $[1 : 0 : 0]$, $[0 : 1 : 0]$, $[0 : 0 : 1]$ is a flex point, hence a Weierstrass point. LMFDB lists these points as cusps; one can check this using the explicit map to the j -line given by LMFDB. Because $\pi : X(7) \rightarrow X(1)$ is a Galois covering and the cusps on $X(7)$ are the points in the fiber $\pi^{-1}(\infty)$, it follows that every cusp is a Weierstrass point and then that the Weierstrass points on $X(7)$ are precisely the cusps, each with weight 1. We will give a second proof of this later on, in Example 8.2.

Example 6.10. The modular curve $X_1(20)$ has 20 cusps, of which 6 are $\mathbb{Q}$ -rational in the canonical model. According to LMFDB, the defining quartic is

$$F(X, Y, Z) = X^3Y - X^2Z^2 + XY^3 - 2XYZ^2 + XZ^3 - Y^2Z^2 + YZ^3.$$

With MAGMA we compute that there are 20 Weierstrass points on $X_1(20)$: 16 of weight 1 and 4 of weight 2. The four weight 2 Weierstrass points are

$$P_1 := [0 : 0 : 1] \text{ and } P_2 := [1 : 1 : 1],$$

which are $\mathbb{Q}$ -rational and

$$P_3 := [-\zeta_{20}^6 + \zeta_{20}^4 : -\zeta_{20}^6 + \zeta_{20}^4 : 1] \text{ and } P_4 := [\zeta_{20}^6 - \zeta_{20}^4 - 1 : \zeta_{20}^6 - \zeta_{20}^4 - 1 : 1],$$

which are $\mathbb{Q}(\zeta_{20})$ -rational. Using the explicit map to the j -line given in the LMFDB, we find that P_1, P_2, P_3, P_4 are all cusps. Moreover, we compute that P_1, P_2, P_3, P_4 are the only $\mathbb{Q}(\zeta_{20})$ -rational Weierstrass points. Since every cusp on $X_1(20)$ is $\mathbb{Q}(\zeta_{20})$ -rational, it follows that P_1, P_2, P_3, P_4 are precisely the cuspidal Weierstrass points on $X_1(20)$. Because $X_1(20)$ has 16 Hall cusps and 4 non-Hall cusps, it follows from Theorem 4.9a) that P_1, P_2, P_3, P_4 are the non-Hall cusps on $X_1(20)$ and that in the case no Hall cusp is a Weierstrass point.

Example 6.11. The modular curve $X_1(3, 12)$ has 20 cusps. Using the LMFDB and MAGMA we compute that there are 20 Weierstrass points, 16 with weight 1 and 4 with weight 2. Using the explicit j -function provided by LMFDB, we check that each of the Weierstrass points is a cusp, so on $X_1(3, 12)$ the Weierstrass points are precisely the cusps.

Example 6.12.

- a) The modular curve $X_{\text{sp}}^+(10)$ has 9 cusps, of which 3 are $\mathbb{Q}$ -rational. According to LMBFDB, the defining quartic is

$$F(X, Y, Z) = X^3Y - 2X^2Y^2 - X^2YZ + XY^3 - XY^2Z + 2XYZ^2 + XZ^3 + YZ^3.$$

We compute that there are 24 Weierstrass points, so not all Weierstrass points are cusps. The three $\mathbb{Q}$ -rational cusps are $[1 : 0 : 0]$, $[0 : 1 : 0]$ and $[0 : 0 : 1]$, of which the first two are Weierstrass points but the third is not, so not all cusps are Weierstrass points.

- b) The modular curve $X_{\text{sp}}^+(13)$ is nonhyperelliptic of genus 3, called the **cursed curve** [BDMT19]. It has 7 cusps, of which 1 is $\mathbb{Q}$ -rational in the canonical model. A computation using the LMFDB and MAGMA shows that the Weierstrass divisor is strongly irreducible: there are 24 Weierstrass points, forming a single $\mathfrak{g}_{\mathbb{Q}}$ -orbit. Since each cusp has degree at most 9, no cusp is a Weierstrass point.

Example 6.13.

- a) The modular curve $X_{\text{ns}}^+(13) = X_{\text{ns}}^*(13) = X_{\text{ns}}^*(26)$ has 6 cusps, none of which are $\mathbb{Q}$ -rational in the canonical model. Again computations show that the Weierstrass divisor is strongly irreducible, and it follows that no cusp is a Weierstrass point. In fact Baran has shown [Ba14] that the curves $X_{\text{sp}}^+(13)$ and $X_{\text{ns}}^+(13)$ are isomorphic over $\mathbb{Q}$, hence the Weierstrass points on $X_{\text{sp}}^+(13)$ and $X_{\text{ns}}^+(13)$ are isomorphic as $\mathfrak{g}_{\mathbb{Q}}$ -sets. This isomorphism lies much deeper than e.g. the isomorphism between $X_{\text{sp}}(N)$ and $X_0(N^2)$ of Remark 3.1: indeed, the curves $X_{\text{sp}}^+(13)$ and $X_{\text{ns}}^+(13)$ don't have the same number of cusps, so the affine curves $Y_{\text{sp}}^+(13)$ and $Y_{\text{ns}}^+(13)$ are not isomorphic (even) over $\mathbb{C}$.
- b) The modular curve $X_{\text{ns}}^+(24)$ has genus 3. It has 4 cusps, none of which are $\mathbb{Q}$ -rational, so clearly not every Weierstrass point is a cusp. For this curve a plane model is not (as of 6/11/26) listed in the LMFDB. We will return to this curve shortly: see Example 6.16.

Example 6.14. The modular curve $X_{S_4}(13)$ has 7 cusps, none $\mathbb{Q}$ -rational but all $\mathbb{Q}(\zeta_{13})$ -rational. Using the LMFDB and MAGMA, we find that there are 24 Weierstrass points, none of which are $\mathbb{Q}(\zeta_{13})$ -rational. So no cusp is a Weierstrass point.

6.3. Modular Forms and Zywin's Algorithm. Let Γ be a finite index subgroup of $\text{PSL}_2(\mathbb{Z})$. There is an injective $\mathbb{C}$ -linear map from meromorphic differentials on $X(\Gamma)$ to meromorphic differentials on $\mathcal{H}$: namely, we restrict from $X(\Gamma)$ to $Y(\Gamma)$ and then pull back via the natural map $\pi : \mathcal{H} \rightarrow Y(\Gamma)$. We use this injection identify meromorphic differentials on $X(\Gamma)$ with expressions $f(z)dz$ for f a meromorphic function on $\mathcal{H}$. Using this, we get a natural isomorphism

$$\mathcal{S}_2(\Gamma) \rightarrow H^0(X, \Omega^1), \quad f \mapsto f(z)dz$$

from the $\mathbb{C}$ -vector space of weight 2 cusp forms on Γ to the $\mathbb{C}$ -vector space of holomorphic differentials on the Riemann surface $X(\Gamma)$ [DS, p. 82].

Let

$$U := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix},$$

so U generates the stabilizer of ∞ in $\text{PSL}_2(\mathbb{Z})$. The **cusp width** at ∞ of Γ is the least $h \in \mathbb{Z}^+$ such that $U^h \in \Gamma$. There is indeed such an h : if $\text{Core}(\Gamma) := \bigcap_{g \in \text{PSL}_2(\mathbb{Z})} g^{-1}\Gamma g$ is the normal core of Γ , then $\text{Core}(\Gamma)$ is a finite index normal subgroup of $\text{PSL}_2(\mathbb{Z})$ – say of index I – and for every $g \in \text{PSL}_2(\mathbb{Z})$ we have $g^I \in \text{Core}(\Gamma)$, hence $g^I \in \Gamma$. (If Γ is a congruence subgroup of level N , then $h \leq N$.) Then h is also the ramification index at ∞ of the natural map $X(\Gamma) \rightarrow X(1)$.

Let $f \in \mathcal{S}_2(\Gamma)$ be a weight 2 cusp form. Let $\pi : \overline{\mathcal{H}} \rightarrow X(\Gamma)$ be the natural map. Setting $q_h := e^{\frac{2\pi iz}{h}}$, there is a local coordinate φ at the cusp at ∞ on $X(\Gamma)$ such that $\varphi \circ \pi = q_h$ [DS, §2.4], and there is a unique holomorphic function F on the open unit disk such that $f(z) = F(q_h)$, and thus there is a complex sequence $\{a_n\}_{n=0}^\infty$ such that $F(q_h) = \sum_{n=0}^\infty a_n q_h^n$. Because $dq_h = \frac{2\pi i}{h} q_h dz$, we have

$$f(z)dz = \frac{h}{2\pi i} F(q_h) \frac{dq_h}{q_h},$$

giving

$$\text{ord}_\infty(f(z)dz) = \text{ord}_{q_h}(F(q_h)) - 1$$

and thus $a_0 = 0$. Let g be the genus of $X(\Gamma)$. If $F_1(q_h), \dots, F_g(q_h)$ are the q_h -expansions at ∞ for a basis of $\mathcal{S}_2(\Gamma)$, then $\frac{F_1(q_h)}{q_h} dq_h, \dots, \frac{F_g(q_h)}{q_h} dq_h$ is a $\mathbb{C}$ -basis for the holomorphic differentials of $X(\Gamma)$. If Γ is a congruence subgroup of level N , then by [Zy, Lemma 3.2], we may choose the basis $F_1, \dots, F_g$ such that for all $1 \leq i \leq g$, we have $F_i(q_h) = \sum_{n=1}^\infty a_{n,i} q_h^n$ with $a_{n,i} \in \mathbb{Q}(\zeta_N)$ for all n .

For functions $u(z), f_1(z), \dots, f_g(z)$, the Wronskian of $uf_1, \dots, uf_g$ is u^g times the Wronskian of $f_1, \dots, f_g$, and thus Theorem 2.9 gives that the Weierstrass weight w_∞ at the cusp at ∞ is

$$\text{ord}_{q_h} \Phi_{\frac{F_1(q_h)}{q_h}, \dots, \frac{F_g(q_h)}{q_h}}(q_h) = \text{ord}_{q_h} (\Phi_{F_1(q_h), \dots, F_g(q_h)}(q_h)/q_h^g) = \text{ord}_{q_h} (\Phi_{F_1(q_h), \dots, F_g(q_h)}(q_h)) - g.$$

Now let $s = \frac{a}{c} \in \mathbb{Q}$, written in lowest terms. There are $b, d \in \mathbb{Z}$ such that $ad - bc = 1$, and then putting $\sigma := \pm \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, we have that $\sigma(\infty) = s$. The cusp width of s in Γ is the least $h \in \mathbb{Z}^+$ such that

$$(\sigma U \sigma^{-1})^h = \sigma U^h \sigma^{-1} = \pm \begin{bmatrix} 1 - hac & ha^2 \\ -hc^2 & 1 + hac \end{bmatrix} \in \Gamma.$$

It is also the cusp width of ∞ in $\sigma^{-1}\Gamma\sigma \subseteq \text{PSL}_2(\mathbb{Z})$ and is the ramification index at s of $X(\Gamma) \rightarrow X(1)$, showing that it is independent of the choice of σ .

Example 6.15.

- a) Let $s = 0 = \frac{0}{1}$, so we may take $\sigma = \pm \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Thus the cusp width of 0 in Γ is the least $h \in \mathbb{Z}^+$ such that $\pm \begin{bmatrix} 1 & 0 \\ h & 1 \end{bmatrix} \in \Gamma$.
- b) For $d \in \mathbb{Z}^+$, let $s = \frac{1}{d}$, so we may take $\sigma = \pm \begin{bmatrix} 1 & -1 \\ d & 1-d \end{bmatrix}$. Thus the cusp width of $\frac{1}{d}$ in Γ is the least $h \in \mathbb{Z}^+$ such that $\pm \begin{bmatrix} 1-hd & h \\ -hd^2 & 1+hd \end{bmatrix} \in \Gamma$.

Again let $f \in \mathcal{S}_2(\gamma)$. Then $f_{s,\sigma} := (f|_2\sigma)(z)$ is a weight 2 modular form on $\sigma^{-1}\Gamma\sigma$ so has a q_h -expansion $f_{s,\sigma}(q_h) = \sum_{n=0}^{\infty} a_n(\sigma)q_h^n$. We analyze the dependence on σ : if $\sigma' \in \text{PSL}_2(\mathbb{Z})$ is another element carrying ∞ to s , then there $m \in \mathbb{Z}$ such that $\sigma' = \sigma U^m$, so

$$a_n(\sigma') = (e^{\frac{2\pi im}{h}})^n a_n(\sigma).$$

This shows that the q_h -expansion at s is unique up to a substitution $q_h \mapsto e^{\frac{2\pi im}{h}} q_h$. Also if $s' = \gamma s$ for some $\gamma \in \Gamma$, then

$$f_{s',\sigma} = (f|_2\gamma\sigma) = ((f|_2\gamma)|_2\sigma) = (f|_2\sigma) = f_{s,\sigma}.$$

Thus as done above for $s = \infty$, if we have access to the q_h -expansions of $f_{s,\sigma}(q_h)$ as f ranges through a $\mathbb{C}$ -basis of $\mathcal{S}_2(\sigma^{-1}\Gamma\sigma)$, then as above, we can compute the Weierstrass weight at the cusp s by subtracting g from the q_h -adic valuation of their Wronskian.

Now let Γ be a congruence subgroup of $\text{PSL}_2(\mathbb{Z})$. In [Zy], Zywinia describes an algorithm to compute a basis for the space of weight 2 cusp forms (and in fact for cusp forms of any even weight $k \geq 2$) on Γ . He also fully implements this algorithm in his MAGMA package found in <https://github.com/davidzywinia/Modular>. For a Zywinia's implementation finds matrices $A_i \in \text{SL}_2(\mathbb{Z})$ such that $\{A_i\infty\}$ is a set of representatives for all cusps on $X(\Gamma)$, finds a basis for the space of cusp forms and computes the q_h -expansion of each element of the basis at all cusps up to a given precision. Thus we can use Zywinia's algorithm to compute Weierstrass weights at all cusps of Γ ...at least in theory. In practice, it can be quite computationally intensive, and the following procedure is preferable:

As above, let s be a cusp of $X(\Gamma)$, of width h , choose $\sigma \in \text{PSL}_2(\mathbb{Z})$ such that $\sigma(\infty) = s$, let $F_1(q_h), \dots, F_g(q_h) \in \mathbb{Q}(\zeta_N)[[q_h]]$ be the q_h -expansions of a $\mathbb{C}$ -basis of $\mathcal{S}_2(\sigma^{-1}\Gamma\sigma)$, and for $1 \leq i \leq g$, write $F_i(q_h) = \sum_{n=1}^{\infty} a(i, n)q_h^n$. Let $A \in M_{g, 2g-1}(\mathbb{Q}(\zeta_N))$ be the matrix with (i, j) entry $a(i, j)$, and let A' be the reduced row echelon form of A . Row reduction of A amounts to replacing the

basis $F_1(q_h), \dots, F_g(q_h)$ by an adapted basis – in the sense of §2.3 – $G_1(q_h), \dots, G_g(q_h)$, and the indices $1 = b_1 < b_2 < \dots < b_g$ of the columns of A' containing pivots are precisely the gaps in the Weierstrass semigroup at s .¹¹ We may truncate the q_h -expansions at q_h^{2g-1} because every Weierstrass gap is at most $2g - 1$, so the matrix A already has full rank g .

Not only does the above procedure yield not just the Weierstrass weight at the cusp s but the full Weierstrass semigroup at s , but it is also considerably faster than computing the Wronskian to sufficient precision. The bottleneck in this computation is now the time it takes Zywin's algorithm to return the q_h -expansions $F_1, \dots, F_g$ at each cusp of $X(\Gamma)$ up to order q_h^{2g-1} .

The following are all examples of successful implementations of Zywin's algorithm to compute Weierstrass weights at all cusps of certain modular curves $X(\Gamma)$ that we will need for future results.

Example 6.16. We list the Weierstrass weights at the cusps of $X_1(M, N)$ for some pairs (M, N) that we will need for future results.

- a) $X_1(4, 12)$ has genus 5 and 24 cusps, all of Weierstrass weight 3. See also Example 9.11.
- b) $X_1(6, 12)$ has genus 9 and 32 cusps, all of weight 9.
- c) $X_1(2, 14)$ has genus 4 and 18 cusps, all of weight 1. See also Example 8.12.
- d) $X_1(7, 14)$ has genus 19 and 48 cusps, all of weight 27.
- e) $X_1(3, 15)$ has genus 9 and 32 cusps, all of weight 9.
- f) $X_1(5, 15)$ has genus 17 and 48 cusps all of weight 12.
- g) $X_1(2, 16)$ has genus 5 and 24 cusps: 16 of weight 2 and 8 of weight 3.
- h) $X_1(4, 16)$ has genus 13 and 40 cusps, all of weight 13.
- i) $X_1(2, 18)$ has genus 7 and 24 cusps: 18 of weight 3 and 6 of weight 5.
- j) $X_1(3, 18)$ has genus 10 and 36 cusps, all of weight 2.
- k) $X_1(2, 20)$ has genus 9 and 32 cusps, all of weight 3.
- l) $X_1(2, 24)$ has genus 13 and 40 cusps, 32 of weight 3 and 8 of weight 17.
- m) $X_1(25)$ has genus 12 and 28 cusps. Each of the 20 Hall cusps has weight 3 and each of the 8 non-Hall cusps has weight 12.

In each of the 12 cases above, the degree of the cuspidal part of $\mathbb{W}(X_1(M, N))$ is less than $g(X_1(M, N))^3 - g(X_1(M, N))$, so there are noncuspidal Weierstrass points on $X_1(M, N)$.

Example 6.17.

- a) $X_{\text{ns}}^+(24)$ has genus 3 and 4 cusps. No cusp is a Weierstrass point.
- b) $X_{\text{ns}}^*(20)$ has genus 2 and 2 cusps. No cusp is a Weierstrass point.
- c) $X_{\text{ns}}^*(36)$ has genus 5 and 3 cusps, all of weight 2.
- d) $X_{\text{ns}}^*(48)$ has genus 9 and 4 cusps, all of weight 1.
- e) $X_{\text{ns}}^*(60)$ has genus 7 and 2 cusps. No cusp is a Weierstrass point.
- f) $X_{\text{ns}}^*(84)$ has genus 13 and 3 cusps, all of weight 9.
- g) $X_{\text{ns}}^*(120)$ has genus 33 and 4 cusps, all of weight 1.

7. APPLICATIONS OF ATKIN–LIMONOV

7.1. Proof of Theorem 1.7. First suppose that $N \equiv 2 \pmod{4}$ and $N \neq 2$. Consider the following maps:

$$\pi : X_{\text{ns}}(N) \rightarrow X_{\text{ns}}(N/2)$$

and

$$\pi^+ : X_{\text{ns}}^+(N) \rightarrow X_{\text{ns}}^+(N/2).$$

¹¹For $1 \leq i \leq g$, if $g_i(z)dz$ is the holomorphic differential corresponding to $G_i(q_h)$, then we have $\text{ord}_s(g_i(z)dz) = \text{ord}_{q_h}(G_i(q_h)) - 1$, but the corresponding Weierstrass gap is $\text{ord}_s(g_i(zdz)) + 1 = \text{ord}_{q_h}(G_i(q_h))$.

Using [HL25, Prop. 4.2 and Table 1] we find that π and π^+ both have degree 2; moreover

$$v_\infty(X_{\text{ns}}(N)) = \varphi(N) = \varphi(N/2) = v_\infty(X_{\text{ns}}(N/2))$$

and

$$v_\infty(X_{\text{ns}}^+(N)) = \frac{\varphi(N)}{2} = \frac{\varphi(N/2)}{2} = v_\infty(X_{\text{ns}}^+(N/2)).$$

Thus every cusp on $X_{\text{ns}}(N)$ (resp. on $X_{\text{ns}}^+(N)$) is totally ramified in the map π (resp. π^+). By Theorem 2.23c), we find that all the cusps on $X_{\text{ns}}(N)$ (resp. on $X_{\text{ns}}^+(N)$) are Weierstrass points as soon as either curve has at least five cusps. Using the above formulas for $v_\infty(X_{\text{ns}}(N))$ and $v_\infty(X_{\text{ns}}^+(N))$ and the fact that $\varphi(N) \geq 10$ for all $N \geq 31$, we arrive at parts a) and b).

Now suppose that $N \equiv 3, 6 \pmod{9}$ and consider the map

$$\pi : X_{\text{ns}}^*(N) \rightarrow X_{\text{ns}}^*(N/3).$$

By [HL25, Cor. 4.6 and Table 1] we find that π has degree 3, and also: if $N \equiv 2 \pmod{4}$, then

$$v_\infty(X_{\text{ns}}^*(N)) = \frac{\varphi(N)}{2^{\omega(N)-1}} = \frac{\varphi(N/3)}{2^{\omega(N/3)-1}} = v_\infty(X_{\text{ns}}^*(N/3)),$$

while if $N \not\equiv 2 \pmod{4}$, then

$$v_\infty(X_{\text{ns}}^*(N)) = \frac{\varphi(N)}{2^{\omega(N)}} = \frac{\varphi(N/3)}{2^{\omega(N/3)}} = v_\infty(X_{\text{ns}}^*(N/3)).$$

Either way, every cusp on $X_{\text{ns}}^*(N)$ is totally ramified in the map π , so by Theorem 2.23c), all cusps on $X_{\text{ns}}^*(N)$ are Weierstrass points as soon as there are at least five cusps. Using the above formulas for $v_\infty(X_{\text{ns}}^*(N))$ and a straightforward calculation that we omit, we find that the set of $N \equiv 3, 6 \pmod{9}$ such that $v_\infty(X_{\text{ns}}^*(N)) \leq 4$ is as given in the statement of Theorem 1.7c).

7.2. Revisiting Atkin's Work on $X_0(N)$. In this section we revisit Atkin's proof of Theorem 4.3. Most of it is a – conceptually, at least – direct application of Theorem 2.23b). Namely, if $N \in \mathbb{Z}^+$ is not squarefree, there is a prime p such that $p^2 \mid N$. The Fricke involution w_N interchanges 0 and ∞ , so ∞ is a Weierstrass point on $X_0(N)$ if and only if 0 is. The natural map

$$f : X_0(N) \rightarrow X_0\left(\frac{N}{p}\right)$$

has degree p , and since for any $n \in \mathbb{Z}^+$, the cusp width of 0 on $X_0(n)$ is n , it follows that f is totally ramified at 0. Using Theorem 2.23b), we get:

Proposition 7.1 (Atkin's Lemma). *Let $N \in \mathbb{Z}^+$, and p be a prime such that $p^2 \mid N$. If*

$$(10) \quad g(X_0(N)) \geq p(g(X_0\left(\frac{N}{p}\right)) + 1),$$

then the cusps 0 and ∞ are Weierstrass points of $X_0(N)$.

Atkin had the genus formula for $X_0(n)$, and he proved every case of Theorem 4.3 except one by checking that (10) holds for some prime p with $p^2 \mid N$. The exceptional case is $N = p^2$ for a prime $p \geq 11$, in which he remarks that the eta quotient $\frac{\eta(p^2\tau)}{\eta(\tau)}$ is a rational function on $X_0(p^2)$ regular away from the cusp 0 and with a pole of order $\frac{p^2-1}{24} > g(X_0(p^2))$ there.

We will analyze the cusps on $X_0(N)$ using Limonov's Theorem (Theorem 2.23c)). Limonov's Theorem is logically weaker than Atkin's Theorem – indeed, we deduced Limonov from Atkin – but, as we are about to see, not by much in this case. The additional information we collect to use Limonov's Theorem allows us to deduce stronger results: in many cases we can show that *every*

cuspidal on $X_0(N)$ is a Weierstrass point. At the end we switch back to Atkin's Theorem and see that this does yield further improvements.

For $n \in \mathbb{Z}^+$ we write $\text{rad}(n) := \prod_{p|n} p$, the product of the distinct prime divisors of n .

Lemma 7.2. *Let $M, N \in \mathbb{Z}^+$ be such that $M \mid N$ and $M \neq N$. We consider the natural map*

$$f : X_0(N) \rightarrow X_0(M).$$

- a) *If M is a Hall divisor of N , then the fiber of f over every cusp of $X_0(M)$ has $v_\infty(X_0(\frac{N}{M}))$ points. In particular, f is not totally ramified at any cusp.*
- b) *The following are equivalent:*
 - (i) *The cusp 0 on $X_0(N)$ is totally ramified under f .*
 - (ii) *Some cusp on $X_0(N)$ is totally ramified under f .*
 - (iii) *We have $\text{rad}(M) = \text{rad}(N)$.*

Proof. a) For $n \in \mathbb{Z}^+$, let $C_0(n)$ be the set of cusps on $X_0(n)$. For $n_1, n_2 \in \mathbb{Z}^+$ we have natural maps $\pi_i : C_0(n_1 n_2) \rightarrow C_0(n_i)$ and thus a natural map $\pi_1 \times \pi_2 : C_0(n_1 n_2) \rightarrow C_0(n_1) \times C_0(n_2)$. As one can see from the description of cusps on $X_0(n)$ given in [Og73, §1], $\pi_1 \times \pi_2$ is a bijection if n_1 and n_2 are coprime. Thus if M is a Hall divisor of N , every fiber of $C_0(N) = C_0(M \cdot \frac{N}{M}) \rightarrow C_0(M)$ has size $v_\infty(X_0(\frac{N}{M}))$, which is greater than 1 since $M < N$.

b) Clearly (i) $\implies$ (ii). For (ii) $\implies$ (iii): going by contrapositive, we suppose that $\text{rad}(M) < \text{rad}(N)$, so there is a prime p that divides N but does not divide M . The map f factors as $X_0(N) \rightarrow X_0(pM) \rightarrow X_0(M)$, so if some cusp is totally ramified under f , its image in $X_0(pM)$ is totally ramified under the map $X_0(pM) \rightarrow X_0(M)$, which has degree $p + 1 \geq 2$. Since M is a Hall divisor of pM , this contradicts part a). (iii) $\implies$ (i): Suppose that $\text{rad}(M) = \text{rad}(N)$. Since 0 has width N on $X_0(N)$ and width M on $X_0(M)$, the ramification index of f at 0 is $\frac{N}{M}$. Since $\text{rad}(M) = \text{rad}(N)$, we have

$$\deg(f) = \frac{\psi(N)}{\psi(M)} = \frac{N \prod_{p|N} (1 + \frac{1}{p})}{M \prod_{p|M} (1 + \frac{1}{p})} = \frac{N}{M},$$

so f is totally ramified at 0. □

If Limonov's Theorem applies to a map of Riemann surfaces $f : X \rightarrow Y$ – i.e., if there are least 5 totally ramified points – then it also applies to any factor $g : X \rightarrow Z$ of f . Combining this with Lemma 7.2, we find that in order to use Limonov's Theorem and maps $X_0(N) \rightarrow X_0(M)$ to produce Weierstrass points on $X_0(N)$, we may restrict to the case in which $p^2 \mid N$ and $M = \frac{N}{p}$.

We digress to introduce some terminology. In §4, for a congruence subgroup $\Gamma(N) \subseteq \Gamma \subseteq \Gamma_0(N)$ and a cusp P of $X(\Gamma)$ represented by a primitive vector (x, y) in $(\mathbb{Z}/N\mathbb{Z})^2$, we defined the invariant $d(P) = \gcd(y, N)$. When $\Gamma = \Gamma_0(N)$, Ogg showed [Og73, §1] that P has a unique representative with $y = d(P)$ and x lying in any given reduced residue system modulo $\gcd(d, \frac{N}{d})$, so for any positive integer $d \mid N$ there are $\varphi(\gcd(d, \frac{N}{d}))$ cusps P on $X_0(N)$ with $d(P) = d$. In view of this, in this case we call $d(P)$ the **denominator** of P . Thus for instance 0 is the unique cusp of denominator 1, represented by $(0, 1)$, and ∞ is the unique cusp of denominator N , represented by $(1, N)$.

Now we can give a complete analysis of the ramification of cusps in $X_0(N) \rightarrow X_0(\frac{N}{p})$:

Theorem 7.3. *Let $N \in \mathbb{Z}^+$, let p be a prime such that $p^2 \mid N$, and let $f : X_0(N) \rightarrow X_0(\frac{N}{p})$ be the natural map. Write $N = p^a M$ with $p \nmid M$. Let P be a cusp of $X_0(N)$, with denominator d , and write $d = p^b m$ with $p \nmid m$, so $0 \leq b \leq a$ and $m \mid M$. Then:*

- a) If $b < \frac{a}{2}$, then P is totally ramified under f .
- b) If $b \geq \frac{a}{2}$, then P is unramified under f .
- c) The number of cusps that are totally ramified under f is $p^{\lceil \frac{a}{2} \rceil - 1} v_\infty(X_0(M))$.

Proof. We prove parts a) and b) at the same time.

Step 0: Let $m \mid n$ be positive integers, consider the map $f : X_0(n) \rightarrow X_0(m)$, and let P be a cusp of $X_0(n)$ of denominator d . Then $f(P)$ has denominator $\gcd(d, m)$. Specializing to $m = p^{a-1}$ and $n = p^a$: if $d = p^b$ with $0 \leq b \leq a-2$, then every cusp on $X_0(p^a)$ that reduces to a cusp of $X_0(p^{a-1})$ of denominator p^b itself has denominator p^b , while a cusp on $X_0(p^a)$ reduces to a cusp of $X_0(p^{a-1})$ of denominator p^{a-1} – i.e., the cusp at ∞ – if and only if it has denominator p^{a-1} or p^a .

Step 1: Suppose that $M = 1$, so $f : X_0(p^a) \rightarrow X_0(p^{a-1})$. Then $d = p^b$ for some $0 \leq b \leq a$.

- Suppose that $b < \frac{a}{2}$, and let $\bar{P}$ be a cusp of $X_0(p^{a-1})$ of denominator p^b . By Step 0, every cusp of $X_0(p^a)$ that reduces to $\bar{P}$ also has denominator p^b , and every cusp of $X_0(p^a)$ of denominator p^b reduces to some cusp of denominator p^b . The number of cusps of denominator p^b on $X_0(p^a)$ is $\varphi(\gcd(p^b, p^{a-b})) = \varphi(p^b)$, while – since $a-1-b \geq b$ – the number of cusps of denominator p^b on $X_0(p^{a-1})$ is $\varphi(\gcd(p^b, p^{a-1-b})) = \varphi(p^b)$. Thus every cusp of denominator p^b on $X_0(p^a)$ is totally ramified under f .

- Suppose that $\frac{a}{2} \leq b \leq a-2$. Then again a cusp P of $X_0(p^a)$ has denominator p^b if and only if $f(P) \in X_0(p^{a-1})$ has denominator p^b , but now the number of cusps of denominator p^b on $X_0(p^a)$ is $\varphi(\gcd(p^b, p^{a-b})) = \varphi(p^{a-b})$, while the number of cusps of denominator p^b on $X_0(p^a)$ is $\varphi(\gcd(p^b, p^{a-b-1})) = \varphi(p^{a-b-1}) = \frac{\varphi(p^{a-b})}{p}$. Thus every cusp of denominator p^b on $X_0(p^a)$ is unramified under f .

- There is one cusp on $X_0(p^{a-1})$ of denominator p^{a-1} : the cusp at ∞ . Its preimage in $X_0(p^a)$ is the set of cusps of denominator either p^{a-1} or p^a . On $X_0(p^a)$ there are $\varphi(p) = p-1$ cusps of denominator p^{a-1} and one cusp of denominator p^a , so f is unramified over ∞ , or equivalently at all cusps of denominator p^{a-1} or p^a .

Step 2: Now suppose $N = p^a M$ with $p \nmid M$. As in the proof of Lemma 7.2 we have

$$C_0(p^a M) = C_0(p^a) \times C_0(M) \text{ and } C_0(p^{a-1} M) = C_0(p^{a-1}) \times C_0(M),$$

and if we denote the map $X_0(p^a) \rightarrow X_0(p^{a-1})$ from Step 1 by f_p , then upon restriction to cusps, the map $f : X_0(N) \rightarrow X_0(\frac{N}{p})$ takes the form $f = f_p \times 1_M$, where $1_M : C_0(M) \rightarrow C_0(M)$ is the identity map. This shows that a cusp P of $X_0(N)$ is totally ramified (resp. unramified) under f if and only if its image in $X_0(p^a)$ is totally ramified under (resp. unramified) under f_p . If a cusp P of $X_0(N)$ has denominator d , its image on $X_0(p^a)$ has denominator $p^{\text{ord}_p(d)}$, so P is totally ramified if and only if $\text{ord}_p(d) < \frac{a}{2}$ and otherwise unramified, completing the proof of parts a) and b).

c) Our analysis shows that the number of totally ramified cusps in $X_0(N) \rightarrow X_0(\frac{N}{p})$ is

$$\left(\sum_{b=0}^{\lceil \frac{a}{2} \rceil - 1} \varphi(p^b) \right) v_\infty(X_0(M)) = p^{\lceil \frac{a}{2} \rceil - 1} v_\infty(X_0(M)). \quad \square$$

To simplify the statement of the next result, let us record here that for $n \in \mathbb{Z}^+$ we have $v_\infty(X_0(n)) \geq 3$ if and only if $\omega(n) \geq 2$ or $(n = p^a \text{ is a prime power with } a \geq 2 \text{ and } p^a \neq 4)$. Moreover, we have $v_\infty(X_0(n)) \geq 5$ if and only if at least one of the following holds: $\omega(n) \geq 3$; $\omega(n) = 2$ and n is not squarefree; or $(n = p^a \text{ is a prime power with } a \geq 2 \text{ and } p^a \notin \{4, 8, 9\})$.

Corollary 7.4. *Maintain the notation of Theorem 7.3.*

- a) The number of cuspidal total ramification points of $f : X_0(N) \rightarrow X_0(\frac{N}{p})$ is at least five in precisely the following cases:

- (i) $a \geq 7$.
- (ii) $a \in \{5, 6\}$ and $p \geq 3$.
- (iii) $a \in \{5, 6\}$, $p = 2$ and $M \geq 2$.
- (iv) $a \in \{3, 4\}$ and $p \geq 5$.
- (v) $a \in \{3, 4\}$, $p = 3$ and $M \geq 2$.
- (vi) $a \in \{3, 4\}$, $p = 2$ and $v_\infty(X_0(M)) \geq 3$.
- (viii) $a = 2$ and $v_\infty(X_0(M)) \geq 5$.
- b) Suppose that f has at least five ramification points. Then every cusp on $X_0(N)$ with denominator d satisfying $\text{ord}_p d \neq \frac{a}{2}$ is a Weierstrass point of $X_0(N)$.
- c) If a is odd, then every cusp on $X_0(N)$ is a Weierstrass point.

Proof. a) This follows directly from Theorem 7.3c).

b) c) Suppose f has at least five ramification points. By Theorem 2.23c), every total ramification point of f is a Weierstrass point. Theorem 7.3a) shows that every cusp on $X_0(N)$ with denominator d satisfying $\text{ord}_p d < \frac{a}{2}$ is a total ramification point of f , hence a Weierstrass point. Applying the Atkin–Lehner involution w_{p^a} to these cusps, we get that all cusps with denominator d satisfying $\text{ord}_p d > \frac{a}{2}$ are also Weierstrass points. The only cusps we have *not* shown to be Weierstrass points have denominator d satisfying $\text{ord}_p d = \frac{a}{2}$, of which there are clearly none if a is odd. $\square$

Still with the notation of Theorem 7.3, all the cuspidal ramification points of f are total ramification points. In Remark 2.24a), we mentioned that parts b) and c) of Theorem 2.23 are equivalent when every ramification point of the map f is a total ramification point. Our analysis shows that this is *almost* the case here, but not quite: $\Gamma_0(\frac{N}{p})$ may have elliptic points, over which the map f may also be ramified (and thus not totally ramified when $p \geq 5$). In the genus formula for $\Gamma_0(n)$, the contribution from the elliptic points of $\Gamma_0(n)$ is small in magnitude but more intricate than the other terms, which leads to the more complicated hypotheses in parts (ii) and (iii) of Theorem 4.3b). We do not wish to revisit this somewhat case-intensive part of Atkin’s analysis here, so we will be content to state our final result in a less than fully explicit form:

Theorem 7.5. *Let $N \in \mathbb{Z}^+$, and let p be a prime such that $p^2 \mid N$ and (10) holds for N and p . Put $a := \text{ord}_p(N)$. Then:*

- a) *Every cusp of $X_0(N)$ with denominator d satisfying $\text{ord}_p(d) \neq \frac{a}{2}$ is a Weierstrass point.*
- b) *If a is odd, then every cusp on $X_0(N)$ is a Weierstrass point.*

Example 7.6. Let p be a prime, and let $a \in \mathbb{Z}^{\geq 2}$. By Theorem 7.3, the map $f : X_0(p^a) \rightarrow X_0(p^{a-1})$ is totally ramified at precisely $p^{\lceil \frac{a}{2} \rceil - 1}$ cusps and unramified at all other cusps.

- a) Suppose $a \geq 3$ and $p \geq 5$. Then there are at least 5 totally ramified cusps, and by Corollary 7.4, all cusps on $X_0(p^a)$ are Weierstrass points except possibly the **middle cusps** – those of denominator $p^{a/2}$ when a is even. By Theorem 2.25, each non-middle cusp has Weierstrass weight at least $\frac{(p-1)(p^{\lceil \frac{a}{2} \rceil - 1} - 3)^2}{8} \gg p^{a-1}$.
- b) Suppose $a \geq 3$ and $p = 3$. Then $X_0(3^a)$ has genus at least 2 if and only if $a \geq 4$, and if $a \geq 5$ the number of totally ramified cusps is at least 9, so when $a \geq 5$ all cusps are Weierstrass points except possibly the middle cusps when a is even, and each non-middle cusp has Weierstrass weight at least $\frac{(3-1)(3^{\lceil \frac{a}{2} \rceil - 1} - 3)^2}{8} \asymp 3^a$.

When $a = 4$, the curve $X_0(81)$ has genus 4 and 12 cusps: 1 of denominator 1, 2 of denominator 3, 6 of denominator 9, 2 of denominator 27 and 1 of denominator 81. Atkin did a modular forms calculation to show that the cusps $0, \infty$ are not Weierstrass points. A calculation using the methods of §6.3 confirms this, shows that the cusps of denominator 3 and 27 are also not Weierstrass points, but that the 6 cusps of denominator 9

are Weierstrass points, each of weight 4. That the middle cusps on $X_0(81)$ are Weierstrass points was surprising to us, but in retrospect we can give a theoretical explanation. Recall from §2.7 that a genus 4 Riemann surface with 6 Weierstrass points $P_1, \dots, P_6$ of weight 4 has a unique trigonal map that is moreover a Galois cover, for which the 6 total ramification points are $P_1, \dots, P_6$. That this applies to $X_0(81)$ was already shown by Hasegawa–Shimura [HS99, pp. 136–137], who found the affine model

$$y^3 = x^6 + 9x^3 + 27.$$

So Schoeneberg–Lewittes implies that the six middle cusps are Weierstrass points.

- c) Suppose $a \geq 3$ and $p = 2$. Then $X_0(2^a)$ has genus at least 2 if and only if $a \geq 6$, and if $a \geq 7$ the number of totally ramified cusps is at least 8, so when $a \geq 7$ all cusps are Weierstrass points except possibly the middle cusps when a is even. In this case, Theorem 2.25 implies that each non-middle cusp has Weierstrass weight at least $\frac{(2^{\lfloor \frac{a}{2} \rfloor - 1} - 2)(2^{\lfloor \frac{a}{2} \rfloor - 1} - 4)}{8} \asymp 2^a$.

When $a = 6$, the curve $X_0(64)$ has genus 3 and 12 cusps: 1 of denominator 1, 1 of denominator 2, 2 of denominator 4, 4 of denominator 8, 2 of denominator 16, 1 of denominator 32 and 1 of denominator 64. As we’ve seen in Example 6.7, *every* cusp has weight 2 and $X_0(64)$ is isomorphic to the Fermat quartic.

- b) Suppose $a = 2$. Then f is totally ramified at 0 and unramified at every other cusp, and moreover (10) never holds. The curve $X_0(p^2)$ has genus at least 2 if and only if $p \geq 11$. As mentioned above, Atkin in [At67] used an explicit modular function to show that $0, \infty$ are Weierstrass points of $X_0(p^2)$. There are $p - 1$ further cusps on $X_0(p^2)$, each with denominator p , and the methods of Schoeneberg–Lewittes–Atkin–Limonov do not seem to apply to determine whether they are Weierstrass points.

8. $X_1(M, N)$

8.1. An Approach to Theorem 4.1. For $N \geq 2$, in the $\mathrm{PSL}_2(\mathbb{Z}/N\mathbb{Z})$ -Galois covering $\pi : X(N) \rightarrow X(1)$, the ramification indices are $2, 3, N$, but moreover the generators of the first two groups H_1 of order 2 and H_2 of order 3 are elliptic elements (absolute value of trace less than 2), while every element of the group H_N of order N is a parabolic element (absolute value of trace equal to 2), so the three nontrivial isotropy groups H_1, H_2, H_3 are pairwise disjoint. Applying Theorem 2.29, we get:

Corollary 8.1. *Let $N \in \mathbb{Z}^{\geq 3}$. Let*

$$\gamma_a := \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \gamma_b := \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \gamma_c := \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix},$$

viewed as elements of $\mathrm{PSL}_2(\mathbb{Z}/N\mathbb{Z})$, and let

$$H_a := \langle \gamma_a \rangle, H_b := \langle \gamma_b \rangle, H_c := \langle \gamma_c \rangle.$$

Then $(\gamma_1, \gamma_b, \gamma_c)$ form a triple in $\mathrm{PSL}_2(\mathbb{Z}/N\mathbb{Z})$, and the map $X(N) \rightarrow X(1)$ is the associated Galois covering of $\mathbb{P}^1$, ramified over $j = 1728, 0, \infty$. The subgroups H_a, H_b, H_c , of orders 2, 3 and N , are the stabilizers of the the images of $\sqrt{-1}, e^{2\pi i/3}$ and ∞ in $X(N)$. Then:

- a) *The number of γ_a -fixed points on $X(N)$ is*

$$\#NH_a/H_a = \frac{N \prod_{p|N} \left(1 - \left(\frac{-4}{p}\right) \frac{1}{p}\right)}{2} \geq \frac{\varphi(N)}{2}.$$

- b) *The number of γ_b -fixed points on $X(N)$ is*

$$\#NH_b/H_b = \frac{N \prod_{p|N} \left(1 - \left(\frac{-3}{p}\right) \frac{1}{p}\right)}{3} \geq \frac{\varphi(N)}{3}.$$

c) The number of γ_c -fixed points on $X(N)$ is

$$\#NH_c/H_c = \frac{\varphi(N)}{2}.$$

Example 8.2. Take $N = 7$ in Corollary 8.1. Then $X(7)$ has genus 3, and there are precisely 3 ramified $\mathrm{PSL}_2(\mathbb{Z}/7\mathbb{Z})$ -orbits on $X(7)$, of sizes 84, 56 and 24; every other orbit has size 168. The cusps form the G -orbit of size 24. The curve $X(7)$ has at most $3^3 - 3 = 24$ Weierstrass points, which are a union of $\mathrm{PSL}_2(\mathbb{Z}/7\mathbb{Z})$ -orbits. Since the cusps are the unique orbit of size 24, we conclude once again that the Weierstrass points on $X(7)$ are precisely the cusps.

Example 8.3. Take $N = 8$ in Corollary 8.1. Then $X(8)$ has genus 5, and there are precisely 3 ramified $G = \mathrm{PSL}_2(\mathbb{Z}/8\mathbb{Z})$ -orbits on $X(8)$, of sizes 96, 64 and 24; every other orbit has size 192. The cusps form the orbit of size 24. Since $\deg \mathbb{W}(X(8)) = 5^3 - 5 = 120$, the Weierstrass points must lie in ramified orbits; letting the common Weierstrass weights in the orbits of sizes 96, 64 and 24 being $a, b, c \in \mathbb{Z}^{\geq 0}$ respectively, we have

$$120 = 96a + 64b + 24c.$$

The only two solutions are $(a, b, c) = (1, 0, 1)$ or $(a, b, c) = (0, 0, 5)$. We rule out the former using the degree two map

$$f : X(8) \rightarrow X_1(4, 8),$$

which genus formulas show is the quotient by a bielliptic involution ι_1 on $X(8)$. The involution ι_1 has $2g(X(8)) - 2 = 8$ fixed points; by Theorem 2.17 each fixed point has weight 3 or 5. Combined with our earlier analysis this shows that ι_1 is totally heavy: each of its eight cuspidal fixed points has Weierstrass weight 5. Because the 24 cusps form a full G -orbit, each cusp is fixed by a unique conjugate of ι_1 . It follows that ι_1 is conjugate to precisely two more bielliptic involutions ι_2 and ι_3 , so the cusps are the $8 + 8 + 8$ fixed points of the three bielliptic involutions, that each has weight 5, and thus comprise all the Weierstrass points on $X(8)$, as advertised in §2.7.

Remark 8.4. Examples 8.2 and 8.3 are instances of a slightly more general phenomenon: if X is a Riemann surface of genus at least 2 and automorphism group G with fewer than $\#G$ Weierstrass points, then every Weierstrass point must lie in a ramified G -orbit of $\pi : X \rightarrow X/G$. In order to apply this without knowing that X has a Weierstrass point of higher weight, we need $\#G > g(X)^3 - g(X)$, as is the case in the two examples above. The largest genus for which this holds is 7, for the **Macbeath curve** M with automorphism group $\mathrm{PSL}_2(\mathbb{F}_8)$. This is a Hurwitz curve: it comes from a triple in $\mathrm{PSL}_2(\mathbb{F}_8)$ with order triple $(2, 3, 7)$. It follows easily that the Weierstrass points on M form a single $\mathrm{PSL}_2(\mathbb{F}_8)$ -orbit of size 252 [LS12, Thm. 11]. From [MSSV02] we see that there are no such curves of genus 6, two such curves of genus 5 ($X(8)$ and the special Humbert curve H), three such curves of genus 4 ($X_1(5, 10)$ and two curves with 72 automorphisms¹²), and five such curves of genus 3 (the modular curves $X_0(48)$, $X_0(64)$, $X(7)$ and $X_1(3, 12)$, and the hyperelliptic curve $y^2 = x^8 - 1$, which has 32 automorphisms).

If $N \geq 31$, then $\frac{\varphi(N)}{2} \geq 5$, so every element of the fibers of π over 1728 and ∞ are Weierstrass points on $X(N)$. If $N \geq 43$, then $\frac{\varphi(N)}{3} \geq 5$, so every element of the fiber of π over 0 is a Weierstrass point on $X(N)$. For the remaining values of N , using the formulas of parts a) and b) of Corollary 8.1 gives parts a) and b) of Theorem 4.1.

¹²See https://www.lmfdb.org/HigherGenus/C/Aut/?genus=4&group_order=72.

8.2. The normalizer of $\Gamma_1(M, N)$ in $\mathrm{PSL}_2(\mathbb{R})$. For $N \in \mathbb{Z}^+$, the normalizer $\mathfrak{N}\Gamma_1(N)$ of $\Gamma_1(N)$ in $\mathrm{PSL}_2(\mathbb{R})$ was computed by Kim–Koo [KK00] and Lang [La01], independently. A more recent paper of Zemel [Ze17] gives a systematic computation of $\mathfrak{N}\Gamma$ for various classes of congruence subgroups Γ . For positive integers $M \mid N$, Zemel computes $\mathfrak{N}\Gamma_1(M, N)$ as a clever deduction from some other normalizer computations: one combines his Theorem 3.1, Proposition 3.4 and Remark 3.6. In fact Kim–Koo’s argument extends straightforwardly from $\Gamma_1(N)$ to $\Gamma_1(M, N)$, as we realized before we learned of Zemel’s work. This yields a more direct proof of the result we need, so we give it here.

Theorem 8.5. (Zemel [Ze17]) *Let $M, N \in \mathbb{Z}^+$ with $M \mid N$ and $(M, N) \notin \{(1, 4), (2, 4)\}$. Then the normalizer of $\Gamma_1(M, N)$ in $\mathrm{PSL}_2(\mathbb{R})$ is $\Gamma_0^*(\frac{N}{M})$, so*

$$\mathrm{Aut}_M X_1(M, N) \cong \Gamma_0^*\left(\frac{N}{M}\right) / \Gamma_1(M, N).$$

Proof. Let us first note that the case $M = N$ is much easier: then $\Gamma_1(M, N) = \Gamma(N)$, which is normal in $\mathrm{PSL}_2(\mathbb{Z})$, hence $\mathfrak{N}\Gamma(N)$ contains $\mathrm{PSL}_2(\mathbb{Z})$. On the other hand $\mathrm{PSL}_2(\mathbb{Z})$ is a maximal Fuchsian group, so the normalizer cannot be any larger than $\mathrm{PSL}_2(\mathbb{Z})$. If $(M, N) = (1, 2)$ then $\Gamma_1(2) = \Gamma_0(2)$ and the result is due to Lehner–Newman [LN64, Thm. 4]. Henceforth we assume that $N \nmid 4$. Once again: what follows is closely modelled on the proof of [KK00, Thm. 1].

Step 1: We show that $\mathfrak{N}\Gamma_1(M, N) \subseteq \Gamma_0^*(\frac{N}{M})$.

Let $m = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathrm{SL}_2(\mathbb{R})$ represent an element of $\mathfrak{N}\Gamma_1(M, N)$. Then

$$A := m \begin{bmatrix} 1 & M \\ 0 & 1 \end{bmatrix} m^{-1} = \begin{bmatrix} 1 - acM & a^2M \\ -c^2M & 1 + acM \end{bmatrix} \in \pm\tilde{\Gamma}_1(M, N)$$

and

$$B := m \begin{bmatrix} 1 & 0 \\ N & 1 \end{bmatrix} m^{-1} = \begin{bmatrix} 1 + bdN & -b^2N \\ d^2N & 1 - bdN \end{bmatrix} \in \pm\tilde{\Gamma}_1(M, N).$$

Both A and B have trace 2, hence trace 2 (mod N); if either A or B lies in $-\tilde{\Gamma}_1(M, N)$ then its trace would be -2 (mod N) and thus $N \mid 4$: contradiction. So $A, B \in \tilde{\Gamma}_1(M, N)$ and

$$(11) \quad ac, c^2 \in \frac{N}{M}\mathbb{Z} \text{ and } a^2, bd, b^2\frac{N}{M}, d^2 \in \mathbb{Z}.$$

Using (11) and $ad - bc = 1$, we find:

- $ab = (ad - bc)ab = a^2bd - \frac{N}{M}b^2\frac{ac}{N} \in \mathbb{Z}$.
- $a^3d = a^2(ad) = a^2(1 + bc) = a^2 + ab \cdot ac \in \mathbb{Z}$.
- $cd = (ad - bc)cd = ac \cdot d^2 - bd \cdot c^2 \in \frac{N}{M}\mathbb{Z}$.
- $ad^3 = (1 + bc)d^2 = d^2 + bd \cdot cd \in \mathbb{Z}$.

Since $ac \in \frac{N}{M}\mathbb{Z}$, we have

$$\gcd(a^3d, \frac{N}{M}) = \gcd(a^2 + ab \cdot ac, \frac{N}{M}) = \gcd(a^2, \frac{N}{M}).$$

Since $bd \in \mathbb{Z}$ and $cd \in \frac{N}{M}\mathbb{Z}$, we have

$$\gcd(ad^3, \frac{N}{M}) = \gcd(d^2 + bd \cdot cd, \frac{N}{M}) = \gcd(d^2, \frac{N}{M}).$$

Since $ad = \frac{a^3d}{a^2} \in \mathbb{Q}$ and $(ad)^2 \in \mathbb{Z}$ we have $ad \in \mathbb{Z}$. Since $ac \in \frac{N}{M}\mathbb{Z}$, we have

$$\gcd(a^2d^2, \frac{N}{M}) = \gcd(ad(1 + bc), \frac{N}{M}) = \gcd(ad + ac \cdot bd, \frac{N}{M}) = \gcd(ad, \frac{N}{M}).$$

Thus

$$\gcd(a^2, \frac{N}{M}) \mid \gcd(a^2 d^2, \frac{N}{M}) = \gcd(ad, \frac{N}{M}) \mid \gcd(a^3 d, \frac{N}{M}) = \gcd(d^2, \frac{N}{M})$$

and

$$\gcd(d^2, \frac{N}{M}) \mid \gcd(a^2 d^2, \frac{N}{M}) = \gcd(ad, \frac{N}{M}) \mid \gcd(d^2, \frac{N}{M}),$$

so we may put

$$Q := \gcd(a^2, \frac{N}{M}) = \gcd(ad, \frac{N}{M}) = \gcd(d^2, \frac{N}{M}).$$

Thus there are $u, v, w, x \in \mathbb{Z}$ with u, v positive and squarefree such that

$$a^2 = Qx^2u, \quad d^2 = Qw^2v$$

and

$$\gcd(x^2u, \frac{N}{QM}) = \gcd(w^2v, \frac{N}{QM}) = 1.$$

There is $l \in \mathbb{Z}$ such that $c^2 = \frac{N}{M}l$, and then

$$a^2c^2 = Qx^2u \frac{N}{M}l = Q^2 \frac{N}{QM}x^2ul.$$

Since $Q^2 \frac{N}{QM}x^2ul$ is a square, x^2u and $\frac{N}{QM}$ are coprime and u is squarefree, we must have $u \mid l$. Also, from

$$\left(\frac{N}{M}\right)^2 \mid (ac)^2 = Q^2 \frac{N}{QM}x^2ul$$

we get

$$\frac{N}{MQ} \mid x^2ul$$

and thus $\frac{N}{MQ} \mid l$. Putting $l = \frac{N}{MQ}ul'$ with $l' \in \mathbb{Z}$, we get

$$(ac)^2 = x^2u^2 \frac{N^2}{M} l'.$$

Thus $l' = z^2$ is a perfect square, so

$$(ac)^2 = (xu \frac{N}{M}z)^2$$

and thus

$$c^2 = \frac{N^2}{QM^2}z^2u.$$

Since $b^2 \in \frac{M}{N}\mathbb{Z}$, there is $t \in \mathbb{Z}$ such that $b^2 = \frac{Mt}{N}$, and thus

$$(bd)^2 = \frac{Mt}{N}Qw^2v = t \left(\frac{N}{MQ}\right)^{-1} w^2v,$$

which since $\frac{N}{MQ}$ and w^2v are coprime implies $\frac{N}{MQ} \mid t$. Thus for some $t' \in \mathbb{Z}$ we may write

$$(bd)^2 = t'w^2v.$$

Since $(bd)^2$ is a perfect square and v is squarefree, there is $y \in \mathbb{Z}$ such that

$$(bd)^2 = (y w v)^2,$$

and thus

$$b^2 = \frac{1}{q}y^2v.$$

In summary, we have:

$$a^2 = Qx^2u, \quad b^2 = \frac{1}{Q}y^2v, \quad c^2 = \frac{N^2}{M^2Q}z^2u, \quad d^2 = Qw^2v.$$

Then from

$$1 = ad - bc = \pm Qxw\sqrt{uv} \pm \frac{N}{MQ}yz\sqrt{uv}$$

we get $\sqrt{uv} \in \mathbb{Q}$. Since u and v are squarefree and positive, we conclude that $u = v$ and then $u \mid 1$, so $u = v = 1$. Finally, from

$$\pm Qxw \pm \frac{N}{MQ}yz = 1$$

we get that Q is a Hall divisor of $\frac{N}{M}$. Thus we have shown (after replacing some of x, y, z, w with their negatives if necessary) that

$$m = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{Q}x}{\frac{N}{M\sqrt{Q}}z} & \frac{\frac{1}{\sqrt{Q}}y}{\sqrt{Q}w} \end{bmatrix} \in \Gamma_0^*\left(\frac{N}{M}\right)$$

(cf. Remark 3.9).

Conversely, let $m \in \Gamma_0^*\left(\frac{N}{M}\right)$. Then we may represent m as $\begin{bmatrix} Qx & y \\ \frac{N}{M}z & Qw \end{bmatrix}$ with Q a Hall divisor of $\frac{N}{M}$ and $x, y, z, w \in \mathbb{Z}$ and $\det m = Q^2xw - \frac{N}{M}yz = Q$. Let

$$\gamma = \begin{bmatrix} Na + 1 & Mb \\ Nc & Nd + 1 \end{bmatrix} \in \tilde{\Gamma}_1(M, N).$$

Then we calculate that

$$m\gamma m^{-1} = \begin{bmatrix} 1 + N(Qawx - bxz + cwy - \frac{dNyz}{MQ}) & M(Qbx^2 - a\frac{N}{M}xy + d\frac{N}{M}xy - \frac{cNy^2}{MQ}) \\ N(Qcw^2 + \frac{N}{M}wz - d\frac{N}{M}wz - \frac{bNz^2}{MQ}) & 1 + N(Qdwx + bxz - cwy - \frac{aNyz}{MQ}) \end{bmatrix}$$

is, since $Q \mid \frac{N}{M}$, an element of $\tilde{\Gamma}_1(M, N)$. $\square$

Remark 8.6. Since the modular curves $X_1(4)$ and $X_1(2, 4)$ have genus 0, we have no need of the excluded cases $(M, N) \in \{(1, 4), (2, 4)\}$. But the computation of $\mathfrak{N}\Gamma_1(4)$ was done by Kim–Koo in [KK00] – it contains $\Gamma_0^*(4)$ as an index 3 subgroup – while Zemel’s work shows $\mathfrak{N}\Gamma_1(2, 4) = \Gamma_0^*(2)$.

8.3. $\text{Aut}_M X_1(M, N)$. The proof of the following result may safely be left to the reader:

Lemma 8.7. *Let $\tilde{\Gamma}_1$ and $\tilde{\Gamma}_2$ be subgroups of $\text{SL}_2(\mathbb{R})$, with $\tilde{\Gamma}_1$ a normal subgroup of $\tilde{\Gamma}_2$. We denote by Γ_1 and Γ_2 the images of $\tilde{\Gamma}_1$ and $\tilde{\Gamma}_2$ in $\text{PSL}_2(\mathbb{R})$.*

- a) *We have $\Gamma_2/\Gamma_1 \cong \tilde{\Gamma}_2/(\tilde{\Gamma}_2 \cap \{\pm I_2\})\tilde{\Gamma}_1$.*
- b) *If either $\tilde{\Gamma}_1$ contains $-I_2$ or $\tilde{\Gamma}_2$ does not contain $-I_2$, then $\Gamma_2/\Gamma_1 \cong \tilde{\Gamma}_2/\tilde{\Gamma}_1$.*
- c) *If $-I_2 \in \tilde{\Gamma}_2 \setminus \tilde{\Gamma}_1$, then $\Gamma_2/\Gamma_1 \cong \tilde{\Gamma}_2/\tilde{\Gamma}_1/(\pm \tilde{\Gamma}_1)$.*

Suppose that $N \neq 4$. By Theorem 8.5 we have a short exact sequence of groups

$$(12) \quad 1 \rightarrow \Gamma_1\left(\frac{N}{M}\right)/\Gamma_1(M, N) \rightarrow \text{Aut}_M X_1(M, N) \rightarrow \Gamma_0^*\left(\frac{N}{M}\right)/\Gamma_1\left(\frac{N}{M}\right) \rightarrow 1.$$

We deduce:

Corollary 8.8. *Let $M \mid N$ be positive integers with $N \geq 5$.*

- a) *We have*

$$\# \text{Aut}_M X_1(M, N) = 2^{\omega(\frac{N}{M})-1} \frac{M\varphi(N)\psi(N)}{\psi(\frac{N}{M})}.$$

b) If $\gcd(M, \frac{N}{M}) = 1$, then

$$\# \operatorname{Aut}_M X_1(M, N) = 2^{\omega(\frac{N}{M})-1} M \psi(M) \varphi(N).$$

When $\gcd(M, \frac{N}{M}) = 1$, we can say more on the structure of $\operatorname{Aut}_M X_1(M, N)$: in this case we have

$$\tilde{\Gamma}_1(M, N) = \tilde{\Gamma}(M) \cap \tilde{\Gamma}_1\left(\frac{N}{M}\right),$$

so the natural map $\tilde{\Gamma}_1(\frac{N}{M}) \rightarrow \operatorname{SL}_2(\mathbb{Z}/M\mathbb{Z})$ is surjective with kernel $\tilde{\Gamma}_1(M, N)$.

Corollary 8.9. *Let $M \mid N$ be positive integers such that $N \nmid 4$ and $\gcd(M, \frac{N}{M}) = 1$.*

a) *If $\frac{N}{M} \leq 2$, we have a short exact sequence*

$$1 \rightarrow \operatorname{PSL}_2(\mathbb{Z}/M\mathbb{Z}) \rightarrow \operatorname{Aut}_M X_1(M, N) \rightarrow \Gamma_0^*\left(\frac{N}{M}\right) / \Gamma_1\left(\frac{N}{M}\right) \rightarrow 1.$$

b) *If $\frac{N}{M} \geq 3$, we have a short exact sequence*

$$1 \rightarrow \operatorname{SL}_2(\mathbb{Z}/M\mathbb{Z}) \rightarrow \operatorname{Aut}_M X_1(M, N) \rightarrow \Gamma_0^*\left(\frac{N}{M}\right) / \Gamma_1\left(\frac{N}{M}\right) \rightarrow 1.$$

Proof. Since $N \geq 3$, we have $-1 \notin \tilde{\Gamma}_1(M, N)$, while $-1 \in \tilde{\Gamma}_1(\frac{N}{M})$ if and only if $\frac{N}{M} \leq 2$. The result now follows from Theorem 8.5 and Lemma 8.7. $\square$

Example 8.10. The modular curve $X_1(5, 10)$ has genus 4. By Corollary 8.9, its modular automorphism group is an extension of $\mathbb{Z}/2\mathbb{Z}$ by $\operatorname{PSL}_2(\mathbb{Z}/5\mathbb{Z}) \cong A_5$ and thus has order 120. As mentioned in Example 2.31, any genus 4 Riemann surface with at least 120 automorphisms is isomorphic to Bring's curve B , and it follows that

$$G := \operatorname{Aut}_M X_1(5, 10) = \operatorname{Aut} X_1(5, 10) \cong \operatorname{Aut} B \cong S_5.$$

Thus the 24 cusps of $X_1(5, 10)$ are stabilized by the G -action. Since by Example 2.31 the smallest G -orbit on $X_1(5, 10)$ has size 24, the cusps form a single G -orbit (this is a special case of part c) of our next result). We saw in Example 2.31 that the Weierstrass points on Bring's curve form a G -orbit of size 60, so we conclude: no cusp of $X_1(5, 10)$ is a Weierstrass point.

8.4. Action of $\operatorname{Aut}_M X_1(M, N)$ on Cusps.

Theorem 8.11. *Let $M \mid N$ be positive integers with $N \neq 4$. Let*

$$\pi : X_1(M, N) \rightarrow X_0\left(\frac{N}{M}\right)$$

be the natural map, and let

$$W\left(\frac{N}{M}\right) := \Gamma_0^*\left(\frac{N}{M}\right) / \Gamma_0\left(\frac{N}{M}\right)$$

be the Atkin-Lehner group. Then:

- a) *The map π induces a bijection from the set of $\operatorname{Aut}_M X_1(M, N)$ -orbits on cusps of $X_1(M, N)$ to the set of $W(\frac{N}{M})$ -orbits on cusps of $X_0(\frac{N}{M})$.*
- b) *The $\operatorname{Aut}_M X_1(M, N)$ -orbit of the cusp at ∞ is the set of **principal cusps**: the cusps P such that $\pi(P)$ is a Hall cusp on $X_0(\frac{N}{M})$. In particular 0 and ∞ lie in the same $\operatorname{Aut}_M X_1(M, N)$ -orbit.*
- c) *$\operatorname{Aut}_M X_1(M, N)$ acts transitively on the cusps of $X_1(M, N)$ if and only if $\frac{N}{M}$ is squarefree.*

Proof. a) It follows from Theorem 8.5 that $\Gamma_1(M, N)$ is a normal subgroup of $\Gamma_0^*(\frac{N}{M})$, hence it is also a normal subgroup of $\Gamma_0(\frac{N}{M})$. Thus the map π is a Galois covering, and its Galois group $H := \Gamma_0(\frac{N}{M})/\Gamma_1(M, N)$ is a subgroup of $G := \text{Aut}_M X_1(M, N)$; indeed H is the kernel of the natural surjective group homomorphism

$$\varphi : G = \Gamma_0^*\left(\frac{N}{M}\right)/\Gamma_1(M, N) \rightarrow W\left(\frac{N}{M}\right) \subseteq \text{Aut}_M X_0\left(\frac{N}{M}\right).$$

Let P_1 and P_2 be cusps of $\Gamma_1(M, N)$ lying in the same $\text{Aut}_M \Gamma_1(M, N)$ -orbit: i.e., there is $g \in G$ such that $gP_1 = P_2$. Then

$$\pi(P_2) = \pi(gP_1) = \varphi(g)\pi(P_1),$$

so $\pi(P_1)$ and $\pi(P_2)$ lie in the same $W(\frac{N}{M})$ -orbit. Conversely, let P_1 and P_2 be cusps of $X_1(M, N)$ such that $\pi(P_1) = \bar{g}\pi(P_2)$ for some $\bar{g} \in W(\frac{N}{M})$. We may choose $g \in G$ such that $\pi(g) = \bar{g}$, and then P_1 and gP_2 lie in the same fiber of π . Since π is an H -Galois covering, there is $h \in H$ that carries gP_2 to P_1 , and thus overall $hgP_2 = P_1$, so P_1 and P_2 lie in the same G -orbit.

b) By part a), the $\text{Aut}_M X_1(M, N)$ -orbit of ∞ is the set of all cusps P such that $\pi(P)$ lies in the same $W(\frac{N}{M})$ -orbit as $\pi(\infty) = \infty$. The Atkin-Lehner group $W(\frac{N}{M})$ acts simply transitively on the set of Hall cusps of $X_0(\frac{N}{M})$, which includes the cusps at 0 and ∞ .

c) Every cusp on $X_0(\frac{N}{M})$ is Hall if and only if $\frac{N}{M}$ is squarefree, so this follows from part b). $\square$

Example 8.12. The curve $X_1(2, 14)$ has genus 4 and 18 cusps. Let $G := \text{Aut}_M X_1(2, 14)$, a group of order 36. Since $\frac{14}{2} = 7$ is squarefree, by Theorem 8.11c) the cusps form a single G -orbit on $X_1(2, 14)$,

and the isotropy group of the cusp at ∞ is generated by $\sigma := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, of order 2. We have $X_1(2, 14)/\langle \sigma \rangle = X_1(14)$, of genus 1, so σ is a bielliptic involution on $X_1(2, 14)$ and thus has 6 fixed points, which are all Weierstrass points. The element σ lies in the subgroup $\text{Aut}(X_1(2, 14)/X_1(7))$ of G , which is normal and isomorphic to $\text{SL}_2(\mathbb{Z}/2\mathbb{Z}) \cong S_3$, so σ has precisely 3 conjugates in G (and thus $X_1(2, 14)$ is triply bielliptic). On the other hand, since G acts transitively on the cusps of $X_1(2, 14)$, every cusp is fixed by some conjugate of σ . It follows that the 6+6+6 fixed points of the three conjugates of σ are precisely the 18 cusps on $X_1(2, 14)$. Each cusp has the same Weierstrass weight, and since $18 \nmid 60 = 4^3 - 4$, these are not all the Weierstrass points on $X_1(2, 14)$. Indeed Example 6.15c) shows that the common Weierstrass weight of the cusps is 1.

Example 8.13. We revisit the curve $X_1(3, 12)$ from Example 6.10. It is nonhyperelliptic of genus 3, whose Weierstrass points are precisely the 20 cusps. From (12) we have $\# \text{Aut}_M X_1(3, 12) = 48$. The only genus 3 curves with more than 48 automorphisms are $X_0(64)$ with 96 automorphisms and $X(7)$ with 168 automorphisms. Since $48 \nmid 168$, $X_1(3, 12)$ is not isomorphic to $X(7)$. Since $X_0(64)$ has 12 Weierstrass points and $X_1(3, 12)$ has 20 Weierstrass points, $X_1(3, 12)$ is not isomorphic to $X_0(64)$. Thus $\# \text{Aut} X_1(3, 12) = 48$, so $\text{Aut}_M X_1(3, 12) = \text{Aut} X_1(3, 12)$.¹³

From [MSSV02, Tables 1 and 2] we find that $X_1(3, 12)$ is up to isomorphism the unique nonhyperelliptic curve of genus 3 with 48 automorphisms – its automorphism group G has group ID (48, 33) – and $\pi : X_1(3, 12) \rightarrow X_1(3, 12)/G$ is a Galois Belyĭ map with order triple (2, 3, 12). Thus $X_1(3, 12)$ has precisely 3 ramified G -orbits, of sizes 24, 16 and 4. By Theorem 8.11b), there are 2 cuspidal G -orbits; since there are 20 cusps on $X_1(3, 12)$, they must be the G -orbits of sizes 16 and 4. Analysis of the map π alone leaves two possibilities for the Weierstrass weights: they could consist of the unique G -orbit of size 24 and all have weight 1, or they could consist of the unique G -orbit of size 16, each with weight 1, together with the unique G -orbit of size 4, each with weight 2. Example 6.10 shows that the latter is the case.

¹³Alternately, because we have defining equations for $X_1(3, 12)$, MAGMA can compute its automorphism group, which indeed has size 48.

8.5. Cuspidal Weierstrass Points I: Principal Cusps.

Theorem 8.14. *Let $M, N \in \mathbb{Z}^+$ with $2 \leq M \mid N$ be such that $g(X_1(M, N)) \geq 2$.*

- a) *No cusp on $X_1(5, 10)$ is a Weierstrass point.*
- b) *If $(M, N) \neq (5, 10)$, then the cusp at ∞ is a Weierstrass point on $X_1(M, N)$. If moreover $\frac{N}{M}$ is squarefree, then all cusps on $X_1(M, N)$ have the same Weierstrass weight, and in particular all cusps are Weierstrass points.*

Proof. The group $\Gamma_1(M, N) = \Gamma(M) \cap \Gamma_1(N)$ is a normal subgroup of $\Gamma_1(N)$, and the quotient $\Gamma_1(N)/\Gamma_1(M, N)$ is cyclic of order M , generated by the class of $\sigma := \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Thus σ defines an order M modular automorphism of $X_1(M, N)$ and $X_1(M, N)/\langle \sigma \rangle = X_1(N)$. Moreover σ fixes the cusp at ∞ . Thus Theorem 2.23 gives that ∞ is a Weierstrass point on $X_1(M, N)$ whenever $g(X_1(M, N)) \geq M(g(X_1(N)) + 1)$. Since $g(X_1(M, N)) = 0$ if $N \leq 4$, we must have $N \geq 5$, and then the desired inequality is

$$\begin{aligned} 1 + \frac{MN^2}{24} \prod_{p \mid N} \left(1 - \frac{1}{p^2}\right) - \frac{1}{4} \sum_{d \mid N} \varphi(d) \varphi\left(\frac{N}{d}\right) \gcd(M, \frac{N}{d}) \\ \geq 2M - \frac{M}{4} \sum_{d \mid N} \varphi(d) \varphi\left(\frac{N}{d}\right) + \frac{MN^2}{24} \prod_{p \mid N} \left(1 - \frac{1}{p^2}\right), \end{aligned}$$

which is equivalent to

$$(13) \quad 8 - \frac{4}{M} \leq \sum_{d \mid N} \varphi(d) \varphi\left(\frac{N}{d}\right) \left(1 - \frac{\gcd(M, \frac{N}{d})}{M}\right).$$

In the right hand side of (13), all terms are non-negative and the $d = N$ term is $\varphi(N)(1 - \frac{1}{M})$. It is easy to see, e.g. using the bound $\varphi(N) \geq \sqrt{\frac{N}{2}}$, that for all $N \geq 43$ we have $\varphi(N) \geq 16$. Thus if $N \geq 43$, then since $M \geq 2$ we have $\varphi(N)(1 - \frac{1}{M}) \geq 8 \geq 8 - \frac{4}{M}$, completing the proof in this case.

For the pairs $2 \leq M \mid N \leq 42$ with $g(X_1(M, N)) \geq 2$, we check directly that (13) holds in all but precisely four cases:

$$(M, N) \in (7, 7), (8, 8), (5, 10), (2, 16).$$

The cases $(7, 7)$, $(8, 8)$, $(5, 10)$ and $(2, 16)$ are treated, respectively, in Example 6.8 (and again in Example 8.2), Example 8.3, Example 8.10 and Example 6.15. Finally, if $(M, N) \neq (5, 10)$ and $\frac{N}{M}$ is squarefree, then it follows from Theorem 8.11c) that every cusp on $X_1(M, N)$ has the same Weierstrass weight as does the cusp at ∞ , hence all cusps are Weierstrass points. $\square$

Remark 8.15.

- a) Example 6.15 shows that when $2 \leq M \mid N$ and $\frac{N}{M}$ is *not* squarefree, it may or may not be the case that all cusps on $X_1(M, N)$ have the same Weierstrass weight.
- b) On $X_1(N)$ it may be that some but not all cusps are Weierstrass points: indeed recall Jeon's Theorem (Theorem 4.9): if $N \geq 18$ and $X_1(N)$ has genus at least 2, then every non-Hall (= non-principal) cusp on $X_1(N)$ is a Weierstrass point, whereas in Examples 6.3 and 6.9 we saw that no Hall cusp on either $X_1(16)$ or $X_1(20)$ is a Weierstrass point. However, our computations support the conjecture that when $M \geq 2$ and $g(X_1(M, N)) \geq 2$, every non-principal cusp is a Weierstrass point. Of course this holds vacuously when $\frac{N}{M}$ is squarefree. In the next section we prove a non-vacuous special case of this conjecture.

8.6. Cuspidal Weierstrass Points II: Non-Principal Cusps. For $n \in \mathbb{Z}^+$, we put

$$X_0^*(n) := \Gamma_0^*(n) \backslash \overline{\mathcal{H}}.$$

In other words, $X_0^*(n)$ is the quotient of the modular curve $X_0(n)$ by the Atkin–Lehner group $W(n) \subseteq (\text{Aut } X_0(n))(\mathbb{Q})$, which gives it a canonical $\mathbb{Q}$ -model.¹⁴ For $n \in \mathbb{Z}^+$, let

$$\bar{v}(n) := v_\infty(X_0^*(n))$$

be the number of cusps on $X_0^*(n)$, which is also the number of orbits under the Atkin–Lehner group $W(n)$ of cusps on $X_0(n)$. We collect some simple facts about this function:

Lemma 8.16.

- a) *The function $\bar{v}$ is multiplicative: if $n_1, n_2 \in \mathbb{Z}^+$ are coprime, then $\bar{v}(n_1 n_2) = \bar{v}(n_1) \bar{v}(n_2)$.*
- b) *Let p be a prime, and let $A \in \mathbb{Z}^{\geq 2}$. We have:*

$$\bar{v}(p) = 1, \quad \bar{v}(p^2) = \left\lfloor \frac{p+1}{2} \right\rfloor, \quad \bar{v}(p^{2A-1}) = p^{A-1}, \quad \bar{v}(p^{2A}) = \frac{p^A + p^{A-1}}{2}.$$

- c) *If $n_1 \mid n_2$, then $\bar{v}(n_1) \leq \bar{v}(n_2)$.*
- d) *We have $\bar{v}(n) = 1$ if and only if n is squarefree.*
- e) *We have $\bar{v}(n) = 2$ if and only if n has one of the following forms:*
 - (i) $n = 4b$ for b odd and squarefree. The cusp $\frac{1}{2}$ represents the non-principal G -orbit.
 - (ii) $n = 8b$ for b odd and squarefree. The cusp $\frac{1}{2}$ represents the non-principal G -orbit.
 - (iii) $n = 9b$ for b prime to 3 and squarefree. The cusp $\frac{1}{3}$ represents the non-principal G -orbit.
- f) *We have $\bar{v}(n) = 3$ if and only if n has one of the following forms:*
 - (i) $n = 16b$ for b odd and squarefree. The cusps $\frac{1}{2}$ and $\frac{1}{4}$ represent the two non-principal G -orbits.
 - (ii) $n = 25b$ for b prime to 5 and squarefree. The cusps $\frac{1}{5}$ and $\frac{2}{5}$ represent the two non-principal G -orbits.
 - (iii) $n = 27b$ for b prime to 3 and squarefree. The cusps $\frac{1}{3}$ and $\frac{2}{3}$ represent the two non-principal G -orbits.

Proof. a) If $n_1, n_2 \in \mathbb{Z}^+$ are coprime, then we have a natural isomorphism $W(n_1 n_2) = W(n_1) \times W(n_2)$ and a natural bijection from the set of cusps on $X_0(n_1 n_2)$ to the Cartesian product of the set of cusps on $X_0(n_1)$ with the set of cusps on $X_0(n_2)$. The result follows.

b) The Atkin–Lehner group $W(p^a)$ has order 2, with nontrivial element w_{p^a} . The element w_{p^a} has a cuspidal fixed point if and only if $p^a = 4$: $v_\infty(X_0(4)) = \{0, \infty, \frac{1}{2}\}$ and w_4 interchanges 0 and ∞ and fixes $\frac{1}{2}$. This proves (i), and (ii) follows using the formula for $v_\infty(X_0(N))$.

c) This is immediate from parts a) and b).

d), e), f) Let $n = p_1^{a_1} \cdots p_r^{a_r}$ for primes $p_1 < \cdots < p_r$. Then $\bar{v}(n) = \prod_{i=1}^r \bar{v}(p_i^{a_i})$. Thus $\bar{v}(n) = 1$ if and only if $\bar{v}(p_i^{a_i}) = 1$ for all i ; a $\bar{v}(n) = 2$ if and only if for exactly one i we have $\bar{v}(p_i^{a_i}) = 2$ and $\bar{v}(p_j^{a_j}) = 1$ for all $j \neq i$; and $\bar{v}(n) = 3$ if and only if for exactly one i we have $\bar{v}(p_i^{a_i}) = 3$ and $\bar{v}(p_j^{a_j}) = 1$ for all $j \neq i$. By the previous parts, we have $\bar{v}(p^a) = 1$ if and only if $a = 1$; $\bar{v}(p^a) = 2$ if and only if $p^a \in \{4, 8, 9\}$; and $\bar{v}(p^a) = 3$ if and only if $p^a \in \{16, 25, 27\}$. The results on the number of Atkin–Lehner orbits follow. For $p^a \in \{4, 8, 9, 16, 25, 27\}$, a direct calculation that we omit yields the action of w_{p^a} on the cusps of $X_0(p^a)$, and the case of $n = p^a b$ follows easily. $\square$

¹⁴For $n \geq 2$, the group $\Gamma_0^*(n)$ is not a subgroup of $\text{PSL}_2(\mathbb{Z})$, so the Riemann surface $X_0^*(n)$ is not a modular curve according to the definition we have taken. That we can leave the class of modular curves by taking the quotient of a modular curve by a group of modular automorphisms suggests that our definition may be a bit restrictive.

Let $p \in \mathbb{Z}^{\geq 2}$, and let $M, N \in \mathbb{Z}^+$ be such that $p^2 M \mid N$. Put

$$G := \text{Aut}_M X_1(M, N).$$

We consider the matrix

$$m(\gamma_p) := \begin{bmatrix} 1 + \frac{N}{p} & -\frac{N}{p^2} \\ N & 1 - \frac{N}{p} \end{bmatrix}.$$

For all $1 \leq i < p$ the matrix $m(\gamma_p)^i$ lies in $\tilde{\Gamma}_1(M, \frac{N}{p}) \cap \tilde{\Gamma}_0(N)$ – hence normalizes $\tilde{\Gamma}_1(M, N)$ – and not in $\pm \tilde{\Gamma}_1(M, N)$, while $m(\gamma_p)^p = \begin{bmatrix} 1 + N & -\frac{N}{p} \\ pN & 1 - N \end{bmatrix}$ lies in $\tilde{\Gamma}_1(M, N)$, so $m(\gamma_p)\Gamma_1(M, N)$ is an order p element of G that we denote by γ_p . We put

$$\tilde{\Gamma}_1^\bullet(M, N) := \langle \tilde{\Gamma}_1(M, N), m(\gamma_p) \rangle.$$

Then we have

$$\tilde{\Gamma}_1(M, N) \subseteq \tilde{\Gamma}_1^\bullet(M, N) \subseteq \widetilde{\Gamma(M)} \cap \tilde{\Gamma}_1\left(\frac{N}{p}\right) \cap \tilde{\Gamma}_0(N) \subseteq \tilde{\Gamma}_1(M, \frac{N}{p})$$

and

$$[\tilde{\Gamma}_1(M, \frac{N}{p}) : \tilde{\Gamma}_1(M, N)] = p^2,$$

$$[\tilde{\Gamma}_1(M, \frac{N}{p}) : \widetilde{\Gamma(M)} \cap \tilde{\Gamma}_1\left(\frac{N}{p}\right) \cap \tilde{\Gamma}_0(N)] \leq [\tilde{\Gamma}_1(M, \frac{N}{p}) : \tilde{\Gamma}_1^\bullet(M, N)] = p.$$

The matrix $\begin{bmatrix} 1 & 0 \\ \frac{N}{p} & 1 \end{bmatrix}$ shows that $[\tilde{\Gamma}(M, \frac{N}{p}) : \widetilde{\Gamma(M)} \cap \tilde{\Gamma}_1\left(\frac{N}{p}\right) \cap \tilde{\Gamma}_0(N)] \geq p$, so it follows that

$$\tilde{\Gamma}_1^\bullet(M, N) = \widetilde{\Gamma(M)} \cap \tilde{\Gamma}_1\left(\frac{N}{p}\right) \cap \tilde{\Gamma}_0(N).$$

We put

$$X_1^\bullet(M, N) := X(\Gamma_1^\bullet(M, N)) = X_1(M, N) / \langle \gamma_p \rangle.$$

Let us now assume that $N \geq 8$ if $p = 2$ and $N \geq 18$ if $p = 3$: indeed, in so doing we are only excluding cases in which $g(X_1(M, N)) \leq 1$. Then we have $\frac{N}{p} \geq 4$. Since $\Gamma_1^\bullet(M, N) \subseteq \Gamma_1(\frac{N}{p})$, the group $\Gamma_1^\bullet(M, N)$ has no elliptic points, and thus every γ_p -fixed point on $X_1(M, N)$ is a cusp. The unique fixed point of $m(\gamma_p)$ on $\overline{\mathcal{H}}$ is $\frac{1}{p}$, so $\frac{1}{p}$ is a fixed point of γ_p . On $X_0(\frac{N}{M})$, the cusp ∞ is a Hall cusp and the cusp $\frac{1}{p}$ is not, so ∞ and $\frac{1}{p}$ lie in different G -orbits on $X_1(M, N)$.

Lemma 8.17. *Let $p \in \mathbb{Z}^{\geq 2}$, and $M, N \in \mathbb{Z}^+$ with $p^2 M \mid N$. If $p = 2$, we assume that $N \geq 8$, and if $p = 3$ we assume that $N \geq 18$. Let $G := \text{Aut}_M X_1(M, N)$. Then:*

- a) $\Gamma_1^\bullet(M, N)$ is a normal subgroup of $\Gamma_0^*(\frac{N}{M})$.
- b) The element γ_p generates a normal subgroup of G of order p .
- c) The fixed point locus of γ_p is a union of cuspidal G -orbits, excluding the principal G -orbit and including the G -orbit on $\frac{1}{p}$.

Proof. a) We will show that $\tilde{\Gamma}_1^\bullet(M, N)$ is normal in $\tilde{\Gamma}_0^*(\frac{N}{M})$ (which contains -1 ; the tilde is just to indicate that we are viewing it as a subgroup of $\text{SL}_2(\mathbb{R})$); this suffices. At the end of the proof of Theorem 8.5, we showed by direct calculation that $\tilde{\Gamma}_1(M, N)$ is normal in $\tilde{\Gamma}_0^*(\frac{N}{M})$. In general, a subgroup N of a group $\mathcal{G}$ is normal if it is generated by a set of elements X such that for all

$x \in X$ and all $g \in \mathcal{G}$ we have $gxg^{-1} \in N$. Since $\tilde{\Gamma}_1^\bullet(M, N)$ is generated by $\tilde{\Gamma}_1(M, N)$ and $m(\gamma_2)$, it suffices to show that $gm(\gamma_2)g^{-1} \in \tilde{\Gamma}_1^\bullet(M, N)$ for all $g \in \tilde{\Gamma}_0^*(\frac{N}{M})$. Let $g \in \tilde{\Gamma}_0^*(\frac{N}{M})$, so

$$g = \begin{bmatrix} Qx & y \\ \frac{N}{M}z & Qw \end{bmatrix}$$

with $x, y, z, w, Q \in \mathbb{Z}$, $MQ \mid N$ and $\det g = Q^2xw - \frac{N}{M}yz = Q$; thus also $Qxw - \frac{N}{MQ}yz = 1$. Using these last two equations to simplify the entries of $gm(\gamma_2)g^{-1}$, we find that $gm(\gamma_2)g^{-1}$ equals

$$\begin{bmatrix} 1 + \frac{N}{p}(Qwx + pwy + \frac{N}{pM}xz + \frac{N}{MQ}yz) & M(\frac{-N}{p^2M}Qx^2 - 2\frac{N}{pM}xy - \frac{N}{MQ}y^2) \\ N(Qw^2 + 2\frac{N}{pM}wz + \frac{N}{p^2M}\frac{N}{MQ}z^2) & 1 + \frac{N}{p}(-Qwx - pwy - \frac{N}{pM}xz - \frac{N}{MQ}yz) \end{bmatrix} \in \tilde{\Gamma}_1^\bullet(M, N).$$

b) Since $G = \text{Aut}_M X_1(M, N) = \Gamma_0^*(\frac{N}{M})/\Gamma_1(M, N)$ and γ_p has order p in G , this is an equivalent restatement of part a).

c) Applying Lemma 2.15 to γ_p , we get that the fixed point locus of γ_p is a union of G -orbits. We already know that all fixed points of γ_p are cusps and that $\frac{1}{p}$ is a fixed point of γ_p . Since ∞ has width M on both $X_1(M, N)$ and $X_1^\bullet(M, N)$, it is not a fixed point of γ_p . $\square$

In the next two results we specialize to taking $p = 2$ and $p \geq 3$ a prime number.

Theorem 8.18. *Let $M, N \in \mathbb{Z}^+$ with $4M \mid N$ and $N \geq 8$.*

a) *Let $G := \text{Aut}_M X_1(M, N)$, and put*

$$\epsilon(M, N) := \begin{cases} 0 & \text{if } \frac{N}{4M} \text{ is even} \\ 1 & \text{if } \frac{N}{4M} \text{ is odd} \end{cases}.$$

Then the G -orbit on the nonprincipal cusp $\frac{1}{2}$ has size

$$(14) \quad \frac{2^{\omega(\frac{N}{M})-2-\epsilon(M, N)} \varphi(N) \psi(N)}{\psi(\frac{N}{M})}.$$

b) *Suppose that $(M, N) \notin \{(1, 8), (2, 8), (4, 8), (1, 12)\}$, so $g(X_1(M, N)) \geq 2$. Then every point in the G -orbit of $\frac{1}{2}$ is a Weierstrass point of $X_1(M, N)$.*

Proof. a) If $G_{\frac{1}{2}}$ is the isotropy subgroup at $\frac{1}{2}$, then the size of the G -orbit on $\frac{1}{2}$ is $[G : G_{\frac{1}{2}}]$. We know the size of $\#G$ by Corollary 8.8, and $\#G_{\frac{1}{2}} = e_1 e_2$, where e_1 is the ramification index of $\frac{1}{2}$ in the map $X_1(M, N) \rightarrow X_0(\frac{N}{M})$ and e_2 is the ramification index of $\frac{1}{2}$ in the map $X_0(\frac{N}{M}) \rightarrow X_0^*(\frac{N}{M})$. By Example 6.14b), for a finite index subgroup Δ of $\text{PSL}_2(\mathbb{Z})$, the width of the cusp $\frac{1}{2}$ on $X(\Delta)$ is the least $h \in \mathbb{Z}^+$ such that $\pm \begin{bmatrix} 1-2h & h \\ -4h & 1+2h \end{bmatrix} \in \Delta$. So we find that the cusp width of $\frac{1}{p}$ in $X_1(M, N)$ is $\frac{N}{p}$ and the cusp width of $\frac{1}{2}$ in $X_0(\frac{N}{M})$ is $\frac{N}{4M}$ (this uses that $4 \mid \frac{N}{M}$), so $e_1 = \frac{N/2}{N/4M} = 2M$. If $\frac{N}{4M}$ is odd, then 4 is a Hall divisor of $\frac{N}{M}$, so $W(\frac{N}{M})$ contains w_4 , which fixes $\frac{1}{2}$. No other nontrivial Atkin–Lehner involution fixes any cusp, so $e_2 = 2$ if $\frac{N}{4M}$ is odd and $e_2 = 1$ if $\frac{N}{4M}$ is even. This computes $\#G_{\frac{1}{2}}$ and completes the proof of part a).

b) By Lemma 8.17c), every point P in the G -orbit of $\frac{1}{2}$ is a fixed point of the involution γ_2 on $X_1(M, N)$, so by Schoeneberg–Lewittes, P is a Weierstrass point if this G -orbit contains at least 5 points. By (14) this G -orbit has size at least $\frac{\varphi(N)}{4}$, so this orbit has size at least 5 for all N such that $\varphi(N) \geq 20$ and thus for all $N \geq 61$. Of the finitely many pairs (M, N) satisfying the hypotheses and with $N \leq 60$, using (14) we find that this G -orbit has size at least 5 unless

$$(M, N) \in \{(3, 12), (1, 16), (1, 20)\}.$$

We have encountered all of these pairs already, in Examples 6.10, 6.3 and 6.9: in each case we saw that all non-principal (a.k.a. non-Hall, when $M = 1$) cusps are Weierstrass points. $\square$

Theorem 8.19. *Let $p \geq 3$ be a prime, and let $M, N \in \mathbb{Z}^+$ with $p^2 M \mid N$ and $N \geq 18$. Put $G := \text{Aut}_M X_1(M, N)$.*

- a) *The G -orbit on the nonprincipal cusp $\frac{1}{p}$ has size $\frac{2^{\omega(\frac{N}{M})-1} \varphi(N) \psi(N)}{p \psi(\frac{N}{M})}$.*
- b) *Suppose that $(M, N) \neq (1, 18)$. Then every point in the G -orbit of $\frac{1}{p}$ is a Weierstrass point of $X_1(M, N)$.*

Proof. a) If $G_{1/p}$ is the isotropy subgroup at $\frac{1}{p}$, then the size of the G -orbit on $\frac{1}{p}$ is $[G : G_{1/p}]$. A formula for $\#G$ is given in Corollary 8.8, while $\#G_{1/p}$ is the ramification index of $\frac{1}{p}$ in the map $X_1(M, N) \rightarrow X_0^*(\frac{N}{M})$. As mentioned above, for $n \in \mathbb{Z}^+$, some cusp is ramified in the map $X_0(n) \rightarrow X_0^*(n)$ if and only if 4 is a Hall divisor of n , in which case the unique ramified cusp is $\frac{1}{2}$. In particular the cusp $\frac{1}{p}$ is unramified in the map $X_0(\frac{N}{M}) \rightarrow X_0^*(\frac{N}{M})$, so $\#G_{1/p}$ is the ramification index of $\frac{1}{p}$ in the map $X_1(M, N) \rightarrow X_0(\frac{N}{M})$, which is the width of $\frac{1}{p}$ in $X_1(M, N)$ divided by the width of $\frac{1}{p}$ in $X_0(\frac{N}{M})$. By Example 6.14b), for a finite index subgroup Δ of $\text{PSL}_2(\mathbb{Z})$, the width of $\frac{1}{p}$ on $X(\Delta)$ is the least $h \in \mathbb{Z}^+$ such that $\pm \begin{bmatrix} 1 - ph & h \\ -p^2 h & 1 + ph \end{bmatrix} \in \Delta$. So we find that the cusp width of $\frac{1}{p}$ in $X_1(M, N)$ is $\frac{N}{p}$ and the cusp width of $\frac{1}{p}$ in $X_0(\frac{N}{M})$ is $\frac{N}{p^2 M}$, so $\#G_{1/p} = pM$.

b) Let us denote by $\mathfrak{c}$ the size of the G -orbit on $\frac{1}{p}$. By Lemma 8.17c), every point P in the G -orbit of $\frac{1}{p}$ is a fixed point of the automorphism γ_p of $X_1(M, N)$, so by Schoeneberg–Lewittes, P is a Weierstrass point if $\mathfrak{c} \geq 5$. Write $N = p^a n$ with $p \nmid n$, so by hypothesis $a \geq 2$. From part a), we have $\mathfrak{c} \geq \frac{\varphi(N)}{p} \geq p^{a-2}(p-1)$, which is at least 6 if either $p \geq 7$ or $a \geq 3$.

• Suppose $p = 3$ and $a = 2$, so we may write $N = 9n$ with $3 \nmid n$. Then $\mathfrak{c} \geq \frac{\varphi(N)}{3} = 2\varphi(n)$, which is at least 6 if $n \geq 7$. In the remaining cases, the formula for $\mathfrak{c}$ given in part a) shows that $\mathfrak{c} \geq 5$ unless $(M, N) = (1, 18)$, which is indeed exceptional: in Example 6.3c) we saw that no cusp on $X_1(18)$ is a Weierstrass point.

• Suppose $p = 5$ and $a = 2$, so we may write $N = 25n$ with $5 \nmid n$. Then $\mathfrak{c} \geq 4\varphi(n)$, which is at least 8 if $n \geq 3$. In the remaining cases, the formula of part a) shows that $\mathfrak{c} \geq 5$ unless $(M, N) = (1, 25)$, where Example 6.15m) shows that every cusp is a Weierstrass point. $\square$

Corollary 8.20. *Let $M, N \in \mathbb{Z}^{\geq 2}$ with $M \mid N$ be such that $g(X_1(M, N)) \geq 2$ and one of the following holds:*

- (i) $N = 4bM$ with b odd and squarefree.
- (ii) $N = 8bM$ with b odd and squarefree.
- (ii) $N = 9bM$ with b indivisible by 3 and squarefree.

Then every cusp on $X_1(M, N)$ is a Weierstrass point.

Proof. By Lemma 8.16, the hypotheses ensure that there are two orbits of $G = \text{Aut}_M X_1(M, N)$ on the cusps. The result follows from Theorems 8.14, 8.18 and 8.19. $\square$

Remark 8.21.

- a) Taking $M = 1$ in Theorem 8.18, we get a result that lies strictly in between Choi’s Theorem (Theorem 4.8) and Jeon’s Theorem (Theorem 4.9). Some of our proof is inspired by Choi, who for an odd prime p shows that $\frac{1}{2}$ is a Weierstrass point by considering ramification in the map $X_1(4p) \rightarrow X_1^\bullet(1, 4p)$. Choi writes down $\frac{p-1}{2}$ ramified cusps “by hand,” hence

he needs $p \geq 11$ to apply Schoenberg–Lewittes. By Theorem 8.18 the size of the G -orbit on $\frac{1}{2}$ in this case is $p - 1$, so the method also works for $p = 7$. (As we saw, when $p = 5$ the proof breaks down but the result is still true.) When $M = 1$ and $4 \mid N$, the involution γ_2 is the diamond operator $\langle \frac{N}{2} + 1 \rangle$. Jeon’s proof makes use of a more general analysis of fixed points of diamond operators due to Jeon–Kim [JK07].

- b) Taking $M = 1$ in Theorem 8.19, we again get a special case of Jeon’s theorem (Theorem 4.9), and again γ_p is a diamond operator: $\langle \frac{N}{p} + 1 \rangle$.
- c) Because $G := \text{Aut}_M X_1(M, N)$ has more than one orbit on the cusps of $X_1(M, N)$ if and only if $\frac{N}{M}$ is divisible by p^2 for some prime p , Theorems 8.18 and 8.19 show that for every pair $(M, N) \neq (1, 18)$ such that $g(X_1(M, N)) \geq 2$ and G has more than one orbit on the cusps of $X_1(M, N)$, there is a non-principal cuspidal G -orbit consisting of Weierstrass points.
- d) As mentioned above, the second author has recently proved a result that together with the results of this section shows: if $M \mid N$ are such that $X_1(M, N)$ has genus at least 2, then every cusp on $X_1(M, N)$ is a Weierstrass point if and only if $(M, N) \notin \{(5, 10), (1, 16)\}$, in which cases, as we’ve already seen in Examples ?? and ??, no cusp is a Weierstrass point. The proof proceeds by identifying a family of diamond operators on $X_1(M, N)$ and showing that every nonprincipal cusp is a fixed point of some diamond operator satisfying the hypotheses of Schoenberg–Lewittes. The new approach is overall less group-theoretic than our present approach, which yields (in the various special cases in which it applies) some further information on sizes of cuspidal $\text{Aut}_M X_1(M, N)$ -orbits.

8.7. Circling Back to the Motivating Result. We asked two questions at the end of §5: suppose $X_1(M, N)$ has genus at least 2. Is the cusp 0 a Weierstrass point on $X_1(M, N)$? Are there any $\mathbb{Q}(\zeta_M)$ -rational cusps on $X_1(M, N)$ that are not Weierstrass points?

The work of this section along with the forthcoming work of the second author answers these questions when $M \geq 2$: when $X_1(M, N)$ has genus at least 2 and $(M, N) \notin \{(5, 10), (1, 16)\}$ then *every* cusp on $X_1(M, N)$ is a Weierstrass point. As we mentioned in §5, we needn’t restrict to cusps: any $\mathbb{Q}(\zeta_M)$ -rational non-Weierstrass point would work as well. But as mentioned above, it seems likely – though out of current reach to show – that there are only finitely many pairs (M, N) from which $Y_1(M, N)(\mathbb{Q}(\zeta_M)) \neq \emptyset$.¹⁵

When $M = 1$, the situation looks quite different: there are many small values of N for which the cusp 0 is *not* a Weierstrass point on $X_1(N)$, and also many small values for which it is. But we have only calculations; we are completely lacking a theorem. Understanding when Hall cusps are Weierstrass points on $X_1(N)$ is perhaps the most intriguing open problem in this area.

9. MODULAR CURVES WITH CUSPIDAL WEIERSTRASS DIVISOR

9.1. The Cusp-Index Ratio. In §6 we saw a few examples of modular curves for which all Weierstrass points are cusps. However, these examples are quite rare, and in this section we will give various classes of modular curves for which, first, only finitely many curves in the family have only cuspidal Weierstrass points and, second, we can find all of these exceptions.

¹⁵It follows from [CCM21, Thm. 1] that there are only finitely many pairs (M, N) such that $X_1(M, N)$ has a **CM** point with residue field any abelian number field, hence certainly also with residue field $\mathbb{Q}(\zeta_M)$.

For a finite index subgroup Γ of $\mathrm{PSL}_2(\mathbb{Z})$, we define the **cusp-index ratio**

$$\mathrm{CI}(\Gamma) := \frac{v_\infty(X(\Gamma))}{\iota(\Gamma)} = \frac{v_\infty(X(\Gamma))}{[\mathrm{PSL}_2(\mathbb{Z}) : \Gamma]}.$$

Lemma 9.1.

a) Let $\Gamma' \subseteq \Gamma \subseteq \mathrm{PSL}_2(\mathbb{Z})$ be finite index subgroups. Then

$$\mathrm{CI}(\Gamma') \leq \mathrm{CI}(\Gamma) \leq \min(1, [\Gamma : \Gamma'] \mathrm{CI}(\Gamma')).$$

b) Let $\Gamma \subseteq \mathrm{PSL}_2(\mathbb{Z})$ be a congruence subgroup such that $X(\Gamma)$ has genus at least 2. If

$$\mathrm{CI}(\Gamma) \leq \frac{325}{16384} < 0.02,$$

then $X(\Gamma)$ has noncuspidal Weierstrass points.

Proof. a) Because the natural map $X(\Gamma') \rightarrow X(\Gamma)$ has degree $\frac{\iota(\Gamma')}{\iota(\Gamma)}$, we have

$$v_\infty(X(\Gamma')) \leq \frac{\iota(\Gamma')}{\iota(\Gamma)} v_\infty(X(\Gamma)),$$

giving the first inequality. In particular we have

$$\mathrm{CI}(\Gamma) \leq \mathrm{CI}(\mathrm{PSL}_2(\mathbb{Z})) = 1.$$

Also

$$\mathrm{CI}(\Gamma) = \frac{v_\infty(X(\Gamma))}{\iota(\Gamma)} \leq \frac{v_\infty(X(\Gamma'))}{\iota(\Gamma')/[\Gamma : \Gamma']} = [\Gamma : \Gamma'] \mathrm{CI}(\Gamma').$$

b) By Riemann–Roch, we have $\gamma(X(\Gamma)) \leq g(X(\Gamma)) + 1$. Using (8), we find that the number of Weierstrass points on $X(\Gamma)$ is at least

$$2g(X(\Gamma)) + 2 \geq 2\gamma(X(\Gamma)) > \frac{325}{16384} \iota(\Gamma) \geq v_\infty(X(\Gamma)). \quad \square$$

Lemma 9.2. Let $\Gamma \subseteq \mathrm{PSL}_2(\mathbb{Z})$ be a congruence subgroup of level N .

a) We have $\mathrm{CI}(\Gamma) \geq \frac{1}{N}$.

b) The following are equivalent:

(i) $\mathrm{CI}(\Gamma) = \frac{1}{N}$.

(ii) Every cusp on $X(\Gamma)$ has width N .

(iii) All the cusps on $X(\Gamma)$ have the same width.

We say Γ is **cuspidally normal** if it satisfies these equivalent conditions.

c) Γ is cuspidally normal if it is normal in $\mathrm{PSL}_2(\mathbb{Z})$, or if it is any of the following Cartan-type groups: $\Gamma_{\mathrm{sp}}(N)$, $\Gamma_{\mathrm{sp}}^+(N)$, $\Gamma_{\mathrm{ns}}(N)$, $\Gamma_{\mathrm{ns}}^+(N)$ or $\Gamma_{\mathrm{ns}}^*(N)$, or if N is prime and $\Gamma = \Gamma_{S_4}(N)$.¹⁶

d) If $N \geq 51$ and Γ is cuspidally normal, then Γ has noncuspidal Weierstrass points.

Proof. a) Since every cusp on $\Gamma(N)$ has width N , we have $\mathrm{CI}(\Gamma(N)) = \frac{1}{N}$. Since $\Gamma(N) \subseteq \Gamma$, by Lemma 9.1 we have $\frac{1}{N} = \mathrm{CI}(\Gamma(N)) \leq \mathrm{CI}(\Gamma)$.

b) Since every cusp on Γ has width dividing N , we have (i) $\iff$ (ii), and clearly (ii) $\implies$ (iii). By a result of Larcher [La82, Thm. 3], Γ has a cusp of width N , so (iii) $\implies$ (ii).

c) Since the cusp widths of $X(\Gamma)$ are the ramification indices over ∞ in the map $\pi : X(\Gamma) \rightarrow X(1)$, if Γ is normal in $\mathrm{PSL}_2(\mathbb{Z})$ then π is a Galois covering, so all such ramification indices are equal. That each Cartan-type subgroups and $\Gamma_{S_4}(N)$ is cuspidally normal follows from the formulas given for the modular index and number of cusps given in [HL25]. $\square$

Proposition 9.3. Let $M, N \in \mathbb{Z}^+$ with $N \geq 5$ and $M \mid N$.

¹⁶We recall that $\Gamma_{\mathrm{ns}}^*(N)$ has level $\frac{N}{2}$ when $N \equiv 2 \pmod{4}$ and $\Gamma_{S_4}(3)$ has level 1.

- a) We have $\text{CI}(\Gamma_0(N)) \leq \frac{1}{\sqrt{N}}$. Thus if $N \geq 2542$, $X_0(N)$ has noncuspidal Weierstrass points.
b) We have $\text{CI}(\Gamma_1(M, N)) \leq \text{CI}(\Gamma_1(N)) \leq N^{-1 + \frac{1.5379 \log 2}{\log \log N}} = O_\epsilon(N^{-1+\epsilon})$.

Proof. a) By Lemma 3.5, we have $\text{CI}(\Gamma_0(N)) \leq \frac{\psi(N)\sqrt{N}}{\psi(N)} = \frac{1}{\sqrt{N}}$. By Lemma 9.1, $X_0(N)$ has a noncuspidal Weierstrass point if $\frac{1}{\sqrt{N}} \leq \frac{325}{16384}$, hence for all $N \geq 2542$.

b) Since $\Gamma_1(M, N) \subseteq \Gamma_1(N)$, by Lemma 9.1a), we have $\text{CI}(\Gamma_1(M, N)) \leq \text{CI}(\Gamma_1(N))$. Moreover

$$\iota(\Gamma_1(N)) = \frac{N^2}{2} \prod_{p|N} \left(1 - \frac{1}{p^2}\right) \leq \frac{N^2}{2}.$$

Using this and Lemma 3.5b), we get the claimed upper bound on $\text{CI}(\Gamma_1(N))$. $\square$

Remark 9.4.

- a) For all $a \in \mathbb{Z}^+$, put $N := 2^{2a}$. Then $\text{CI}(\Gamma_0(N)) = \frac{1}{\sqrt{N}}$. Thus Proposition 9.3a) is sharp.
b) Since $\iota(\Gamma_1(N)) \approx N^2$ and $\limsup_N \frac{v_\infty(X_1(N))}{N} = \infty$, we *do not have* $\text{CI}(X_1(N)) = O(1/N)$. Thus the upper bound on $\text{CI}(\Gamma_1(N))$ given by Proposition 9.3 is asymptotically close to being sharp. However, it is not a great bound for small N : combining with Lemma 9.1, we get that $X_1(N)$ has a noncuspidal Weierstrass point for all $N \geq 3069$. We can do better than this, as seen in the next result.

Lemma 9.5. *Let $N \geq 855$, let Γ be a finite index subgroup of $\Gamma_1(N)$, and let $X(\Gamma)$ be the corresponding modular curve. Then $X(\Gamma)$ has a noncuspidal Weierstrass point.*

Proof. Let $N \in \mathbb{Z}^{\geq 855}$. One checks that

$$\frac{N^2}{2\pi^2} - \frac{1}{2}N^{1 + \frac{1.5379 \log 2}{\log \log N}} > \frac{1}{2}N^{1 + \frac{1.5379 \log 2}{\log \log N}}.$$

Let $d = [\Gamma_1(N) : \Gamma]$, so the map $X(\Gamma) \rightarrow X_1(N)$ has degree d . By Riemann–Hurwitz and Proposition 3.2b) we have

$$\begin{aligned} 2g(X(\Gamma)) + 2 &\geq 2g(X(\Gamma)) - 2 \geq d(2g(X_1(N)) - 2) \\ &\geq d \left(\frac{N^2}{2\pi^2} - \frac{1}{2}N^{1 + \frac{1.5379 \log 2}{\log \log N}} \right) > d \left(\frac{1}{2}N^{1 + \frac{1.5379 \log 2}{\log \log N}} \right) \geq v_\infty(X(\Gamma)). \end{aligned}$$

Thus not every Weierstrass point on $X(\Gamma)$ can be a cusp. $\square$

9.2. The Answer in Our Named Families. In the previous section we saw that in the “named families” of modular curves that appear in the LMFDB, only finitely many curves could have only cuspidal Weierstrass points. In this section we will determine *all* curves in each of these families with only cuspidal Weierstrass points.

Theorem 9.6. *Let $N \in \mathbb{Z}^+$ be such that $g(X_0(N)) \geq 2$. If $N \neq 64$, then there are noncuspidal Weierstrass points on $X_0(N)$.*

Proof. By Proposition 9.3a), if $N \geq 2542$ then $X_0(N)$ has noncuspidal Weierstrass points. The values of $N < 2542$ for which $X_0(N)$ has genus at least 2 and $2g(X_0(N)) + 2 \leq v_\infty(X_0(N))$ are: 28, 30, 40, 45, 48, 50, 54, 64, 72, 75, 81, 98, 100.

Step 1: For $N \in \{28, 30, 50\}$, the modular curve $X_0(N)$ is hyperelliptic and [Og74, Thm. 2] the hyperelliptic involution is an Atkin–Lehner involution w_d with $d \neq 4$, so by [Og74, Prop. 3] none of the Weierstrass points on $X_0(N)$ are cusps.

Step 2: By Example ??, no cusp on either $X_0(40)$ or $X_0(48)$ is a Weierstrass point.

Step 3: For $N \in \{45, 75, 98\}$, the curve has exactly $2g(X_0(N)) + 2$ cusps but is not hyperelliptic, so there is a noncuspidal Weierstrass point.

Step 4: The modular curves $X_0(54)$ and $X_0(81)$ each have genus 4, 12 cusps and are not hyperelliptic. From (4) we find that a nonhyperelliptic curve of genus 4 has at least 16 Weierstrass points, so each of $X_0(54)$ and $X_0(81)$ has a noncuspidal Weierstrass point.

Step 5: The modular curve $X_0(72)$ curve has genus $g = 5$ and 16 cusps, so the number of cusps exceeds $2g + 2$. However, this curve is not hyperelliptic. In §2.9 we saw that a nonhyperelliptic curve of genus 5 has at least 24 Weierstrass points. Thus there is a noncuspidal Weierstrass point.

Step 6: The modular curve $X_0(100)$ has genus 7, 18 cusps and is not hyperelliptic. From (4) we find that a nonhyperelliptic curve of genus 7 has at least 25 Weierstrass points. Thus there is a noncuspidal Weierstrass point. $\square$

Corollary 9.7. *Let $N \in \mathbb{Z}^+$ be such that $g(X_{\text{sp}}(N)) \geq 2$. Then there are noncuspidal Weierstrass points on $X_{\text{sp}}(N)$ if and only if $N \neq 8$.*

Proof. This is immediate from Theorem 9.6 and Remark 3.1. $\square$

Theorem 9.8. *Let $N \in \mathbb{Z}^+$ be such that $g(X_1(N)) \geq 2$. If $N \neq 16$, then there are noncuspidal Weierstrass points on $X_1(N)$.*

Proof. Step 1: For all $N \in [37, 855]$, we check that there are fewer than $2g(X_1(N)) + 2$ cusps on $X_1(N)$, hence there is a noncuspidal Weierstrass point. We have $N \leq 36$ and $g(X_1(N)) \geq 2$ if and only if $N \in \{13, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 27, 28, 30, 36\}$.

Step 2: The curve $X_1(N)$ is hyperelliptic if and only if it has genus 2 if and only if $N \in \{13, 16, 18\}$. As mentioned above, these cases were analyzed by Jeon in [Je15] and also treated above. We saw that if $N \in \{13, 18\}$ then no cusp on $X_1(N)$ is a Weierstrass point, while for $N = 16$ the six Weierstrass points on $X_1(16)$ are precisely the 6 non-Hall cusps, so 16 is a genuine exception.

Step 3: Let $N = 20$, $X_1(N)$ has genus 3. As we saw above, precisely 4 of the 20 cusps are Weierstrass points, and there are 16 noncuspidal Weierstrass points.

Step 4: Let $N \in \{21, 30\}$, so N is squarefree. In these cases a modular forms calculation shows that ∞ is not a Weierstrass point on $X_1(N)$, and since N is squarefree, $\text{Aut}_M X_1(N)$ acts transitively on cusps and thus $X_1(N)$ has no cuspidal Weierstrass points.

Step 5: Let $N = 24$. Then $X_1(24)$ has genus 5, 24 cusps and is not hyperelliptic, so has at least 24 Weierstrass points. Thus if all Weierstrass points are cuspidal then conversely every cusp must be a Weierstrass point, but a modular forms calculation shows that ∞ is not a Weierstrass point on $X_1(24)$, so there are noncuspidal Weierstrass points.

Step 6: Let $N \in \{19, 22, 23, 25, 26, 27, 28, 36\}$. Since $X_1(N)$ is not hyperelliptic, (4) shows that there are more Weierstrass points than cusps on $X_1(N)$. $\square$

Theorem 9.9 (Schoeneberg). *Let $N \geq 7$, so $X(N)$ has genus at least 3. The following are equivalent:*

- (i) *Every Weierstrass point on $X(N)$ is a cusp.*
- (ii) *The Weierstrass points on $X(N)$ are precisely the cusps.*
- (iii) *$N \in \{7, 8\}$.*

Proof. This follows from Theorem 4.1, using the existence of Weierstrass points with j -invariants 1728 or 0: cf. §8.1. We will give another proof using the method of this section.

Step 1: The curve $X(7)$ is treated in Example 6.8 and again in 8.2, while the curve $X(8)$ is treated in Example 8.3.

Step 2: Let $N \geq 12$. We have

$$2g(X(N)) + 2 = 4 + \frac{N^2(N-6)}{12} \prod_{p|N} \left(1 - \frac{1}{p^2}\right)$$

and

$$v_\infty(X(N)) = \frac{N^2}{2} \prod_{p|N} \left(1 - \frac{1}{p^2}\right).$$

It follows that $2g(X(N)) + 2 > v_\infty(X(N))$, and thus there is a noncuspidal Weierstrass point.

Step 3: Since $\text{Aut } X(N)$ acts transitively on the cusps, all cusps of $X(N)$ have the same Weierstrass weight, so if every Weierstrass point is a cusp then we must have

$$v_\infty(X(N)) \mid g(X(N))^3 - g(X(N)).$$

For $N \in \{9, 10, 11\}$ this divisibility does *not* hold, so there is a noncuspidal Weierstrass point. $\square$

Theorem 9.10. *Let $M \mid N$ be positive integers such that $1 < M < N$ and $g(X_1(M, N)) \geq 2$. The following are equivalent:*

- (i) *Every Weierstrass point on $X_1(M, N)$ is a cusp.*
- (ii) *The Weierstrass points on $X_1(M, N)$ are precisely the cusps.*
- (iii) *$(M, N) = (3, 12)$.*

Proof. By Example 6.10, on $X_1(3, 12)$ the Weierstrass points are precisely the cusps, while by Example 8.10 no cusp on $X_1(5, 10)$ is a Weierstrass point.

Since $X_1(M, N)$ covers $X_1(N)$, by Corollary 9.5 we need only check (M, N) with $N \leq 855$. The pairs $(M, N) \notin \{(5, 10), (3, 12)\}$ with $N \leq 855$ such that $2g(X_1(M, N)) + 2 \leq v_\infty(X_1(M, N))$ are

$$(15) \quad (4, 12), (6, 12), (2, 14), (7, 14), (3, 15), (5, 15), (2, 16), (4, 16), (2, 18), (3, 18), (2, 20), (2, 24).$$

Since we have at least $2(g(X_1(M, N)) + 2)$ Weierstrass points on $X_1(M, N)$, for all pairs *not* on the above list, there is a noncuspidal Weierstrass point on $X_1(M, N)$. Example 6.15 shows that for each of the pairs (M, N) in (15), the curve $X_1(M, N)$ has noncuspidal Weierstrass points. $\square$

Example 9.11. The curve $X_1(4, 12)$ has genus 5 and 24 cusps. The map $\pi : X_1(4, 12) \rightarrow X_1(2, 12)$ is the quotient by a bielliptic involution ι , whose 8 fixed points are cuspidal Weierstrass points, including the cusp at ∞ . Let P be a cusp of $X_1(4, 12)$. By Theorem 8.11c), there is $\sigma \in \text{Aut}_M X_1(4, 12)$ such that $\sigma\infty = P$, so P is fixed by the bielliptic involution $\sigma\iota\sigma^{-1}$. Thus $X_1(4, 12)$ is triply bielliptic, and the 24 cusps are the $8 + 8 + 8$ fixed points of these bielliptic involutions. By Example 6.15a), the common Weierstrass weight at each cusp is 3.

Theorem 8.5 gives $\#\text{Aut}_M X_1(4, 12) = 96$. It follows that all automorphisms of $X_1(4, 12)$ are modular: if not, we would have $\#\text{Aut } X_1(4, 12) \geq 192$, and as mentioned above the only genus 5 Riemann surface with more than 160 automorphisms is the Wiman curve $X(8)$, which has 192 automorphisms. Because on $X(8)$ all Weierstrass points have weight 5 and $X_1(4, 12)$ has 24 points of Weierstrass weight 3, these two curves are not isomorphic. By [MSSV02] there is a unique curve of genus 5 with 96 automorphisms; it has automorphism group $G \cong \text{GL}_2(\mathbb{Z}/4\mathbb{Z})$.¹⁷ The map $X_1(4, 12) \rightarrow X_1(4, 12)/G$ is a Galois Belyĭ map, with order triple $(2, 4, 6)$. Thus $X_1(4, 12)$ has precisely 3 ramified G -orbits, of sizes 16, 24 and 48. The orbit of size 24 consists of the cuspidal Weierstrass points, contributing 72 to $\deg \mathbb{W}(X_1(4, 12))$. The remaining Weierstrass points contribute to $\mathbb{W}(X_1(4, 12))$ with degree 48, so there are just two possibilities: either there are 48 more Weierstrass points, each of weight 1, or there are 16 more Weierstrass points, each of weight 3. At present we don't know which is the case.

Theorem 9.12.

- a) *Let $N \in \mathbb{Z}^{\geq 10}$, so $g(X_{\text{sp}}^+(N)) \geq 2$. Then $X_{\text{sp}}^+(N)$ has a noncuspidal Weierstrass point.*

¹⁷See <https://faculty.fordham.edu/dswinarski/RiemannSurfaceAutomorphisms/genus-5.htm> for generators for G and equations for the curve.

- b) Let $N \in \{9\} \cup \mathbb{Z}^{\geq 11}$, so $g(X_{\text{ns}}(N)) \geq 2$. Then $X_{\text{ns}}(N)$ has a noncuspidal Weierstrass point.
- c) Let $N \in \{13, 15, 16, 17\} \cup \mathbb{Z}^{\geq 19}$, so $g(X_{\text{ns}}^+(N)) \geq 2$. Then $X_{\text{ns}}^+(N)$ has a noncuspidal Weierstrass point.
- d) Let $N \in \mathbb{Z}^+$ be such that $g(X_{\text{ns}}^*(N)) \geq 2$. Then $X_{\text{ns}}^*(N)$ has a noncuspidal Weierstrass point.

Proof. In all cases, Proposition 9.3b) implies that we only need to consider $N \leq 50$.

a) For $13 \leq N \leq 50$, we find that $v_{\infty}(X_{\text{sp}}^+(N)) < 2g(X_{\text{sp}}^+(N)) + 2$, so for these values of N there is a non-cuspidal Weierstrass point. We have that $v_{\infty}(X_{\text{sp}}^+(12)) = 12 = 2g(X_{\text{sp}}^+(12)) + 2$, but the curve $X_{\text{sp}}^+(12)$ is not hyperelliptic, so it must have more than 12 Weierstrass points and thus at least one noncuspidal Weierstrass point. Example 6.4 shows that $X_{\text{sp}}^+(11)$ has a noncuspidal Weierstrass point, while Example 6.11a) shows that $X_{\text{sp}}^+(10)$ has a noncuspidal Weierstrass point.

b) For $12 \leq N \leq 50$, we find that $v_{\infty}(X_{\text{ns}}(N)) < 2g(X_{\text{ns}}(N)) + 2$, yielding a non-cuspidal Weierstrass point. We have $v_{\infty}(X_{\text{ns}}(11)) = 10 = 2g(X_{\text{ns}}(11)) + 2$, but the curve $X_{\text{ns}}(11)$ is not hyperelliptic, so it has more than 10 Weierstrass points and thus a noncuspidal Weierstrass point. Example 6.4 shows that no cusp on $X_{\text{ns}}(9)$ is a Weierstrass point.

c), d) We write $X_{\text{ns}}^{\bullet}(N)$ for either $X_{\text{ns}}^+(N)$ or $X_{\text{ns}}^*(N)$. For all $N \leq 50$ such that $g(X_{\text{ns}}^{\bullet}(N)) \geq 2$, we find that $v_{\infty}(X_{\text{ns}}^{\bullet}(N)) < 2g(X_{\text{ns}}^{\bullet}(N)) + 2$, yielding a noncuspidal Weierstrass point. $\square$

Theorem 9.13. *Let $p > 2$ be a prime number such that the modular curve $X_{S_4}(p)$ has genus at least 2: that is, $p \geq 13$. Then $X_{S_4}(p)$ has a noncuspidal Weierstrass point.*

Proof. By Proposition 9.3c), for all $p \geq 53$, the curve $X_{S_4}(p)$ has a noncuspidal Weierstrass point. Using [HL25, Table 1], we find: for all $13 \leq p \leq 47$, we have $2g(X_{S_4}(p)) + 2 > v_{\infty}(X_{S_4}(p))$.¹⁸ $\square$

9.3. A Crude Calculation. We will make a crude calculation that points to an open problem.

Example 9.14. For a prime number p , let

$$\mathfrak{d}(p) := \sum_{\text{cusps } P \text{ on } X(p)} w_P$$

be the degree of the cuspidal part of the Weierstrass divisor on $X(p)$. We will give upper and lower bounds on $\mathfrak{d}(p)$. For reasons that will soon become clear, we will suppress constant factors: by $f(p) \ll g(p)$ we mean there is a fixed $C > 0$ such that $f(p) \leq Cg(p)$ for all p ; by $f(p) \gg g(p)$ we mean $g(p) \ll f(p)$; and by $f(p) \asymp g(p)$ we mean $f(p) \ll g(p)$ and $f(p) \gg g(p)$. We have

$$g(X(p)) \asymp p^3,$$

so

$$\deg \mathbb{W}(X(p)) = g(X(p))^3 - g(X) \asymp p^9.$$

Since

$$v_{\infty}(X(p)) = \frac{p^2 - 1}{2} \asymp p^2$$

and for any $P \in X(p)$ we have

$$w_P \ll g(X(p))^2 \asymp p^6,$$

it follows that

$$(16) \quad \mathfrak{d}(p) \ll p^2 p^6 = p^8,$$

¹⁸When $p = 3$, *loc. cit.* says that the number of cusps on $X_{S_4}(p)$ is $\frac{1}{3}$, which is clearly incorrect. In fact the corresponding congruence subgroup $\Gamma_{S_4}(3)$ is just $\text{PSL}_2(\mathbb{Z})$, so $X_{S_4}(3) = X(1)$.

and thus

$$\lim_{p \rightarrow \infty} \frac{\mathfrak{d}(p)}{\deg \mathbb{W}(X(p))} = 0.$$

On the other hand, by Corollary 8.1 the $\frac{p^2-1}{2}$ cusps of $X(p)$ are partitioned into $p+1$ subsets of size $\frac{p-1}{2}$, each subset being the fixed points of an order p automorphism of $\mathrm{PSL}_2(\mathbb{F}_p)$. Using Theorem 2.25, it follows that each cusp P has Weierstrass weight $w_P \gg p^3$, and thus

$$(17) \quad \mathfrak{d}(p) \gg p^2 p^3 = p^5.$$

Our conclusion: we can give nontrivial upper and lower bounds on $\mathfrak{d}(p)$. Our upper bound shows that not only is $\mathbb{W}(X(p))$ not entirely cuspidal for sufficiently large p , but as p grows the “proportion” of $\mathbb{W}(X(p))$ that is cuspidal tends to 0. Our lower bound shows that both the number and weight of cuspidal Weierstrass points on $X(p)$ grows steadily with p . However, our lower and upper bounds do not have the same orders of magnitude.

One can do similar calculations for other families of modular curves on which we’ve produced cuspidal Weierstrass points: again we can prove nontrivial upper and lower bounds on the degree of the cuspidal part of the Weierstrass divisor, and again these bounds will have different orders of magnitude. Thus a more precise understanding of the degree of the cuspidal part of the Weierstrass divisor on a modular curve remains open.

9.4. A Conjecture on the Cusp-Index Ratio. The results of §9.1 suggest the following:

Conjecture 9.15. *The cusp-index ratio $\mathrm{CI}(\Gamma)$ tends to 0 in the family of all congruence subgroups of $\mathrm{PSL}_2(\mathbb{Z})$: for any $\epsilon > 0$, there are only finitely many congruence subgroups Γ with $\mathrm{CI}(\Gamma) \geq \epsilon$.*

Via Lemma 9.1, Conjecture 9.15 implies that as we range over all modular curves, only finitely many can have only cuspidal Weierstrass points. In fact, it implies that “most Weierstrass points on most modular curves are noncuspidal,” as we worked out for the curves $X(p)$ in Example 9.14.

In addition to the named families considered here, Lemma 9.2 shows that Conjecture 9.15 holds for all congruence subgroups Γ that are cuspidally normal. This includes the class of normal congruence subgroups, which is however quite small. Normal congruence subgroups of $\mathrm{PSL}_2(\mathbb{Z})$ were classified by Newman [Ne63] and McQuillan [McQ65]. In particular, if $\gcd(N, 6) = 1$, then a congruence subgroup Γ of level N is normal if and only its image in $\mathrm{PSL}_2(\mathbb{Z}/N\mathbb{Z})$ lies in the center Z_N , which is the subgroup of scalar matrices. Since Z_N is isomorphic to the 2-torsion subgroup of $(\mathbb{Z}/N\mathbb{Z})^\times / \{\pm 1\}$, if N is moreover a prime power, then $\Gamma = \Gamma(N)$.

Here is another partial affirmative result towards the conjecture:

Theorem 9.16. *Let $p \geq 61$ be prime, let $a \in \mathbb{Z}^+$, and let $\Gamma \subseteq \mathrm{PSL}_2(\mathbb{Z})$ be a congruence subgroup of level p^a . Then:*

a) *We have*

$$\mathrm{CI}(\Gamma) \leq \max \left(\frac{1}{\sqrt{p}}, \frac{60}{p} \right).$$

b) *If $p \geq 3037$, the modular curve $X(\Gamma)$ has a noncuspidal Weierstrass point.*

Proof. a) Step 1: Suppose that $a = 1$, i.e., $\Gamma(p) \subseteq \Gamma \subseteq \mathrm{PSL}_2(\mathbb{Z})$. Let $q : \mathrm{PSL}_2(\mathbb{Z}) \rightarrow \mathrm{PSL}_2(\mathbb{Z}/p\mathbb{Z})$ be the natural map. Since Γ has level p , we have $\Gamma = q^{-1}(G)$ for a proper subgroup G of $\mathrm{PSL}_2(\mathbb{Z}/p\mathbb{Z})$. Let $\tilde{G}$ be a maximal subgroup of $\mathrm{PSL}_2(\mathbb{Z}/p\mathbb{Z})$ containing G . By a theorem of Dickson – see e.g. [Su, Thm. 6.25] – $\tilde{G}$ is either a Borel subgroup, the normalizer of a Cartan subgroup, or is isomorphic to A_4 , S_4 or A_5 . In the first case we have that (up to conjugacy)

$\Gamma \subseteq \Gamma_0(p)$, hence $\text{CI}(\Gamma) \leq \text{CI}(\Gamma_0(p)) \leq \frac{1}{\sqrt{p}}$ by Lemma 9.3a). In the second case we have that up to conjugacy Γ is contained in either $\Gamma_{\text{sp}}^+(p)$ or $\Gamma_{\text{ns}}^+(p)$, hence $\text{CI}(\Gamma) \leq \text{CI}(\Gamma_{\text{sp}}^+(p)) = \text{CI}(\Gamma_{\text{ns}}^+(p)) = \frac{1}{p}$ by Lemma 9.2. In the third case we have $[\Gamma : \Gamma(p)] \leq 60$, so by Lemmas 9.1 and 9.2 we have $\text{CI}(\Gamma) \leq 60 \text{CI}(\Gamma(p)) = \frac{60}{p}$.

Step 2: To complete the proof of part a), it suffices to show: if $p \geq 5$ and Γ is a maximal subgroup of $\text{PSL}_2(\mathbb{Z})$ containing $\Gamma(p^a)$, then Γ contains $\Gamma(p)$. For $1 \leq a < b$, let $q_a : \text{PSL}_2(\mathbb{Z}) \rightarrow \text{PSL}_2(\mathbb{Z}/p^a\mathbb{Z})$ and $q_{a,b} : \text{PSL}_2(\mathbb{Z}/p^b\mathbb{Z}) \rightarrow \text{PSL}_2(\mathbb{Z}/p^a\mathbb{Z})$ be the natural maps. There is a maximal subgroup G of $\text{PSL}_2(\mathbb{Z}/p^a\mathbb{Z})$ such that $\Gamma = q_a^{-1}(G)$. It follows from [Se, Lemmas 3.4.2 and 3.4.3] that $\underline{G} := q_{a,1}(G)$ is a proper subgroup of $\text{PSL}_2(\mathbb{Z}/p\mathbb{Z})$,¹⁹ hence it is contained in a maximal subgroup $\underline{H}$ of $\text{PSL}_2(\mathbb{Z}/p\mathbb{Z})$. Thus Γ is a maximal subgroup of $\text{PSL}_2(\mathbb{Z})$ contained in the proper subgroup $q^{-1}(\underline{H})$, so $\Gamma = q^{-1}(\underline{H})$ and thus Γ contains $\Gamma(p)$.

b) If $p > 3037$ then

$$\text{CI}(\Gamma) \leq \max\left(\frac{1}{\sqrt{p}}, \frac{60}{p}\right) \leq \frac{325}{16384},$$

so by Lemma 9.1b) the curve $X(\Gamma)$ has noncuspidal Weierstrass points. $\square$

Since there are only finitely many congruence subgroups of level less than 3037, in particular Conjecture 9.15 holds when restricted to the class of prime level modular curves.

9.5. Counterexamples for Noncongruence Subgroups. We end by showing that if we remove the restriction to congruence subgroups of $\text{PSL}_2(\mathbb{Z})$, then Conjecture 9.15 – and even its implication on the finiteness of modular curves with only cuspidal Weierstrass points – becomes false.

Example 9.17. The modular curve $X(2)$ has no elliptic elements, has three cusps, represented by 0, 1 and ∞ , and has genus 0, hence [K, p. 98] the Poincaré presentation of $\Gamma(2)$ is

$$\Gamma(2) = \langle x, y, z \mid xyz = 1 \rangle,$$

so $\Gamma(2)$ is freely generated by x and y . Let $G = \langle X, Y \rangle$ be a finite group generated by an element X , of order a , together with an element Y , of order b , and let c be the order of XY . There is then a unique surjective group homomorphism from $\Gamma(2)$ to G mapping x to X and y to Y ; let $\Gamma(G)$ be the kernel of this map, and let $X(G) := \Gamma(G) \backslash \mathcal{H}$ be the corresponding Riemann surface. Thus $\Gamma(G)$ is a finite index subgroup of $\text{PSL}_2(\mathbb{Z})$ but in general *not* a congruence subgroup. We have

$$\iota(\Gamma(G)) = \iota(\Gamma(2)) \cdot \#G = 6 \cdot \#G.$$

The natural map

$$\pi : X(G) \rightarrow X(2)$$

is a Galois covering with Galois group G . The ramification indices over the cusps 0, 1, ∞ are a, b, c respectively, so the number of cusps on $X(G)$ is

$$\#\pi^{-1}(0) + \#\pi^{-1}(1) + \#\pi^{-1}(\infty) = \#G \cdot \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right),$$

and thus the cusp-index ratio of $\Gamma(G)$ is

$$\text{CI}(\Gamma(G)) = \frac{1}{6} \cdot \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right).$$

¹⁹Here we use that $p \geq 5$; when $p \in \{2, 3\}$ this is known not to always hold.

We now specialize to the following situation: for an integer $g \geq 2$, let G be the additive group $\mathbb{Z}/4g\mathbb{Z}$ and take $X = 1 \pmod{4g}$, $Y = 2g - 1 \pmod{4g}$ and $Z = -2g \pmod{4g}$. Then $X + Y + Z = 0$ and $G = \langle X \rangle = \langle X, Y \rangle$. Moreover $a = b = 4g$ and $c = 2$, so

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{2g} + \frac{1}{2} > \frac{1}{2},$$

and thus $\text{CI}(\Gamma(G)) > \frac{1}{12}$.

Still with this choice of G , X and Y , the fibers $\pi^{-1}(0)$ and $\pi^{-1}(1)$ each consist of a single point, while the fiber $\pi^{-1}(\infty)$ consists of $2g$ points. By Riemann–Hurwitz we have $g(X(G)) = g$. Moreover the elements of $\pi^{-1}(0)$, $\pi^{-1}(1)$ and $\pi^{-1}(\infty)$ are all fixed points of the involution Z (the latter because G is commutative, so the decomposition groups of the elements of the fiber are not just conjugate but equal), so Z has at least $2g + 2$ fixed points and therefore is a hyperelliptic involution on $X(G)$. The Weierstrass points on $X(G)$ are precisely the cusps.

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